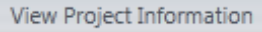
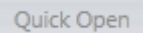
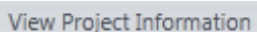
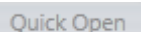
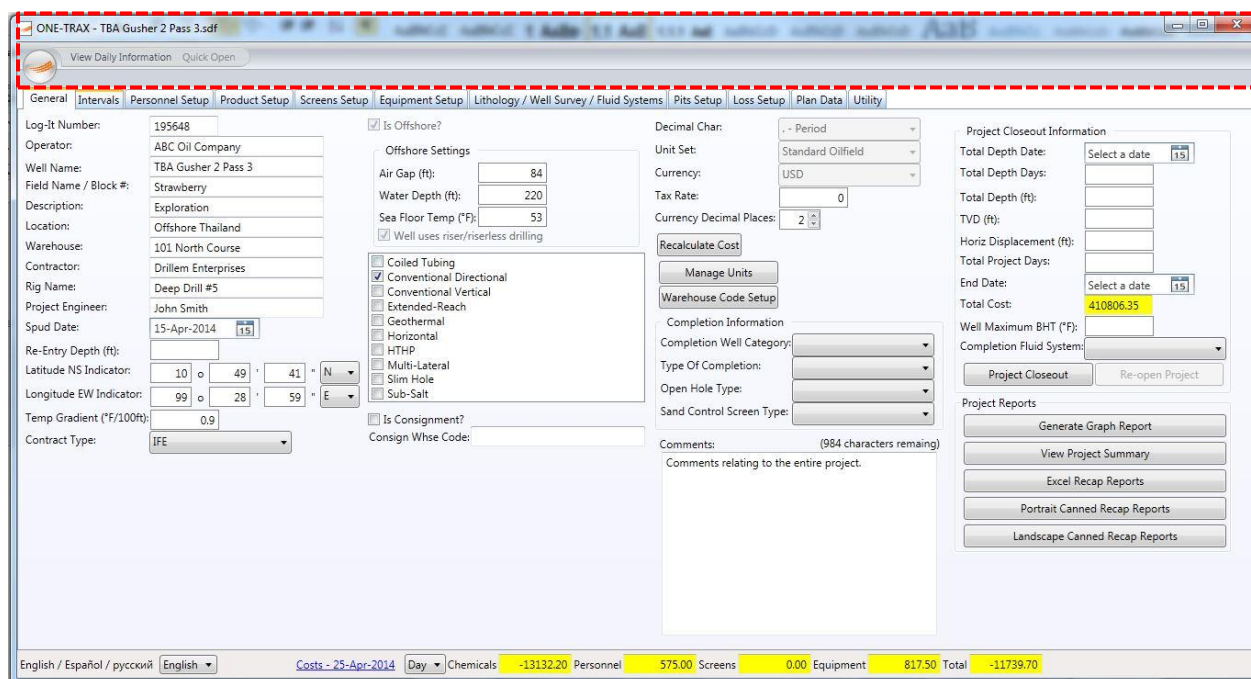


2.2 UPPER WELL INFORMATION RIBBON

The Upper Well Information Ribbon outlined in red, is always present when ONE-TRAX is running. The information and buttons on it are:

- The name of the active well
- The main menu button 
- The control to move between the two views of ONE-TRAX, the project information view and the daily information view. The project view contains the main project information, the information used to set up your well file; the daily view is used to enter your daily operational information.
-  Clicking on the button will change to that view of the program.
-  opens the last active well file and is inactive when a well file is active
-  moves the location of the  and  buttons.
- The standard Windows Minimize, Maximize, and Exit controls; .



ONE-TRAX - TBA Gusher 2 Pass 3.sdf

View Daily Information Quick Open

General Intervals Personnel Setup Product Setup Screens Setup Equipment Setup Lithology / Well Survey / Fluid Systems Pits Setup Loss Setup Plan Data Utility

Log-It Number: 195648
Operator: ABC Oil Company
Well Name: TBA Gusher 2 Pass 3
Field Name / Block #: Strawberry
Description: Exploration
Location: Offshore Thailand
Warehouse: 101 North Course
Contractor: Drillem Enterprises
Rig Name: Deep Drill #5
Project Engineer: John Smith
Spud Date: 15-Apr-2014
Re-Entry Depth (ft):
Latitude NS Indicator: 10 49 41 N
Longitude EW Indicator: 99 28 59 E
Temp Gradient (°F/100ft): 0.9
Contract Type: IFE

☒ Is Offshore?
Offshore Settings
Air Gap (ft): 84
Water Depth (ft): 220
Sea Floor Temp (°F): 53
☒ Well uses riser/riserless drilling

☐ Coiled Tubing
☒ Conventional Directional
☐ Conventional Vertical
☐ Extended-Reach
☐ Geothermal
☐ Horizontal
☐ HTHP
☐ Multi-Lateral
☐ Slim Hole
☐ Sub-Salt

☐ Is Consignment?
Consign Whse Code:

Decimal Char: -- Period
Unit Set: Standard Oilfield
Currency: USD
Tax Rate: 0
Currency Decimal Places: 2
Recalculate Cost
Manage Units
Warehouse Code Setup

Completion Information
Completion Well Category:
Type Of Completion:
Open Hole Type:
Sand Control Screen Type:

Comments: (984 characters remaining)
Comments relating to the entire project.

Project Closeout Information
Total Depth Date: Select a date [15]
Total Depth Days:
Total Depth (ft):
TVD (ft):
Horiz Displacement (ft):
Total Project Days:
End Date: Select a date [15]
Total Cost: 410806.35
Well Maximum BHT (°F):
Completion Fluid System:
Project Closeout Re-open Project

Project Reports
Generate Graph Report
View Project Summary
Excel Recap Reports
Portrait Canned Recap Reports
Landscape Canned Recap Reports

English / Español / русский [English]
Costs - 25-Apr-2014 Day Chemicals -13132.20 Personnel 575.00 Screens 0.00 Equipment 817.50 Total -11739.70

2.3 LOWER WELL INFORMATION RIBBON

The Lower Well Information Ribbon outlined in red, is always present when ONE-TRAX is running. The information displayed here is:

- Language
- Current date active in the well file.
- What portion of the well are the costs displayed for:
 - Day
 - Cumulative
 - Interval
 - Project
- Costs: subtotaled by type and totaled for the selected period.

From the ribbon you may change the language used or the portion of the well costs displayed.

2.4 SORTING AND FILTERING TABLES

Sorting Columns in Tables

It is possible to sort many of the tables in ONE-TRAX. Simply click on the name of the column and it will be sorted.

Click Here to Sort Column

Before Sorting

Product Name	Alt Name	Item No	Locator
M-I BAR BULK	Bar	M0000003	Domestic
M-I GEL BULK	Gel	M0000321	Domestic
DF-1 BASE FLUID	DF1	M0001993	Domestic
CALCIUM CHLORIDE PELLETS 94-97 PERCENT	CaCl2	M0003212	Domestic
CALCIUM CHLORIDE BRINE S.G. 1.40	CaCl2 Brine	M0009017	Domestic
CAUSTIC SODA	Soda	M0004024	Domestic
DUO-VIS	DUO VIS	M0002978	Domestic
J-GUAR 308NB-GUAR GUM	Guar Gum	M0005469	Domestic
LIME	LIME	M0004111	Domestic
VERSAMUL I	Versamul	M0003127	Domestic
VERSACLEAN SYSTEM 11.1 - 11.5 PPG	SBM	M0016822	Domestic
VERSACOAT NS	Versacoat	M0006343	Domestic
VERSAGEL	VersaGel	M0002679	Domestic
NUT PLUG	Nut Plug	M0006408	Domestic
POLYPAC UL	PolyPac	M0002205	Domestic
POTASSIUM CHLORIDE BRINE-(9.6POUNDS/GAL)	KCL Brine	M0001664	Domestic
POTASSIUM CHLORIDE (KCL)	KCL Bagged	M0003641	Domestic

This symbol shows whether the column is sorted Ascending or Descending.

After Sorting

Product Name	Alt Name	Item No	Locator
CALCIUM CHLORIDE BRINE S.G. 1.40	CaCl2 Brine	M0009017	Domestic
CALCIUM CHLORIDE PELLETS 94-97 PERCENT	CaCl2	M0003212	Domestic
CAUSTIC SODA	Soda	M0004024	Domestic
DEEPCLEAN NS	deepclean	M0016425	Domestic
DF-1 BASE FLUID	DF1	M0001993	Domestic
DIATOMACEOUS EARTH	DE	M0002172	Domestic
DUO-VIS	DUO VIS	M0002978	Domestic
FILTER CARTRIDGE 2 MICRON ABSOLUTE 40 IN LONG	2 micro ABS Cartridge	M0000333	Domestic
J-GUAR 308NB-GUAR GUM	Guar Gum	M0005469	Domestic
KCL / POLYMER PREMIX	KCL Premix	M0003239	Domestic
LIME	LIME	M0004111	Domestic
M-I BAR BULK	Bar	M0000003	Domestic
M-I GEL BULK	Gel	M0000321	Domestic
NUT PLUG	Nut Plug	M0006408	Domestic
POLYPAC UL	PolyPac	M0002205	Domestic
POTASSIUM CHLORIDE (KCL)	KCL Bagged	M0003641	Domestic
POTASSIUM CHLORIDE BRINE-(9.6POUNDS/GAL)	KCL Brine	M0001664	Domestic

Any column may be used to sort the table.

Filtering Tables

On some of the tables in ONE-TRAX you will see the icon meaning the column can be filtered to limit the values it displays, . By clicking on the button with the cursor, a list of values on which you can filter the column on will be displayed. Click on one of the values on and only items with that value will be shown. An icon, , with a solid triangle indicates that there is an active filter on that column.

Clicking (Custom) will bring up an additional dialogue box which will enable you to add ranges or multiple entries to filter on.

Qty From Ticket	Actual Qty
0.00	(All)
0.00	(Custom)
0.00	(Blanks)
0.00	(NonBlanks)
0.00	Above Average
0.00	Below Average
0.00	Top 10
0.00	Top 10 percentile
0.00	Bottom 10
0.00	Bottom 10 percentile
0.00	0.00
0.00	40.00
0.00	45.00
0.00	80.00
0.00	120.00
0.00	0.00
0.00	0.00
0.00	0.00
40.00	40.00
40.00	40.00

In the following example, a conditional filter is added so that only items with a quantity greater than zero will be displayed.

Custom Filter Selection

For Field 'Qty From Ticket'

+ Add Condition - Remove Condition(s) Group Selected: 'And' Group 'Or' Group Toggle Ungroup

Operator	Operand
> Greater than	0.00

> '0'

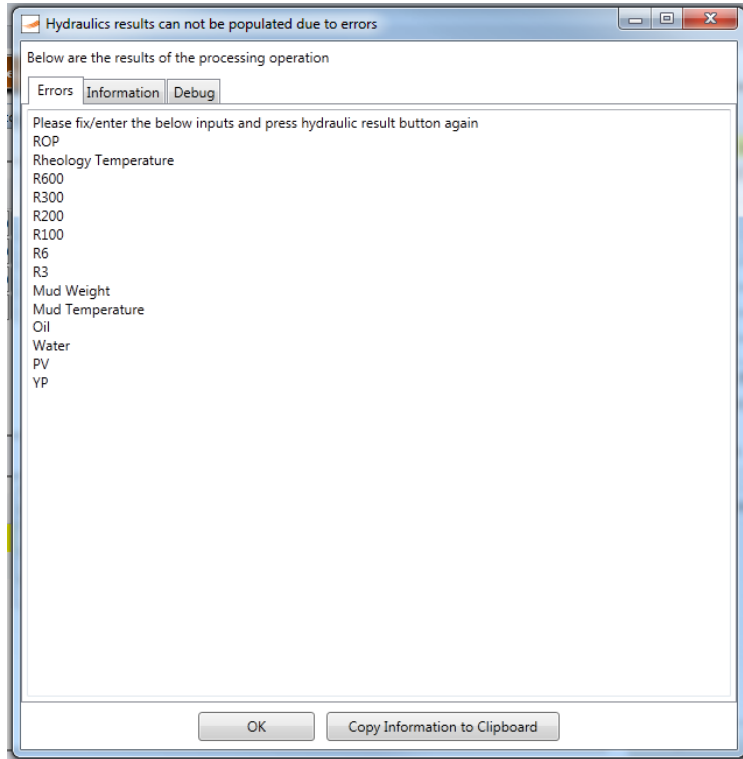
■ 'And' Group ■ 'Or' Group

OK Cancel

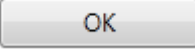
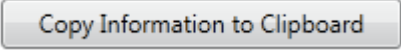
2.5 ERROR MESSAGES

Error Messages

When you activate a button in ONE-TRAX occasionally an error message window will open. Many of the more common errors will contain information on how to correct the problem. Once the problems are corrected, the function will work.



In the example to the left, ONE-TRAX cannot run the requested function because the ROP and rheology information have not been entered.

Press  and the error window will close. If the error message does not have useful information, you may press . This will allow you to paste the error message in an email or document so you may request assistance.

2.6 ADDING AND DELETING ROWS IN TABLES

Red “X” and Green “+”

On many tables in ONE-TRAX you will see there are Red “Xs”  or Green “+s”  at the end of the rows.

Drill String Geometry							
Type	Length ft	Pipe Od in	Pipe Id	Tool Jt OD in	Tool Jt ID	T J Length in	
Drill Pipe	7947.36	5.5000	4.276	7.0000	4	18	
Sub	14.64	8.3740	2.25	0.0000			
							

Pressing  will delete the Row from the Table.  is shown at the end of an empty row. After updating the data, pressing  will save that row. Clicking anywhere outside of the current cell will also save any changes made on the row.

2.7 APPROVED AND UN-APPROVED ITEMS

Add as Approved Item

and

Add as Un-Approved Item

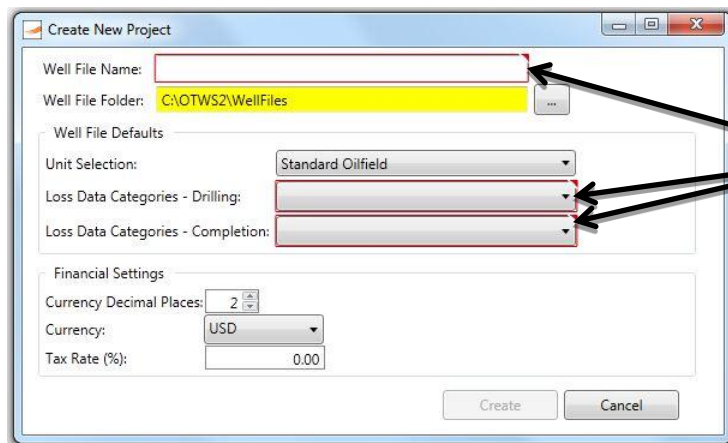
The terms “Approved” and “Un-Approved” items are used in referring to products, screens, and equipment in ONE-TRAX. An “Approved” item has not had any entries changed from those in the master data file. These are the fields used to describe the item to Flex-Billing, usually Item Number and Description. In General, an “Approved” item will be one manufactured by M-I SWACO and an “UN-Approved” item will not.

HELPFUL HINT

When you need to add an item, piece of equipment, or screen that is not in the [Application Data List](#). Start by adding a similar M-I product using the Add as an Un-Approved Item Button and changing the necessary fields.

2.8 RED OUTLINED CELLS

Fields outlined in **Red** indicate a required entry field or a field with a data problem, i.e. values outside of parameters you define or negative results in a calculated data field that should have a positive value.



Graphic showing cells outlined in Red that indicate required entry cells.

Graphic showing cell outlined in Red indicating erroneous data. In this example, the NTU is outside the parameters defined in the daily property specifications.

Property	Unit	Check 1	Check 2	Check 3	Check 4
Sample From		Active 1	Flowline		
Funnel Viscosity	s/qt				
Solids	%	.01	.15		
pH		8.3	8.3		
Temperature	°F	85	139		
Fluid SG	sg	1.38	1.39		
Measured Density	lb/gal				
Corrected Density	lb/gal				
TCT	°F	40	40		
NTU's		30	120		
Iron Content					
SG of Fluid In Hole	sg	1.38	1.38		
Temp of Fluid In Hole	°F	184	184		
Measured Density In Hole	lb/gal				
Corrected Density In Hole	lb/gal				

2.9 YELLOW BACKGROUND CELLS

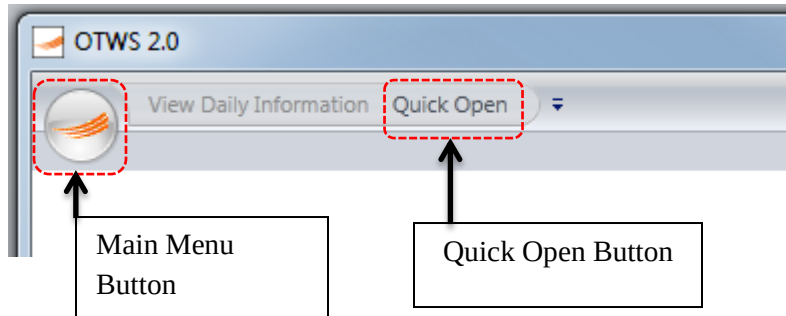
A field with a **Yellow** background is a display only field and will not allow data to be entered.

Property	Unit	Check 1	Check 2	Check 3	Check 4
Uncorr Ret Solids	%vol	19	20		
NAF Base Fluid	%vol	56	56		
Uncorr Water	%vol	25	24		
CI Whole Mud	mg/L	76000	72000		
Retort Mud Wt@Room Temp	lb/gal	11.6	11.6		
Base Fluid SG	sg	.8	.8		
Wt Material SG	sg	4.2	4.2		
Drill Solids SG	sg	2.6	2.6		
Salt w	%wt	32.24	31.95		
Salt Concentration	lb/bbl	41.65	39.46		
Adjusted Solids	%vol	15.8	16.97		
NAF Base Fluid/Water		69/31	70/30		
Avg SG Solids	sg	3.64	3.49		
LGS	%	5.49	7.56		
LGS Concentration	lb/bbl	49.94	68.78		
HGS	%	10.31	9.41		
HGS Concentration	lb/bbl	151.45	138.29		
Water Phase Salinity	ml/L	269513	266444		

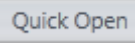
Graphic showing Yellow Highlighted cells, in this example these cells contain calculated data and will not accept entry from the keyboard.

2.10 INITIAL WINDOW

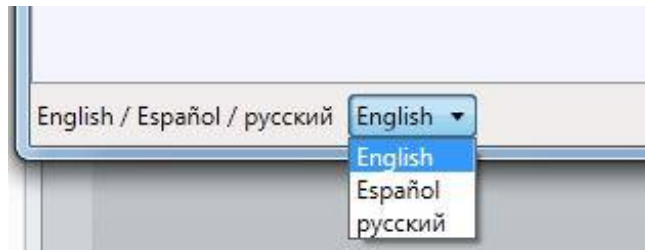
After loading the ONE-TRAX application there are three active functions on the initial window: main menu button, quick open button, and language selection dropdown.



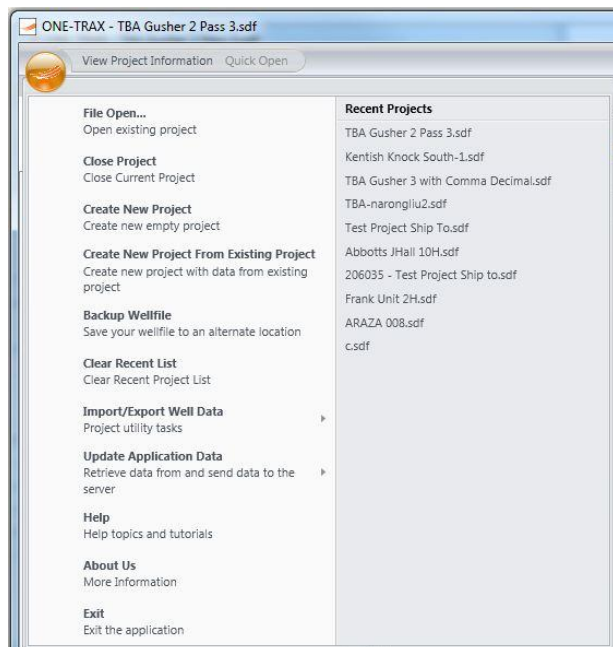
The main menu button  will open and close the main menu that controls the functions of the well files and Application data files.

 button opens the last well file that was open in ONE-TRAX.

At the bottom of the Initial Window, there is a dropdown that will allow you to choose which language ONE-TRAX will use. You may also change the language at any time during the project, this drop down is always visible.



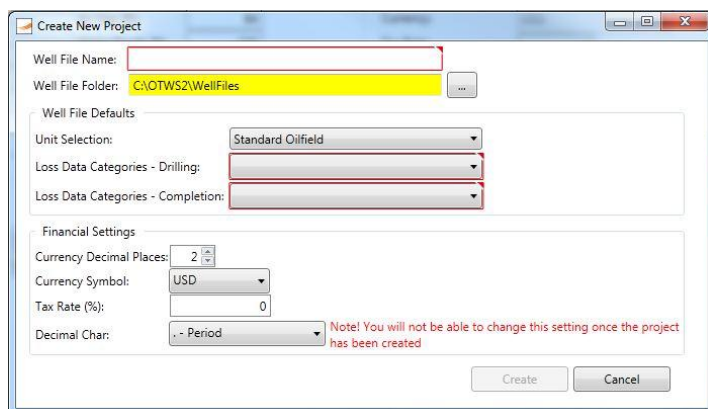
2.11 MAIN MENU



File Open opens a folder where all the well files in this computer are saved by default (System(C:)\OTWS2\Wellfiles). You can choose any well file to open from this folder or navigate to any other folder with the well file you want to open. Well files can also be opened by clicking on a file name in the recent projects table, on the left side of the main menu.

Close Project closes the currently opened project and leaves only the ONE-TRAX initial window open.

Create New Project will create a new project file. You use this command if you are **not** copying information from any other well to create a new well file. Enter the information required in the **Create New Project** window that will open after you click **Create New Project**.



Before you create a well file name, familiarize yourself with any rules or standards that might exist in your region on how the well file should be named.

Well file Folder shows the default folder where your file will be saved. If you want to save a well file in

3 PROJECT INFORMATION VIEW

The Project Information View is where data relating to the overall project is entered. This view is where the items that are used to create the daily records are entered. This information is usually not updated on a daily basis and that is why ONE-TRAX opens in the daily view once a daily record has been created. This section of the manual will discuss how to enter descriptive well information, personnel, products, and other information used in the project.

3.1 GENERAL INFORMATION TAB

General Tab – Project View

This tab that will open when a new well file is created and if a project is opened that does not have any daily records. It is where information for identifying and tracking the well is entered. When opened on a newly created project, only the information from the create new project window will be filled in. All other information will need to be entered in the spaces provided.

Below is the tab, for the purpose of this chapter we will discuss the tab by dividing it into 5 sections.

The screenshot shows the ONE-TRAX software interface with the General tab selected. The interface is divided into several sections, each highlighted by a red dashed box and labeled with an arrow and a box:

- Well Description Section:** Includes fields for Log-It Number (195648), Operator (ABC Oil Company), Well Name (TBA Gusher 2 Pass 3), Field Name / Block # (Strawberry), Description (Exploration), Location (Offshore Thailand), Warehouse (101 North Course), Contractor (Drillm Enterprises), Rig Name (Deep Drill #5), Project Engineer (John Smith), Spud Date (15-Apr-2014), Re-Entry Depth (ft) (10), Latitude NS Indicator (49), Longitude EW Indicator (28), Temp Gradient (°F/100ft) (0.9), and Contract Type (IFE).
- Project Settings Section:** Includes fields for Is Offshore? (checked), Offshore Settings (Air Gap (ft) 84, Water Depth (ft) 220, Sea Floor Temp (°F) 53), Coiled Tubing (checked), Conventional Directional (checked), Extended-Reach (checked), Geothermal (checked), Horizontal (checked), Multi-Lateral (checked), Slim Hole (checked), Sub-Salt (checked), Is Consignment? (checked), and Consign Whse Code.
- Project Closeout Section:** Includes fields for Decimal Char (Period), Unit Set (Standard Oilfield), Currency (USD), Tax Rate (0), Currency Decimal Places (2), Recalculate Cost, Manage Units, Warehouse Code Setup, Completion Information (Completion Well Category: Producer – Oil, Type Of Completion: Case Hole-Perforations, Open Hole Type: Sand Control Screen Type), and Project Closeout Information (Total Depth Date: 22-Apr-2014, Total Depth Days: 7, Total Depth (ft): 8002, TVD (ft): 6880, Horiz Displacement (ft): 1000, Total Project Days: 10, End Date: 24-Apr-2014, Total Cost: 410806.35, Well Maximum BHT (°F): 184, Completion Fluid System: Calcium Chloride).
- Completion Information Section:** Includes fields for Completion Well Category (Producer – Oil), Type Of Completion (Case Hole-Perforations), Open Hole Type (Sand Control Screen Type), and Comments (984 characters remaining).
- Comments Section:** Includes a text area for Comments relating to the entire project.
- Reports Section:** Includes buttons for Generate Graph Report, View Project Summary, Excel Recap Reports, Portrait Canned Recap Reports, and Landscape Canned Recap Reports.

The bottom of the screen shows a status bar with the following information: English / Español / русский / English, Costs - 22-Apr-2014, Day, Chemicals -13132.20, Personnel 575.00, Screens 0.00, Equipment 817.50, Total -11739.70.

3.1.1 Well Description Section

These fields contain data that describes the well. Once the data has been entered it is unlikely that it will need to be changed. If you do not know any of the required data, contact the office to see if they can remedy this.

Log-It Number is a field of primary importance, this is the unique number that M-I SWACO assigns to the project and by which it is tracked by the company.

The temperature data are also fields that should not be skipped. This information is required by the API 5th Edition rules and is used to provide more accurate results in hydraulics calculations. If these figures are incorrect, the annular pressure loss will be wrong.

Surface Temp (°F):	68.00
Temp Gradient (°F/100ft):	0.90

HELPFUL HINT

ONE-TRAX will only accept Latitude and Longitude in Degree Minutes Seconds format. If you are given this information in decimal format, a quick internet search using “convert decimal lat long to degrees” will give links to websites which will convert these figures.

☒ Is Offshore?
Offshore Settings
Air Gap (ft): 84.00
Water Depth (ft): 220.00
Sea Floor Temp (°F): 53.00
☒ Uses Riser

The upper right of the well description section contains fields that are important for offshore projects. The checkboxes can only be entered or changed prior to the creation of the first daily record. After that they are locked. The depths and sea floor temperature fields will not be locked and can be updated at any time in the project.

Sea floor temperature is also used in the API 5th Edition rules.

Accurate depth measurements for the air gap and water depth are important. All depths on an offshore project are reported as depth below the drilling floor (RKB).

The ☒ Uses Riser box tells ONE-TRAX that the project will use a riser at some point during the drilling process. **IMPORTANT: If your well will use a Riser at any time during drilling, check this box!** When drilling riserless, there is another checkbox on the well geometry tab that informs ONE-TRAX of this fact.

- ☐ Coiled Tubing
- ☐ Conventional Directional
- ☐ Conventional Vertical
- ☐ Extended-Reach
- ☐ Geothermal
- ☐ Horizontal
- ☐ HTHP
- ☐ Multi-Lateral
- ☐ Slim Hole
- ☐ Sub-Salt

Below the Offshore information fields there is a boxed area containing classifications for the well type. You are not limited to choosing only one type to describe the well. The type is used in planning future projects when searching for offset wells.

Coiled Tubing – the well is drilled with mud motors and bit on the end of coiled tubing. Conventional drill pipe is not used.

Conventional Directional – a well with the target not lying directly under the surface location of the well and not falling into a more specific category.

Conventional Vertical – a well with the target lying directly under the surface location of the well and not falling into a more specific category.

Extended Reach – a well with horizontal displacement greater than the vertical depth or which has a vertical depth greater than 25,000 ft.

Geothermal – a well drilled to produce energy form the heat energy present beneath the Earth’s crust.

Horizontal – a well in which the wellbore exceeds 80° from the vertical.

HTHP – a well with undisturbed bottom hole temperature (BHT) greater than 300°F and a pore pressure in excess of 15.3 lb. /gal; or requiring a BOP with a rating greater than 10,000 psi.

Multi-Lateral- a well with more than one branch radiating out from the main wellbore.

Slim Hole – a general term for a well with a bore of 6 inches (153 mm) or less.

Sub-Salt – a well with the target below a layer of salt.

Consignment Warehouse - The area contains information used when the project will be set up using a consignment warehouse. Check the ☐ **Is Consignment?** box and type in the warehouse name in the field provided.

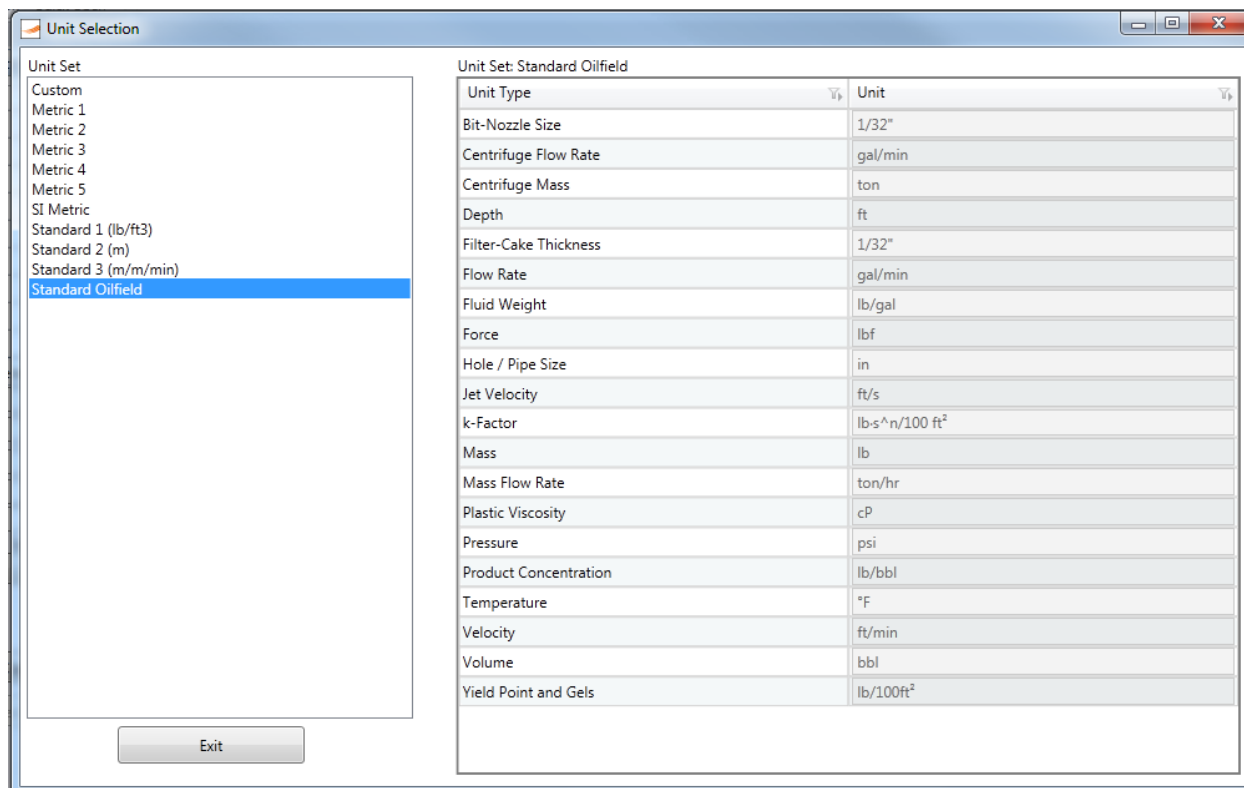
3.1.2 Project Settings Section

This section displays data that was entered when the project was created. The decimal character, unit set, and currency will be locked for the project once the first daily record is created. You may change the tax rate or number of decimal places at any time during the project. If you change the tax rate or decimal places, press the **Recalculate** button to have ONE-TRAX reprocess all monetary transactions with the new information.

Manage Units allows you to choose the unit set you wish for the project. **You may only change unit sets prior to the first daily record being created.** When clicked, a window will open displaying the active unit set, below is a screenshot of the Unit Selection window prior to the creation of the first day. You chose which

unit set you wish to use by clicking on it, the background will turn blue showing it is the active set. The right side will display which units each parameter is recorded in. If the custom unit set is selected, the right hand column, unit, will have a white background and you may choose what value you want that information to be

recorded in. Clicking  will record the project as using that unit set.

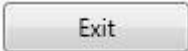


After a daily record has been created, the unit set table on the left will have a gray background and you will no longer be able to change the set you are currently using.  will close the window and not make any changes at that time.

 opens the warehouse setup window. Here is where you will enter the warehouses that your project will receive deliveries from. Enter the code and name of your warehouse(s).

Warehouse Cd	Warehouse Name
101	North Course
102	Aberdeen

Exit

 will close the window and not make any changes.

3.1.3 Project Completion Information Section

This section is used to enter the project completion information. The drop down menus are controlled from the application data and it is not a requirement that all be filled in. This will depend on the type of completion of the well.

Completion Information

Completion Well Category: Producer - Oil

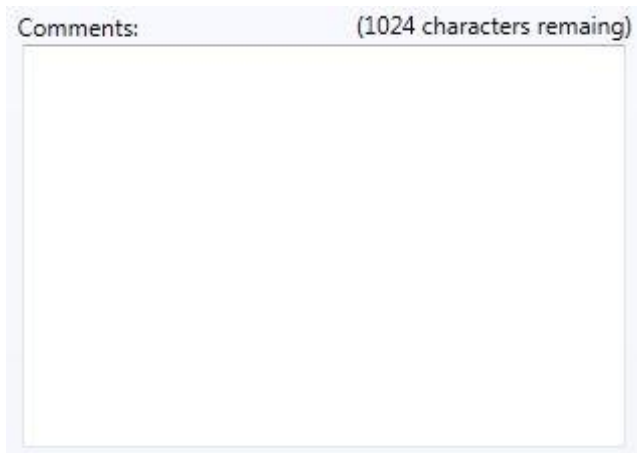
Type Of Completion: Case Hole-Perforations

Open Hole Type:

Sand Control Screen Type:

3.1.4 Comments Section

The lower portion consists of an area to enter any comments pertaining to the overall project. Above the entry area on the right, a display shows how many characters remain before the field is full.

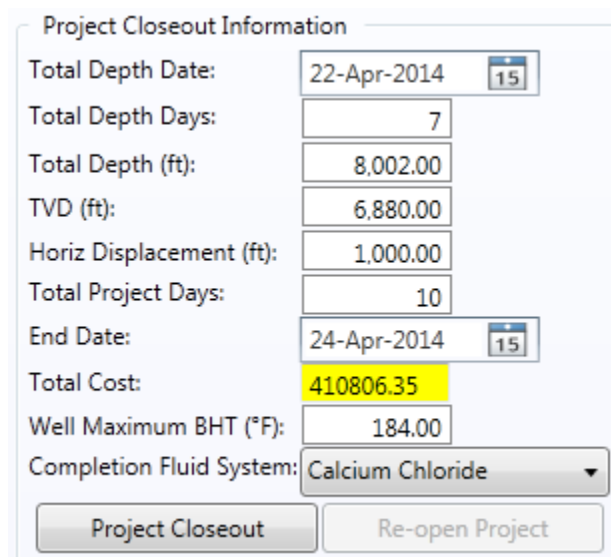


Comments: (1024 characters remaining)

3.1.5 Project Closeout Section

This column contains the information relating to the closeout of the well. You will enter the relevant data in the fields provided. Most of the data is self-explanatory, but there has been some confusion on the difference between total depth date and end date. Total depth date is the date when the well reaches TD and end date is the date when services provided by M-I SWACO are halted. In some cases the dates are the same, but they can be different if completion operations are conducted or some days are used for rigging down equipment.

In wells where M-I SWACO does not provide completion services, there will be no completion fluid system to be chosen in the drop down menu.



Project Closeout Information

Total Depth Date:	22-Apr-2014	15
Total Depth Days:	7	
Total Depth (ft):	8,002.00	
TVD (ft):	6,880.00	
Horiz Displacement (ft):	1,000.00	
Total Project Days:	10	
End Date:	24-Apr-2014	15
Total Cost:	410806.35	
Well Maximum BHT (°F):	184.00	
Completion Fluid System:	Calcium Chloride	

Project Closeout Re-open Project

Project Closeout

is used to convert the project status from opened to closed. This will prevent any changes to or transactions being added to the well file. It will only operate if, on the daily information view, you are on the last day of the project. When first pressed it will run a series of checks to insure the well file is ready for close out. If there is any information missing an error message dialogue box will open informing you for the missing data. Enter the missing data and retry.

HELPFUL HINT

Plan Data is required if the project was 15 days or longer, or if the project is billed in US dollars and the cost was greater than \$150,000.00.

When the well file is free of errors a server login window will open. **You will need to be connected to the Schlumberger intranet for the process to continue.**

Once you have logged in a project closeout window will open. If you have not entered the project close out information on the general tab you will prompted to do so here. Fill in any missing data and check the checkbox certifying the information is correct. If a person on the project does not have a valid LDAP recorded, a second check will appear asking you to verify this.

Closeout Screen with Errors

Project Closeout

Total Depth Date:

Total Depth Days:

Total Depth (ft):

TVD (ft):

Horiz Displacement (ft):

Total Project Days:

End Date:

Total Cost?:

Well Maximum BHT (°F):

Completion Fluid System:

Log-It Number:

Operator:

Well Name:

Field / Area:

Contractor:

Rig Name:

Latitude NS Indicator: ° ' "

Longitude EW Indicator: ° ' "

Project Personnel

Name	LDAP ID	Email	Service Line	Description	Charge Item	Daily Charge Amount
Derek McCoy	dmccoy		Drilling Solutions	Mud Engineer	R0088110	\$0.00
Jeff Clark	Jclark09		Drilling Solutions	Mud Engineer	R0088110	\$800.00
Joan Gray	Jgray2		Drilling Solutions	Mud Engineer	R0088110	\$0.00

You must validate the LDAP IDs before you can go to the next step

☐ I, pabbott, certify that the information above is complete and accurate, according to the project details and M-I SWACO data quality requirements

Closeout Screen without Errors

Project Closeout

Total Depth Date: 22-Jul-2014
 Total Depth Days: 27
 Total Depth (ft): 12,728.00
 TVD (ft): 8,160.00
 Horiz Displacement (ft):
 Total Project Days: 28
 End Date: 22-Jul-2014
 Total Cost?: 157448.50
 Well Maximum BHT (°F): 184.00
 Completion Fluid System:

Log-It Number: 202042
 Operator: Rex Energy Corporation
 Well Name: Abbotts JHall 10H
 Field / Area: Utica Shale Horizontal
 Contractor: Sidewinder Drilling
 Rig Name: Sidewinder 63
 Latitude NS Indicator: 35 ° 59 ' 52 " N
 Longitude EW Indicator: 81 ° 15 ' 37 " W

Project Personnel

Name	LDAP ID	Email	Service Line	Description	Charge Item	Daily Charge Amount
Derek McCoy	dmccoy		Drilling Solutions	Mud Engineer	R0088110	\$0.00
Jeff Clark	Jclark09		Drilling Solutions	Mud Engineer	R0088110	\$800.00
Joan Gary	Jgary2		Drilling Solutions	Mud Engineer	R0088110	\$0.00

☒ I, pabbott, certify that the information above is complete and accurate, according to the project details and M-I SWACO data quality requirements

Next Cancel

Validate LDAP IDs

Cancel

will close the window without closing the project and return to the general tab.

Next

will only operate after the checkbox verification is active. When pressed, a 2nd project closeout window will open showing a well summary and any operational or fluid problems entered. Another verification checkbox will be displayed.

Project Closeout

Well Overview

Interval #	Type	Casing OD (in)	Hole Size (in)	Depth MD/TVD (ft)	Default Fluid System	Max Density (lb/gal)	Max Inclination (°)	Max BHT (°F)	Length Drilled (ft)	Cost (USD)	Days	Losses Surface (bbl)	Losses Subsurface (bbl)
1	Casing	9.62	12.25	7000/7000	MEGADRIL (w)	1.00			710.00	46962.79	19.00	0.00	0.00
2	Open Hole		8.50	12728/8160	MEGADRIL (w)	10.80			5,728.00	99233.55	8.00	340.90	16.10

Project Issues

Operational Problems				Fluid Problems			
Date	Issue	Duration (hrs)		Date	Issue	Duration (hrs)	
29-Jun-2014	Stuck Pipe		\$7.00				
30-Jun-2014	Stuck Pipe		\$8.00				

☒ I, pabbott, certify that the information above is complete and accurate, according to the project details and M-I SWACO data quality requirements

Previous Next Cancel

Previous

will return to the 1st project closeout window.

Cancel

will close the window without closing the project and return to the general tab.

Next

will open a verification window if the operational problems or fluid problems table are blank, you will need to acknowledge that there were no problems encountered on this project to proceed. Next a 3rd project closeout window will open presenting the plan data.

Planned Well Total Days: Planned Well Total Costs (USD):

Plan Data

Interval #	Fluid System 1	BitSize	TD (ft)	TVD (ft)	Days	Total Cost (USD)
2	MEGADRIL (w/MEGAMUL)	8.50	12,500.00	8,000.00	2	\$15,000.00

☒ I, pabbott, certify that the information above is complete and accurate, according to the project details and M-I SWACO data quality requirements

Previous Close Project Cancel

Previous

will return to the 2nd project closeout window.

Cancel

will close the window without closing the project and return to the general tab.

Close Project

will mark the project as closed and prevent any changes being made to the well file.

Re-open Project

will become active after the project has been closed. You will need to press the button if, for any reason, you will need to modify the well file. A verification window will open to inform you that once the project is re-opened the entire project closeout process will have to be performed to reclose it.

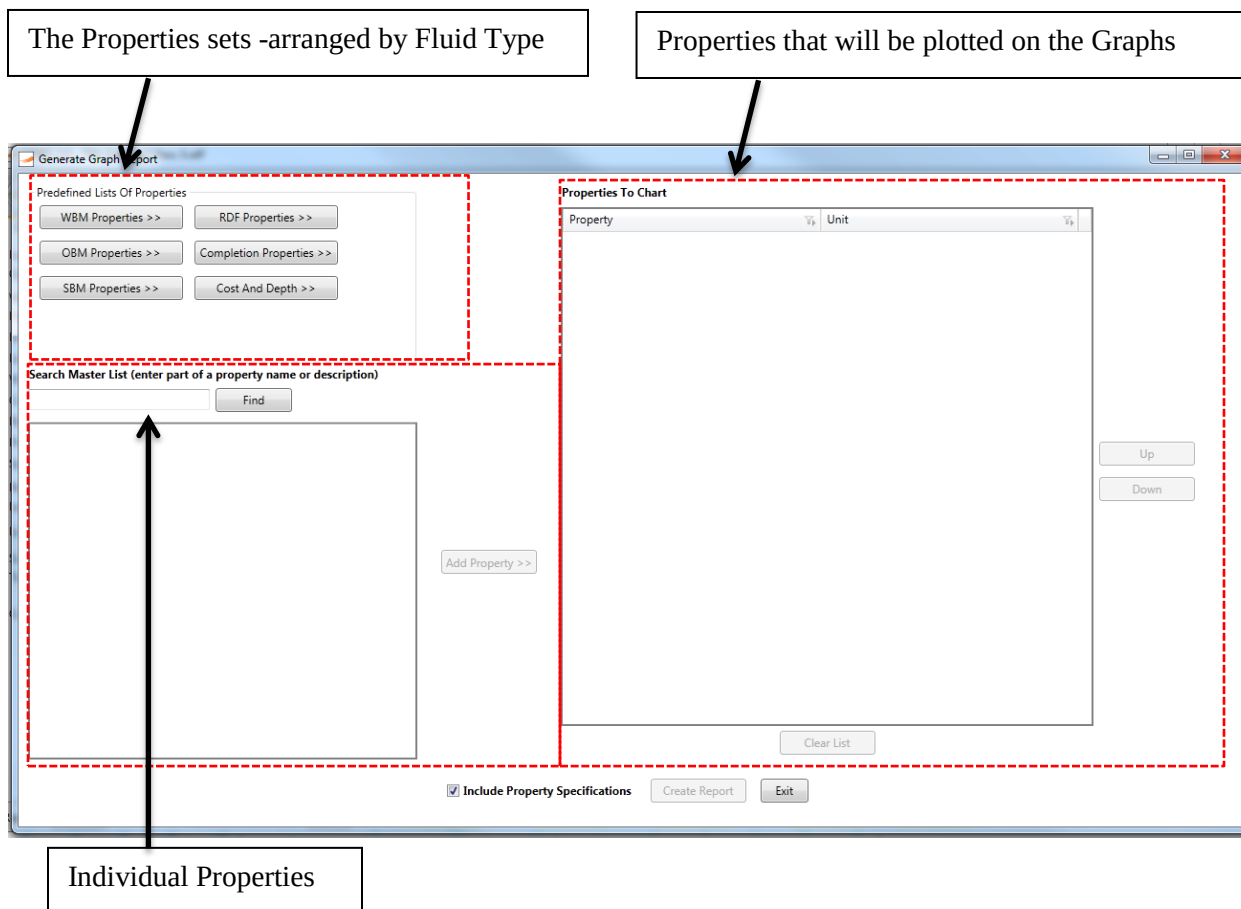
Want To Re-open Project

Do you wish to re-open the project? If you click 'Yes' then you will need to repeat the entire project closeout process.

Yes No

3.1.6 Reports Section

Generate Graph Report is used to create an Excel graph plotting the desired property against measured depth. The generate graph report window that is opened by this button is divided into three sections.



The Property Sets are a quick way to select only the fluid properties associated with that type of fluid. Click on the type of fluids you want to graph and that set will be moved to the properties to chart section.

Individual Properties are added by first performing a search for the desired property and then adding it to the properties to chart. The first step is to enter the name of the property you wish to chart into the search field.

Press the **Find** button and a list of properties containing the search term will be listed. Highlight the one you wish to add by clicking on it, and press **Add Property >>**. The highlighted property will be added to the properties to chart section.

Search Master List (enter part of a property name or description)

Property	Property Description	Unit
API Fluid Loss		cc/30min
Base Fluid SG		sg
Filtration Fluid Property	Used as a description p	
Fluid SG		sg
Fluid Temp		°F
Fluid Type		
Fluid Weight		lb/gal

Properties to Chart is a list of the properties that ONE-TRAX will create a graph for. Each property will be plotted against measured depth. The charts will be made in Excel, 4 charts across and 5 charts high, on each worksheet. The properties will be plotting in the order that they appear in the properties to chart section,

however the order can be modified by using the buttons. Activate the property you wish to move by clicking on it and press the button to move it in the direction you wish. If you press the

button, the properties to chart section will be cleared.

Properties To Chart

Property	Unit	
Depth vs Days		
Cost		
TVD	ft	
Fluid Weight	lb/gal	
Fluid Temp	°F	
Funnel Viscosity	s/qt	
Rheology Temp	°F	
R600		
R300		
R200		
R100		
R6		
R3		
PV	cP	
YP	lb/100ft ²	
10s Gel	lb/100ft ²	
10m Gel	lb/100ft ²	
30m Gel	lb/100ft ²	
API Fluid Loss	cc/30min	
HTHP Fluid Loss	cc/30min	
HTHP FL Temp	°F	
Gel API	lb/100ft ²	

The ☒ **Include Property Specifications** checkbox lets ONE-TRAX know whether or not to plot the daily property specifications you entered on the chart. See [Daily Properties Specifications](#) for more information on this subject.

opens a dialogue box allowing you to name and choose a location to store the Excel file that will be created.

close the window and returns to the general tab.

View Project Summary

will open a tabular view displaying project information. It is displayed on a daily basis with subtotals by interval.

Drop down menus and date selection fields allowing you display only portions of the project.

Filter

will create the table and show only those days relating to your selected criteria.

Project Summary

Interval: All Date From: Select a date 15 Date To: Select a date 15

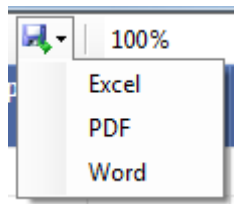
Operational Mode: All Fluid Type: All Fluid System: All

Filter Exit

Interval	Days	Date	Default Interval	Operational Mode	Depth	TVD	Daily Activity	Bit Size	Primary Fluid System	Fluid Type	Density	Surface Loss	Subsurface Loss	Total Cost	Total Cum. Cost
2	1	15-Apr-2014	1	Drilling	334	334	Drilling 26" Hole	26	Spud Mud	WBM - Drilling	8.6	0	0	\$9033.00	\$9033.00
	2	16-Apr-2014	1	Drilling	800	800	Circulate for Casing	26	Spud Mud	WBM - Drilling	8.6	0	1025	\$3140.00	\$12173.00
Subtotal:												0	1025	\$12173.00	
3	1	17-Apr-2014	2	Drilling	800	800	Waiting on Cement	26	KCL/Polymer	WBM - Drilling	10.1	375	0	\$31689.50	\$43862.50
	2	18-Apr-2014	2	Drilling	1202	1199	Running Riser and Beginning 17.5" Section	17.5	KCL/Polymer	WBM - Drilling	9.8	125.72	96.65	\$56720.00	\$100582.50
	3	20-Apr-2014	2	Drilling	5002	4747	Circulating prior to running Casing	17.5	KCL/Polymer	WBM - Drilling	10.5	693.49	2048.01	\$100957.50	\$201540.00
Subtotal:												1194.21	2144.66	\$189367.00	
2	1	21-Apr-2014	3	Drilling	2023	6188	Drilling With LTOBM	12.25	VERSACLEAN	OBM - Drilling	11.6	3108.6	47.08	\$281712.50	\$483252.50
	2	22-Apr-2014	3	Drilling	4002	6880	Displacing Cement	12.25	VERSACLEAN	OBM - Drilling	11.5	329.8	0	-\$23801.45	\$459451.05
Subtotal:												3438.4	47.08	\$257911.05	
3	1	23-Apr-2014	4	Completion	8002	6880	Preparing to Displace	12.25	Calcium Chloride	Completion Brine		0	0.23	\$10432.50	\$469883.55
	2	24-Apr-2014	4	Completion	8002	6880	Preparing to Displace	12.25	Calcium Chloride	Completion Brine		652	1.96	-\$47337.50	\$422546.05
	3	25-Apr-2014	4	Completion	8002	6880	Back Loading Surface Brine	12.25	Calcium Chloride	Completion Brine		0	246.6	-\$11739.70	\$410806.35
Subtotal:												652	248.79	-\$48644.70	

These control buttons

allow you to print this report and save it in a

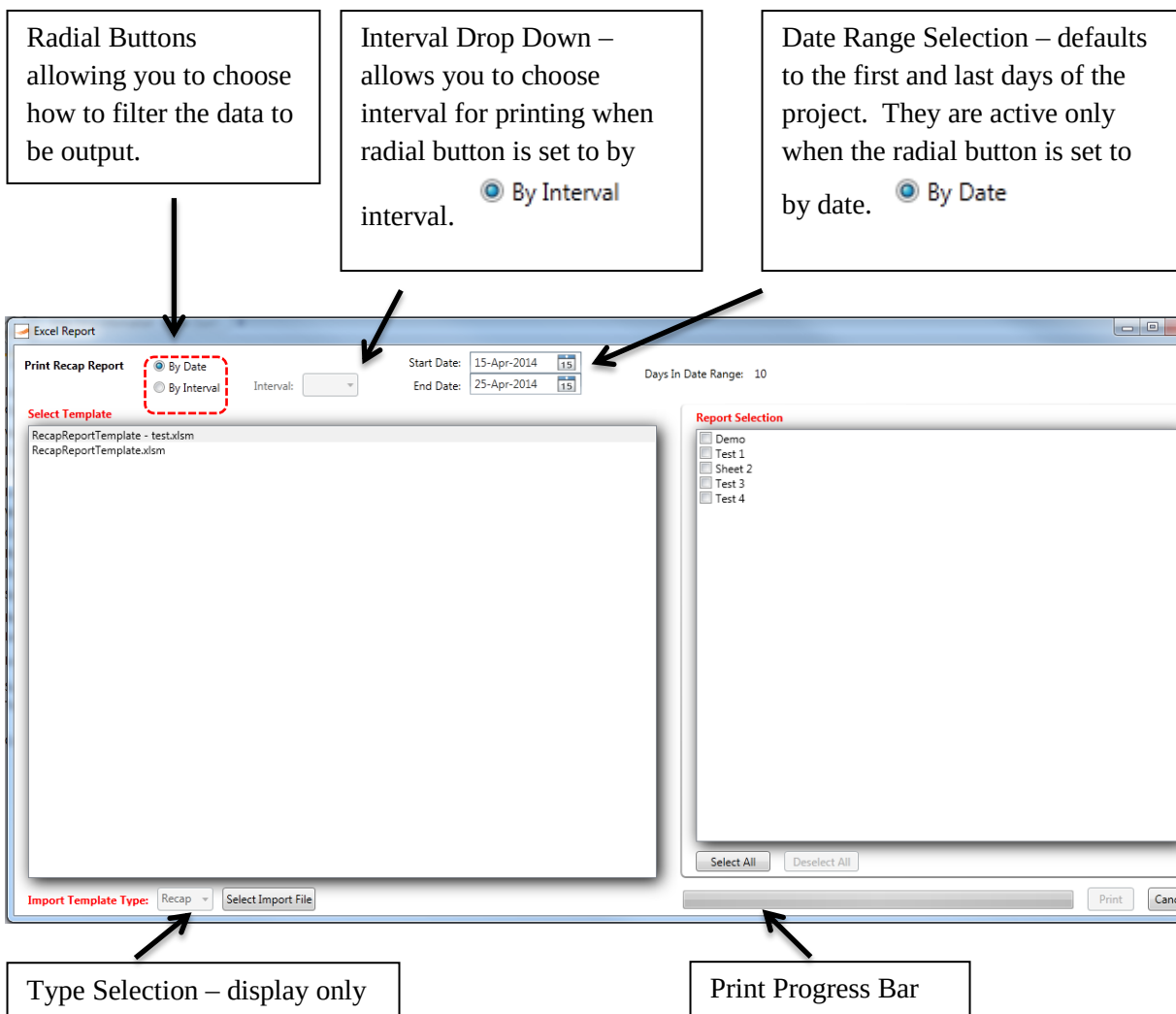


choice of formats.

Exit

The button will close the window and return to the general tab.

Excel Recap Reports opens the dialogue window called Excel report. This window allows you to print customized reports created using Excel templates. This window is used in other places in ONE-TRAX so you will notice that some of the information is locked. You cannot change the template type.

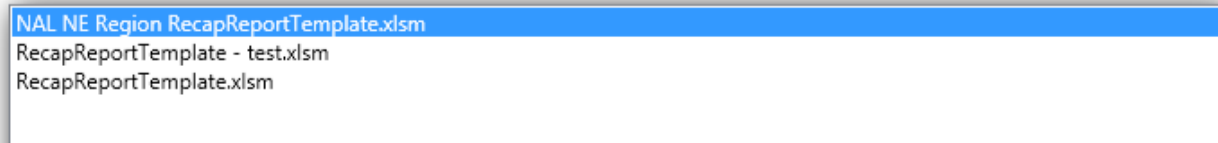


You can import additional templates, if you receive a template that your project engineer requires you use for this well; it can be stored so that ONE-TRAX can use it by pressing **Select Import File**. This will open up a dialogue box where you will need to find and highlight the new template. Once the file is highlighted, pressing **Open** will place it in the select template section and it can then be used to create reports.

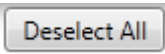
How to make a Recap Report using Templates - This example is for a specific template but gives the user a good overview of the process.

1. Choose the report template for the report you wish to print. You do this by clicking on the template name in the select template section. This will activate this template and indicate this by highlighting it.

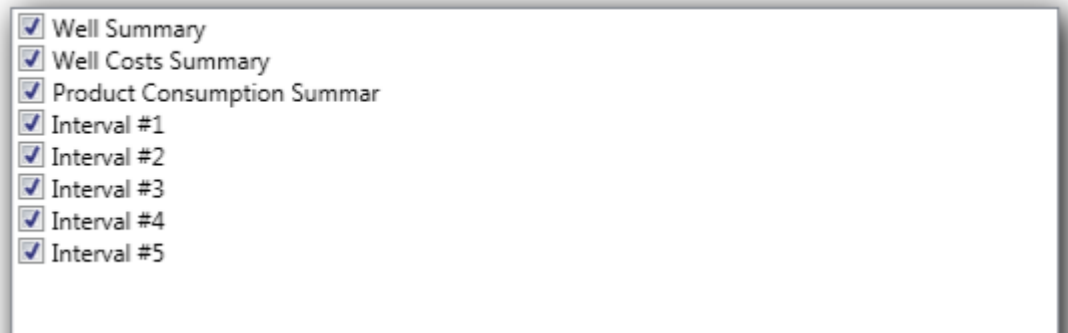
Select Template



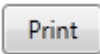
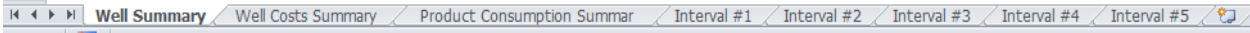
NAL NE Region RecapReportTemplate.xlsm
RecapReportTemplate - test.xlsm
RecapReportTemplate.xlsm

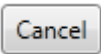
2. This will show which reports are available in this template in the report selection section. You may choose which reports to create by checking the boxes. You may select all by clicking . You may remove all check marks by clicking .

Report Selection

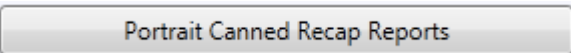


☒ Well Summary
☒ Well Costs Summary
☒ Product Consumption Summar
☒ Interval #1
☒ Interval #2
☒ Interval #3
☒ Interval #4
☒ Interval #5

3. Click  and Excel will open two Workbooks. The Recap_Workbook contains the information that populates the fields in your report workbook. The report workbook is made up of pieces of information about the well.
 4. Activate the Worksheet you want to view or work on clicking on its' name tab.
- 
5. Clean up the reports.
 6. Save your reports.



will close the Excel report window and return to the general tab.



will open a window allowing you to choose which of the portrait oriented reports you wish to print. Check the boxes next to the reports you wish to create, and press

Below is the Interval Tab showing its' three sections.

Interval Table

Interval Details Section

Control Buttons

3.2.1 Interval Table

The interval table is a display only table that shows the basic data of the interval at a glance. The table will extend past the width of its section, so to view it in its entirety you will have to use the scroll bar. The fields are populated from several different tabs, and it gives a good summary view of the project.

Interval Num	Operational I	Type	Interval Leng	Casing OD	Casing ID	Depth	TVD	Top Of Line/ Sidetrack	Bit Size	Max Inclinati	Frac Grad
0	Drilling										
1	Drilling	Casing	800.00	20.00	19.12	800.00	800.00		26.00	0.25	
2	Drilling	Casing	4,202.00	13.38	12.62	5,002.00	4,747.00		17.50	35.00	
3	Drilling	Casing	3,000.00	9.62	8.54	8,002.00	6,880.00		12.25	45.00	14.90
4	Completion	Casing	0.00	9.62	8.54	8,002.00	6,880.00		12.25	45.00	

3.2.2 Interval Detail Section

This section shows more detailed information about the interval. It is a combination of updateable and display fields. The fields will retrieve data from the tabs in Daily View with the exception with the of fracture gradient, this value can only be entered here.

The screenshot shows the 'Interval Details - Interval - 3' form. It contains various input fields for operational parameters, a section for fluid services, and two large text areas for comments and observations. Callouts provide additional context for several elements:

- Operational Mode and Type:** Drop-down menus for 'Drilling' and 'Casing'.
- Depth and TVD:** Input fields for 'Depth (ft): 8002' and 'TVD (ft): 6880'.
- Top Of Liner/ (ft):** Input field for 'Top Of Liner/ (ft): 8002'.
- Bit Size:** Input field for 'Bit Size (in): 12.25'.
- Max Inclination:** Input field for 'Max Inclination: 45'.
- Max Density:** Input field for 'Max Density: 11.6'.
- Frac Grad:** Input field for 'Frac Grad: 14.9'.
- Interval Days:** Input field for 'Interval Days: 2'.
- Interval Costs:** Input field for 'Interval Costs: 211145.25'.
- Max Circ. BHT:** Input field for 'Max Circ. BHT: 184'.
- Time First Used:** Two drop-down menus for time selection.
- Fluids provided by M-I SWACO?** Checkboxed.
- Is RDF?** Unchecked.
- Is Non operational Interval?** Unchecked.
- Update Fields from Daily:** Button.
- Comment for Recap:** Text area with '(2905 characters remaining)'.
- Observations and Recommendations:** Text area with '(2858 characters remaining)'.

Callout boxes provide further details:

- Top Left:** These drop downs are filled in when the interval is created and updated when the next interval is created. They can be changed here if necessary.
- Bottom Left:** These Fields are brought over from the Daily View when 'Update Fields from Daily' is pressed, only, Frac Grad: is not. In order for the Fracture Gradient to be tracked you must enter it.
- Bottom Right:** Use these drop downs to set the time of day that the interval changes. It will give more accurate calculations of costs per Interval.

The three checkboxes below the time first used drop downs are used to indicate data about the interval. You should check the appropriate ones that reflect that interval.

1. Check ☒ **Fluids provided by M-I SWACO?** when we are providing fluid service.
2. The ☐ **Is RDF?** checkbox is used to indicate if you are drilling the interval with a special reservoir drill in fluid.
3. The ☒ **Is Non operational Interval** is used to show that no services were provided by M-I SWACO and so there should be no charges for the interval.

The boxes labelled "Comment for Recap" and "Observations and Recommendations", are used for any data that you feel applies to the overall interval. These fields will not be printed on the daily report but only on the recap.

If the interval type is completion, extra fields will become active in the middle of the interval details section. Displayed here will be the perforation information, which can be entered when the interval with type completion was created, and a field for you to enter the bottom hole formation pressure.

Interval Details - Interval - 4

Operational Mode:	Completion	Type:	Casing
Casing OD (in):	9.625	Casing ID (in):	8.535
Depth (ft):	8,002.00	TVD (ft):	6,880.00
Top Of Liner/ (ft):		Bit Size (in):	12.25
Sidetrack		Max Density:	0
Max Inclination:	45	Interval Days:	3
Frac Grad:		Max Circ. BHT:	184
Interval Costs:	-1878.90	BH Formation Pressure:	
Time First Used:	00 00		

☒ Fluids provided by M-I SWACO? ☐ Is RDF?

☐ Is Non operational Interval

Update Fields from Daily

Comment for Recap

Perforations	
From (ft)	To (ft)

Bottom Hole Formation Pressure

Perforation Information

3.2.3 Control Buttons Section

Create New Interval is how all intervals but interval “0” are created, when pressed it will open the create new interval dialogue. In the upper section, previous interval, confirm or update the information about the finished interval.

In the middle section, new interval, enter the information about the new interval. The fields for casing OD and casing ID will only be active if you choose casing, liner, or side track as the type. The Top of Liner/Sidetrack will be active when those two types are chosen. Click the ☐ Is RDF? box on if the interval will be drilled using reservoir drill-in fluid.

If the new interval type is completion, a section to enter perforation information will also become active underneath the new intervals type.

Create New Interval

will save the data you have entered and create the new interval.

Cancel

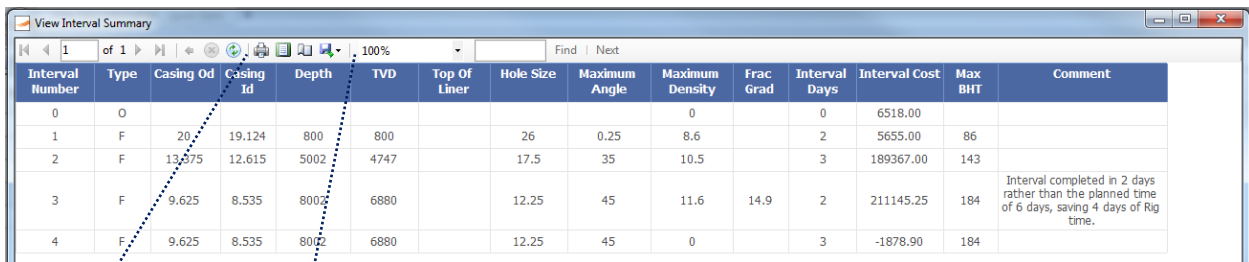
will close the dialogue and abandon your changes.

Delete Selected Interval

will remove the highlighted interval. This button will only be active if the interval has no days or costs associated with it. You are also unable to delete Interval “0”.

View Interval Summary

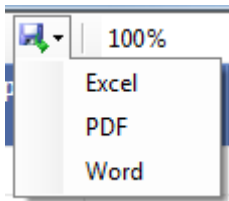
will open up the interval summary window. This window will display the summary information from all intervals.



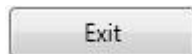
Interval Number	Type	Casing Od	Casing Id	Depth	TVD	Top Of Liner	Hole Size	Maximum Angle	Maximum Density	Frac Grad	Interval Days	Interval Cost	Max BHT	Comment
0	O								0		0	6518.00		
1	F	20	19.124	800	800		26	0.25	8.6		2	5655.00	86	
2	F	13.575	12.615	5002	4747		17.5	35	10.5		3	189367.00	143	
3	F	9.625	8.535	8002	6880		12.25	45	11.6	14.9	2	211145.25	184	Interval completed in 2 days rather than the planned time of 6 days, saving 4 days of Rig time.
4	F	9.625	8.535	8002	6880		12.25	45	0		3	-1878.90	184	



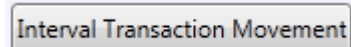
These control buttons allow you to Print this report and save it



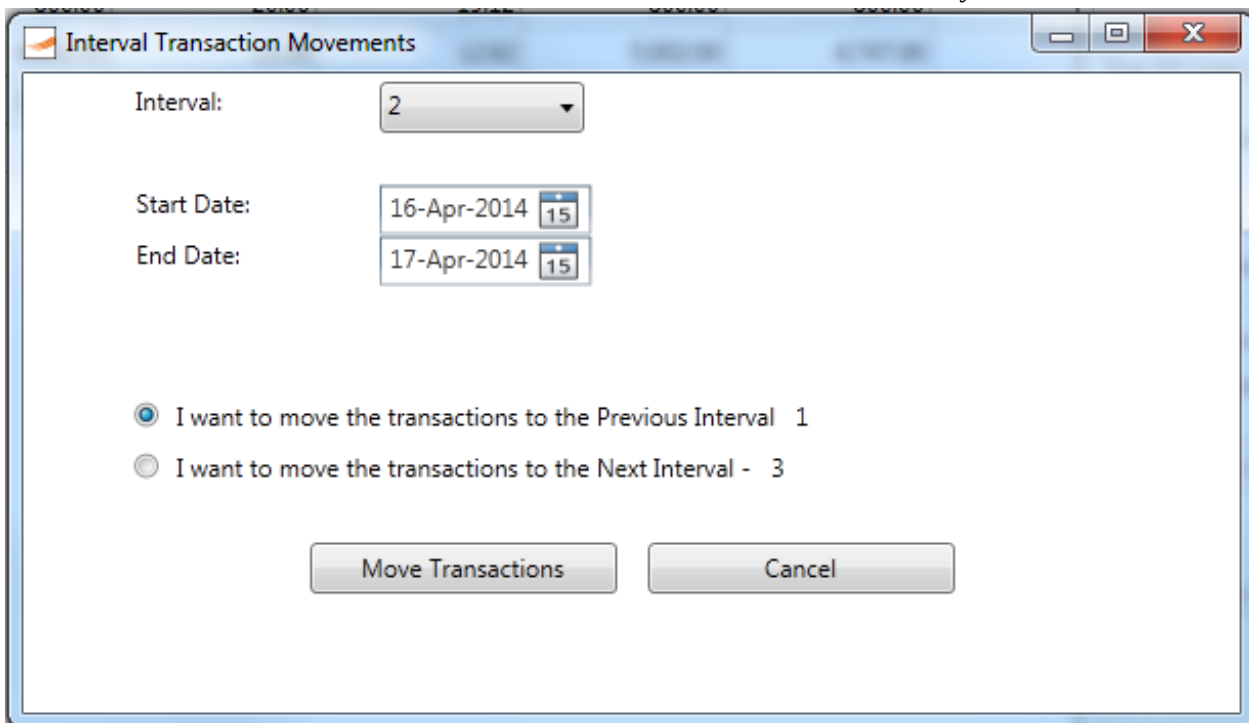
in a choice of formats.



closes the window and returns to the intervals tab.



button allows you to move transactions from one interval to another. It opens a window, the interval transaction movement window, where this is done. You may move all the transactions



Interval: 2

Start Date: 16-Apr-2014 15

End Date: 17-Apr-2014 15

☒ I want to move the transactions to the Previous Interval - 1

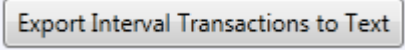
☐ I want to move the transactions to the Next Interval - 3

Move Transactions Cancel

from that interval by using the drop down or a date range by setting the start and end dates. Once you chose which interval you wish to move the transactions too using the radial buttons,  will

become active. Pressing this button will assign all volume accounting transactions in the date range to the chosen interval. It will not change the interval on the date itself.  will close the window without processing the request.

 will open Excel and export the active interval's volume accounting transactions.

 will open Notebook and export the active interval's volume accounting transactions. The text export will include the details of the add chemical transactions where the Excel will not.

3.3 PERSONNEL SETUP TAB

Information for all personnel involved in the project needs to be entered in the personnel setup tab. Information in this table is used to generate a dropdown menu on the general tab, daily information view, M-I SWACO representatives table.

Name	LDAP ID	Email	Description	Service Line	Charge Item	Daily Charge Amount
David Cabral	dcabral		Senior Solids Control Engineer	Environmental Solutions	R0081546	625.00
David Jones	djill		Mud Engineer	Drilling Solutions	R0073102	500.00
Enrico Balzarelli	ebalzarelli		Completion Engineer	Workover/Production	R0088115	575.00
George Strait	gstrait		IFE Engineering Service	Drilling Solutions	R0088114	500.00
Josemaurys Villenilla	jmendozavallenilla	jmendozavallenilla@slb.com	WP Engineer	Drilling Solutions	R0088115	0.00
Phillip Abbott	pabbott	pabbott@slb.com	Mud Engineer	Drilling Solutions	R0073102	550.00

To add a person to a Personnel table:

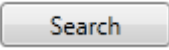
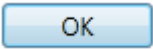
- Enter the employee's full name in the name column
- Enter the employee's LDAP ID, if employee has one, in the LDAP ID column
- Enter the employee's email address in the email column. Entering @slb.com email address is preferred.
- Enter the employee's position description (ex. Mud Engineer, Senior Solids Control Engineer, etc.)
- Choose employee's service line from a drop down menu
- Charge Item is a code assigned for engineering charges by local M-I SWACO office. The number should be provided by the project engineer. The charge number cannot be typed in; it has to be retrieved from the master data list. To find a charge number, click . The search charge number window will open.

Search Charge Number

Search Criteria: R0081546 [Search]

Number ▾	Name ▾	Unit Size ▾	Unit ▾	Package ▾
R0081546	SENIOR SOLIDS CONTROL ENGINEER	1 DAY	N/A	

[OK] [Cancel]

Enter the charge number or a description in the search criteria field and click . The table will be populated with entries matching the information entered in the search criteria field. Activate the record you want to add by clicking on it, click  and the charge number will be added to the personnel setup table.

- Enter the daily charge amount for the employee's services.

Validate SLB User Names

To validate M-I SWACO employees' LDAP ID click . Note: a SLB intranet connection is required for ID validation. LDAPs ID that are confirmed to be a correct ID will be highlighted green; if a LDAP ID is not found in the server, it will be highlighted red. If the LDAP is not highlighted red or green, it means a validation has not been performed on that person.

LINK – need to determine if the validation works will full, basic, or both access.

HELPFUL HINT

If having trouble validating LDAP IDs, check the ID field for extra spaces at the end of the ID or misspellings.

General	Intervals	Product Setup	Screens Setup	Equipment Setup	Pe
Name	LDAP ID	Email			
David Cabral	dcabral				
David Jones	djIII				
Enrico Balzarelli	ebalzarelli				
George Strait	gstrait				
Josemaurys Villenilla	jmendozavallenilla				
Phillip Abbott	pabbott				
Eric Hamington	Ehammington2				

3.4 PRODUCT SETUP TAB

The screenshot shows the ONE-TRAX Product Setup Tab. It features a top menu bar with tabs: General, Intervals, Personnel Setup, Product Setup (selected), Screens Setup, Equipment Setup, Lithology / Well Survey / Fluid Systems, Pits Setup, Loss Setup, Plan Data, and Utility. Below the menu is a toolbar with buttons: View Daily Information, Quick Open, and a search icon. The main area is divided into three sections:

- Active Items Table:** A table listing items with columns: Item Code, Item Name, Alt Name, Unit Size, Unit, and Unit Price. The table contains 27 items, with the first item highlighted in blue.
- Dare Range Price Table:** A table with columns: Date From, Date To, and Price. It is currently empty.
- Item Information Section:** A form for editing an item. It includes fields for Item Code, Item Name, Alt Name, Unit Size, Unit, Packing, Price, Discount(%), SG, Show Conc?, Inventory Item, Product Group, Cost Code, CF Cost Code, Price Per Unit, and Starting Qty. There are also buttons for Save, Cancel, and Unlock.

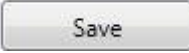
At the bottom of the window, there is a status bar showing various costs and totals. Below the screenshot, three labels with arrows point to the respective sections:

- Active Items Table
- Dare Range Price Table
- Item Information Section

This picture shows the product setup tab with red boxes outlining the three main sections of the tab.

3.4.1 Active Items Table

This table is exactly what its' name implies, it is the list of items that have been set up to be used in this well file. The very top of the table displays the count of active products in the well file. The fields that have a gray background cannot be changed while those with a white background can. Updates to the item record and adding products are done in the item information section, discussed elsewhere; and the same gray/white colors will be used there as well.

To change the information of an item in the active products table, click on the row of the item you wish to modify; and the item information section will be filled with the information for that product. Merely change the cell(s) you need to update and click .

Number of items - 26

Item Co	Item Name	Alt Name	Unit Size	Unit
M0000003	M-I BAR BULK	Bar	1 Ton	
M0000321	M-I GEL BULK	Gel	1 Ton	
M0000333	FILTER CARTRIDGE 2 MICRON ABSOLUTE 40 IN LONG	2 micro ABS Cartridge	1 Not A	
M0001664	POTASSIUM CHLORIDE BRINE-(9.6POUNDS/GAL)	KCL Brine	1 Barrel	
M0001686	SAFE-VIS E	Safe Vis	5 Gallor	
M0001993	DF-1 BASE FLUID	DF1	1 Barrel	
M0002172	DIATOMACEOUS EARTH	DE	50 Pounc	
M0002205	POLYPAC UL	PolyPac	50 Pounc	
M0002679	VERSAGEL	VersaGel	50 Pounc	
M0002978	DUO-VIS	DUO VIS	50 Pounc	
M0003047	SAFE-CARB 40	SC 40	50 Pounc	
M0003127	VERSAMUL I	Versamul	200 Liter	
M0003212	CALCIUM CHLORIDE PELLETS 94-97 PERCENT	CaCl2	100 Pounc	
M0003239	KCL / POLYMER PREMIX	KCL Premix	1 Barrel	
M0003641	POTASSIUM CHLORIDE (KCL)	KCL Bagged	50 Pounc	
M0004024	CAUSTIC SODA	Soda	50 Pounc	
M0004111	LIME	LIME	50 Pounc	
M0004181	SODA ASH	Soda Ash	100 Pounc	
M0005469	J-GUAR 308NB-GUAR GUM	Guar Gum	50 Pounc	
M0006343	VERSACOAT NS	Versacoat	200 Liter	
M0006408	NUT PLUG	Nut Plug	50 Kilogr	
M0008169	SAFE-CARB 30	SC 30 Bulk	1 MT	
M0009017	CALCIUM CHLORIDE BRINE S.G. 1.40	CaCl2 Brine	1 Barrel	
M0016425	DEEPCLEAN NS	deepclean	55 Gallor	

Delete Item

will remove the active item from the project. You will only be able to delete the item if it has not been used.

Import

will bring in an outside item list into the project. The file to be imported must have certain characteristics:

1. It must be a Microsoft Excel file.
2. It must have a worksheet called ProjectItems
3. The column headers of that worksheet must be in the first row and in the correct format

It is easiest to import a file that has been exported from another ONE-TRAX well file. Pressing the button starts the process by opening a navigation window allowing you to locate the workbook you wish to import.

Export

will export the active items table as a Microsoft Excel workbook. Pressing it will open a navigation dialogue allowing you name and store the list as you desire.

Quick Update

opens a window allowing you to enter certain cells in the active items table as if it were a spreadsheet. The items will be shown in two sub-tables, approved and un-approved items. Once the fields

Save

have been updated the button will write the changes and the

Cancel

will leave the tab without rewriting the records.

Approved Items						
#	Item Code	Item Name	Alt Name	UOM	Price	Starting Qty
1	M0000003	M-I BAR BULK	Bar	1 TN N/A	105	162
29	M0000321	M-I GEL BULK	Gel	1 TN N/A	154	0
3	M0001993	DF-1 BASE FLUID	DF1	1 BL N/A	85	240
4	M0003212	CALCIUM CHLORIDE PELLETS 94-97 PPG	CaCl2	100 LB BG	25.5	20
5	M0009017	CALCIUM CHLORIDE BRINE S.G. 1.40	CaCl2 Brine	1 BL N/A	22.6	3984
6	M0004024	CAUSTIC SODA	Soda	50 LB BG	25.5	64
7	M0002978	DUO-VIS	DUO VIS	50 LB BG	320	279
8	M0005469	J-GUAR 308NB-GUAR GUM	Guar Gum	50 LB BG	75	40
30	M0004111	LIME	LIME	50 LB BG	6.5	0
10	M0003127	VERSAMUL I	Versamul	200 LT DM	485	17
11	M0016822	VERSACLEAN SYSTEM 11.1 - 11.5 PPG	SBM	1 BL N/A	90	0
31	M0006212	VERSACLEAN	Versaclean	200 LT DM	510	0

Unapproved Items						
#	Item Code	Item Name	Alt Name	UOM	Price	Starting Qty
35	M0000060-001	METHANOL	Methanol	1 KG N/A	10	0
36	M0000140-001	IRON CHLORIDE	FeCl	100 LT DM	15	0
37	M0000190-001	HYDROCARBON ABSORBANT MOSS	Absorbant	25 KG BG	25.25	0

Change Prices on Same Date

is a quick way to change the price for several items on the same date. Clicking will open a tab displaying the active items table. At the top will be the date the price change is effective, this defaults to active date but can be changed by typing a new date in the start date field or by clicking . Enter the new price in the price column for the items that had a price change and press

Save

. A new date price range table entry for the items with new prices changed will be created

Cancel

will close the tab without updating.

Change Prices On Same Date

Start Date: 05-Aug-2014

#	Item Code	Item Name	Alt Name	UOM	Old Price	Price
1	M0000003	M-I BAR BULK	Bar	1 TN N/A	105	105
29	M0000321	M-I GEL BULK	Gel	1 TN N/A	154	154
3	M0001993	DF-1 BASE FLUID	DF1	1 BL N/A	85	85
4	M0003212	CALCIUM CHLORIDE PELLETS 94-97 PER	CaCl2	100 LB BG	25.5	25.5
5	M0009017	CALCIUM CHLORIDE BRINE S.G. 1.40	CaCl2 Brine	1 BL N/A	22.6	22.6
6	M0004024	CAUSTIC SODA	Soda	50 LB BG	25.5	25.5
7	M0002978	DUO-VIS	DUO VIS	50 LB BG	320	320
8	M0005469	J-GUAR 308NB-GUAR GUM	Guar Gum	50 LB BG	75	75
30	M0004111	LIME	LIME	50 LB BG	6.5	6.5
10	M0003127	VERSAMUL I	Versamul	200 LT DM	485	485
11	M0016822	VERSACLEAN SYSTEM 11.1 - 11.5 PPG	SBM	1 BL N/A	90	90
31	M0006343	VERSACOAT NS	Versacoat	200 LT DM	510	510
13	M0002679	VERSAGEL	VersaGel	50 LB BG	71	71
14	M0006408	NUT PLUG	Nut Plug	50 KG BG	18.5	18.5
35	M0000060-001	METHANOL	Methanol	1 KG N/A	10	10
36	M0000140-001	IRON CHLORIDE	FeCl 3	100 LT DM	15	15
37	M0000190-001	HYDROCARBON ABSORBANT MOSS	Absorbant	25 KG BG	25.25	25.25
28	M0016425	DEEPCLEAN NS	deepclean	55 GA DM	2500	2500
19	M0003047	SAFE-CARB 40	SC 40	50 LB BG	7.5	7.5
21	M0004181	SODA ASH	Soda Ash	100 LB BG	25.5	25.5
22	M0003239	KCL / POLYMER PREMIX	KCL Premix	1 BL N/A	52.5	52.5
23	M0001686	SAFE-VIS E	Safe Vis	5 GA CN	210	210
24	M0002172	DIATOMACEOUS EARTH	DE	50 LB BG	7.5	7.5
25	M0000333	FILTER CARTRIDGE 2 MICRON ABSOLUTE	2 micro ABS Cartridge	1 N/A EA	30	30
26	M0016822	VERSACLEAN SYSTEM 11.1 - 11.5 PPG	Used Versaclean Mud	1 BL N/A	70	70

Save Cancel

View Price History

displays the price and the effective date for each item. There is one row for each price table record.

View Price History

Product Price History List By Product Name

#	Product	Description	Price	Start Date
6	M0002920	CALCIUM CHLORIDE POWDER 94-97 PERCENT	22.4	03-Dec-2012
7	M0007942	CAUSTIC SODA	46.26	03-Dec-2012
33	M0007438	CIRCAL 1000	25.67	03-Dec-2012
9	M0006565	CITRIC ACID	57.94	03-Dec-2012
10	M0019517	DEEPCLEAN	15.75	03-Dec-2012
11	M0003617	DEFOAM A	99.78	03-Dec-2012
12	M0002944	DUO-VIS	166.38	03-Dec-2012
13	M0007336	ECOTROL RD	438.59	03-Dec-2012
14	M0008115	G-SEAL FINE	86.51	03-Dec-2012
15	M0006781	GUAR GUM 3500 CPS	76.53	03-Dec-2012
1	M0001588	KCL SUBSTITUTE BRINE	65.51	03-Dec-2012
16	M0000467	KLEEN UP	80.12	03-Dec-2012
17	M0006805	LIME	15.54	03-Dec-2012
2	M0006008	M-I BAR BULK	325.85	03-Dec-2012
3	M0006050	M-I GEL BULK	382.45	03-Dec-2012
18	M0018824	NOVATEC F	5.47	03-Dec-2012
19	M0019515	NOVATEC P	5.31	03-Dec-2012
20	M0019516	NOVATEC S	5.11	03-Dec-2012
5	M0002200	NOVATEC SYSTEM	320.25	03-Dec-2012
5	M0002200	NOVATEC SYSTEM	318.21	03-Dec-2012
5	M0002200	NOVATEC SYSTEM	350	03-Dec-2012
22	M0007441	OMYACARB 40	19.39	03-Dec-2012
21	M0012123	OMYACARB 50	18.05	03-Dec-2012

Exit Print

Novatec had three prices during this project.

Exit

will close the window and

Print

will open Excel and export the price book.

3.4.2 Item Information Section

Items are added through the search feature of the master data list. The search term is entered in the master list filter field and the **Search** button is pressed. The more defined the search term is the fewer results will be returned for you to choose among. To add a product, you need to activate it by clicking on its' row in the search table.

Master List Filter: mega Search

Item Code	Item Name	Unit Size	Unit	Packing
M0014024	MEGAMUL	55	GA	DM
M0014040	MEGAMUL	1	GA	
M0014045	MEGADRIL SYSTEM	1	BL	
M0014722	MEGATROL	25	KG	BG
M0014749	MEGADRIL SYSTEM	1	CM	
M0016770	MEGAMUL	1000	LT	TK
M0017459	BIODEGRADABLE DEGREASER MEGAS	200	LT	DM

☐ Include old/obsolete item codes in the search

Add as Approved Item
Add as Un-Approved Item

The Blue Highlight shows when the row is Active

☐ Include old/obsolete
item codes in the search

The check box, ☐ Include old/obsolete item codes in the search, will allow you to search for products that are marked “inactive” in the master data list. Generally these are the products that are no longer actively manufactured. This facility is included so that you can add these products to the project if it is necessary.

Then press or , pressing either of them will fill in the data cells of the item information table with the information from the master data list. You need to enter any information that is missing in the cells and pull-downs. **It is preferable to use “Approved” items whenever possible.**

Adding item as Approved Item

Item Code: M0000001	Item Name: M-I BAR	Alt Name:
Unit Size: 100	Unit: LB	Packing: BG
Price: 12	Discount(%): 0	Taxable: ☐
SG: 4.2	Show Conc? ☒	Inventory Item ☒
Product Group: Weight Materi	Cost Code: Chemicals	CF Cost Code: N/A
Price Per Unit: ☐	Locator: Domestic	Starting Qty: 0

HELPFUL HINT

Checking the show concentration box will have ONE-TRAX calculate the concentration of that product when it is used in the drilling fluid

When you , you will be able to modify the cells that are locked in by an add approved item transaction:

- Item Code – for approved items this is the M-I SWACO product number
- Item Name – for approved items this is the official M-I SWACO name.
- Unit Size
- Unit
- Packing

By changing these fields the item will no longer match the master data list and you may slow down the billing process at the close of the well.

Fields that are updateable in both approved and un-approved item adds:

- Alt. Name
- Price
- Discount – How much of a discount off the price is the client receiving

- Taxable – will M-I SWACO collect taxes on the item
- SG – specific gravity of the item, if not provided by the application data, it can be found in the MSDS or SDS sheet.
- Show Concentration Checkbox (See Helpful Hint)
- Inventory Item Checkbox - this checkbox is used to show if the item is one that has counted inventory that must be tracked. Certain items are not inventory items, for example, Freight Pass Through or shrink wrapping. These are items that are charged for but not delivered using a ticket and therefore will normally result in an negative ending inventory. When the inventory checkbox is cleared, ONE-TRAX will ignore negative ending quantities. See graphic.

Freight is not an inventory item, Therefore there is no value for the final count.

Product Name	Alt Name	Item No	Gross Price	Used For Fluid	Used Other	Final
FREIGHT - PASS-THROUGH	Freight	R0088304	1	0	2004.5	N/A
G-SEAL PLUS	G Seal	M0008069	84.52	0	0	35

Save

will store the item record as shown to the active items table.

Cancel

clears the item information section's data fields.

Unlock

will change the item from "Approved" to "Un-Approved" and can only be used on an item already added to the active items table. The item can be changed back but only if none of the previously locked fields have been changed.

Product Group – Some items have several uses, please chose the use that the item will have on your project, from the application data.

Cost Code – what category the cost for this item will be added into.

CF (Completion Fluid)
Cost Code – what category the cost will be added into when the well is using Completion fluids.

Product Group: Base Fluid

Cost Code: Base Fluid

CF Cost Code: N/A

Price Per Unit: ☐

Locator: Domestic

Starting Qty: 0

Price Per Unit – Checking this box will have the item changed per unit rather than per pack, normally not checked.

Locator – Used to track if the item is imported or domestic.

Starting Qty – if the item has stock that will not be received through a delivery, place the amount here.

3.4.3 Product Date Range Price Table

This feature allows you to update the prices on products without effecting transactions on prior dates. To enter a new price you first activate the product by clicking on it in the active items table and then pressing the

Add Date Range Price

button. The system will create a date record using the active date as when the new price will start. You may change the start date for the price change by either typing in a new date or pressing  and navigating to the date you wish. Enter the new price and during that period that product will be

charge at that price. You must press **Recalculate** before you can save the price record, this will

recalculate all item costs in order to insure all records are current. You must press **Save** to record these changes.

Date From	Date To	Price
23-Apr-2014 	23-May-2014	90

Add Date Range Price
Delete Date Range Price

The **Delete Date Range Price** button is used to remove an entry from the table. First you must activate the row by clicking on it. The active row will have a field with a blue background  or a blue outline .

If the table has multiple records in it, deleting a record will cause the price to revert the prior time period's price.

In the example below, notice how the first entries' date to field adjusted to take over the date range of the second when it is removed.

Before

Date From	Date To	Price
15-Apr-2014	04-Aug-2014	210
05-Aug-2014	04-Sep-2014	220
05-Sep-2014	05-Oct-2014	230
06-Oct-2014	05-Nov-2014	240

After

Date From	Date To	Price
15-Apr-2014	04-Sep-2014	210
05-Sep-2014	05-Oct-2014	230
06-Oct-2014	05-Nov-2014	240

3.4.4 Control Buttons

Warehouse Codes

will bring up the warehouse maintenance tab, see section on [general tab, project view](#) for details.

Product Groups

will open maintenance for the product group pull-down list in the item information section. This allows you to add more product groups to the ones provided by the application data. Any group you add may be used in this well file and on any well using it as the source well file in the **Create New Project from Existing Project** function.

To create a new product group, enter the name in the group name field and press

Add New Group

. The group is now able to be used.

To delete a group, highlight it in the group name table by clicking on it and press

Delete Selected Group

and it will be deleted.

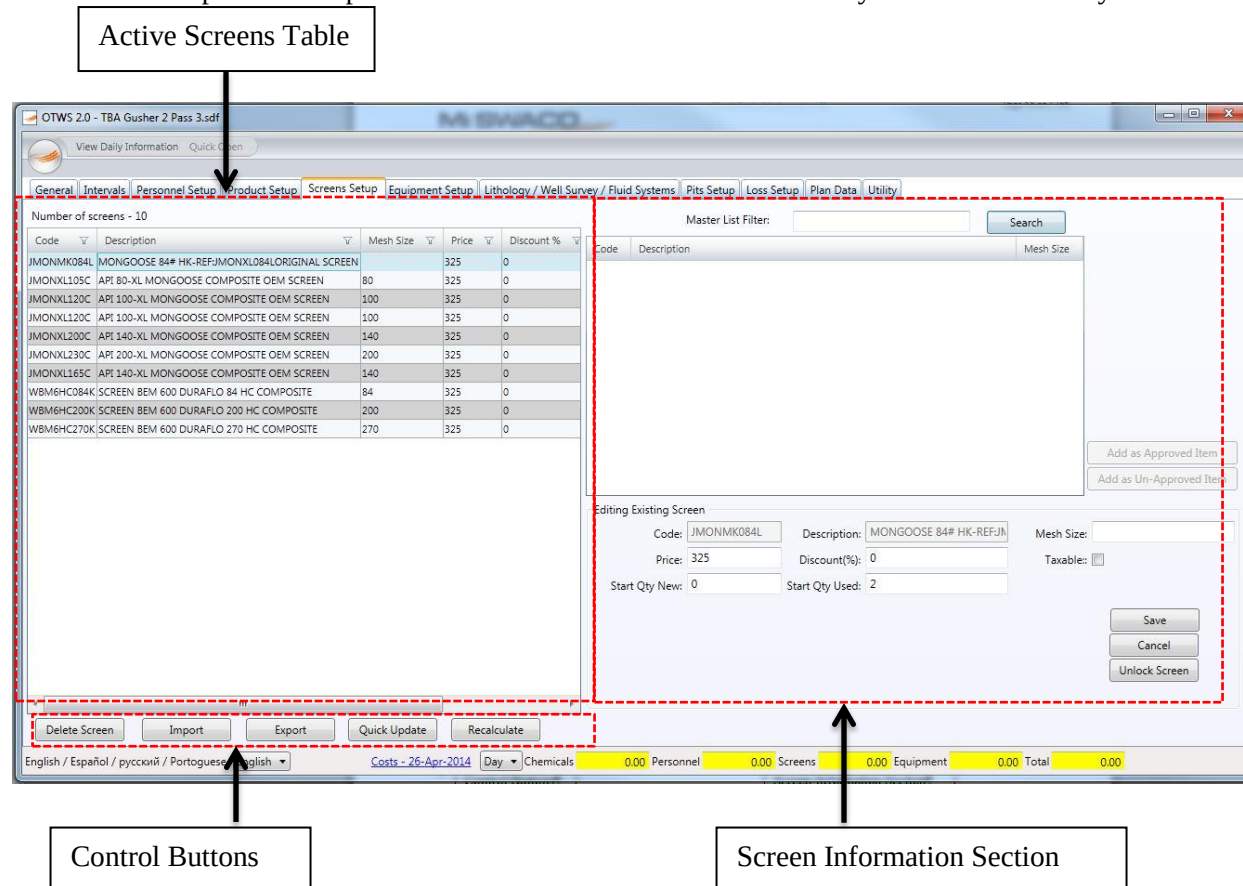
You can only delete the groups added and cannot delete the groups from the application data, these product groups have a gray background.

Done

closes the window.

3.5 SCREENS SETUP TAB

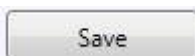
The screens setup tab is where the information about the shaker screens you will use on your project is entered. It is similar to the product setup tab and has much of the same functionality but is limited solely to screens.



3.5.1 Active Screens Table

This table is a list of the screens that have been set up to be used in this well file. The very top of the table displays the count of the number of screens setup in the well file.

To change the information of a screen, click on the row you wish to modify; and the screen information section will be filled with the information for that screen. Merely change the cell(s) you need to update and click



and your changes will be updated.

3.5.2 Screen Information Section

Screens are added through the search feature of the master data list. The search term is entered in the master list filter field and the  button is pressed. The more defined the search term is the fewer results

will be returned for you to choose among. To add a screen, you need to activate it by clicking on its' row in the search table.

Master List Filter:

Code	Description	Mesh Size
J2X6MG200D	API 200-MG SWACO 2x6 UNBONDED OEM SCR	200
J380XL230M	API 200-XL D285P/380 MTL BACK REPLACEMEN	200
J380XR325M	API 200-XR D285P/380 MTL BACK REPLACEMEN	200
JBHXC325C	API 200-HC BHX COMPOSITE REPLACEMENT SC	200
JBHXXL230C	API 200-XL BHX COMPOSITE REPLACEMENT SC	200
JBHXXL270C	API 200-XL BHX COMPOSITE REPLACEMENT SC	200
IBM3XL230I	API 200-XL BEM3 PLST BACK OEM SCREEN	200

The Blue Highlight shows when the row is Active

Then press or , pressing either of them will fill in the data cells of the screen information table with the information from the master data list. You need to enter any information that is missing in the cells and pull-downs. It is preferable to use "Approved" items whenever possible.

Code:	JMONXL105C	Description:	API 80-XL MONGOOSE CO	Mesh Size:	80
Price:	325	Discount(%):	0	Taxable::	☐
Start Qty New:	1	Start Qty Used:	2		

When you , you will be able to modify the cells that are locked in by an add approved item transaction:

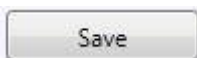
- Code – for approved screens this is the M-I SWACO product number
- Description – for approved items this is the official M-I SWACO name.

By changing these fields the item will no longer match the master data list and you may slow down the billing process at the close of the well.

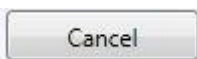
Fields that are updateable in both approved and un-approved item adds:

- Price

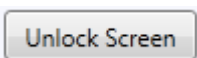
- Discount – How much of a discount off the price is the client receiving
- Taxable – will M-I SWACO collect taxes on the item
- Start Qty New – how many screens of this type on the rig are new.
- Start Qty Used – how many screens of this type on the rig are used.



will store the item record as shown in the active screens table.

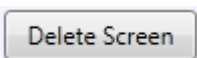


clears the item information section's data fields and makes no changes to the item.

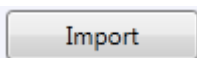


will change the screen from "Approved" to "Un-Approved" and can only be used on an item already added to the active screens table. The screen can be changed back but only if none of the previously locked fields have been changed.

3.5.3 Control Buttons



will remove the active screen from the project. You will only be able to delete the screen if it has not been used.



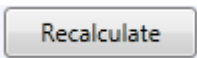
will bring in an outside screen list to the project. The file to be imported must have certain characteristics:

1. It must be a Microsoft Excel file.
2. It must have a worksheet called ProjectScreens
3. The Column headers of that worksheet must be in the first row and in the correct format

It is easiest to import a file that has been exported from another ONE-TRAX well file. Pressing the button starts the process by opening a navigation window allowing you to locate the workbook you wish to import.



will export the active screen table as a Microsoft Excel workbook. Pressing it will open a navigation dialogue allowing you a name and storage location.



will have ONE-TRAX recalculate all transactions dealing with screens. It will calculate cost and inventory.

Quick Update

opens a window allowing you to enter certain cells in the active screen table as if it were a spreadsheet.

Screens Setup - Quick Update

Approved Screens

#	Code	Description	Mesh Size	Price	Start Qty New	Start Qty Used
1	JMONMK084L	MONGOOSE 84# HK-REFJ		325	0	2
2	JMONXL105C	API 80-XL MONGOOSE CO80		325	1	2
3	JMONXL120C	API 100-XL MONGOOSE C100		325	6	2
4	JMONXL120C	API 100-XL MONGOOSE C100		325	2	2
5	JMONXL200C	API 140-XL MONGOOSE C140		325	4	2
6	JMONXL230C	API 200-XL MONGOOSE C200		325	3	3
7	JMONXL165C	API 140-XL MONGOOSE C140		325	2	2
8	WBM6HC084K	SCREEN BEM 600 DURAFLK84		325	0	0
9	WBM6HC200K	SCREEN BEM 600 DURAFLK200		325	0	0
10	WBM6HC270K	SCREEN BEM 600 DURAFLK270		325	0	0

Unapproved Screens

#	Code	Description	Mesh Size	Price	Start Qty New	Start Qty Used
---	------	-------------	-----------	-------	---------------	----------------

Save Cancel

The items will be shown in two sub-tables, approved and un-approved items. Once the fields have been updated the **Save** button will write the changes and the **Cancel** will leave the tab without rewriting the records.

3.6 EQUIPMENT SETUP TAB

The equipment setup tab is where the information about the equipment you will use on your project is entered. It is similar to the other setup tabs and has much of the same functionality but only deals with equipment. You may enter equipment rented from M-I SWACO to the project; or equipment owned by the rig or other companies for which it is desired to track statistics for.

Active Equipment Table

The screenshot shows the 'Equipment Setup' tab in the OTWS 2.0 software. The 'Active Equipment Table' label points to the table listing equipment items. The 'Equipment Information Section' label points to the 'Edit Existing Equipment' form. The 'Control Buttons' label points to the buttons at the bottom of the table.

Serial No.	Code	Description	Alt Name	Rental	Stand-by	Taxable
50258	62-8411	518 Centrifuge	Centrifuge 1	350	230	N
50138	62-8411	518 Centrifuge	Centrifuge 2	350	230	N
V120	NC-019	100 HP Vacuum Unit	Vacuum	600	250	Y
VG-4568	NC-059	Verti-G (01) High Capacity	Verti G	850	650	N
A12345	NC-065	Auger	Auger	50	20	N
F2424	NC-085	DE Filter Press Units (1200 sqft)	DE Filter	140	30	N
C0000	NC-066	Cuttings Boxes	Cuttings Box	20	20	N
12005	NC-016	Mongoose Shaker	Shaker 1	0	0	N
12025	NC-016	Mongoose Shaker	Shaker 2	0	0	N
12173	NC-016	Mongoose Shaker	Shaker 3	0	0	N
21056	62-8442	Mud Cleaner 8T4 w/o pump	Mud Cleaner	0	0	N
HVD938	NC-030	Horizontal Vacuum D-GASSER	D-Gasser	0	0	N
10KUC02	NC-028	20K Choke	Choke	0	0	N
3408	NC-066-001	Storage Container	Storage Container	20	20	N
2456	NC-066-001	Storage Container	Storage Container	20	20	N
2346	NC-066-001	Storage Container	Storage Container	20	20	N
345	NC-066-001	Storage Container	Storage Container	20	20	N
25866	NC-002	BEM 600 Shaker	BEM Shaker #3	0	0	N
F1023	NC-086	(Duplex) Cartridge Units	Guard Filter	140	70	N

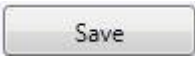
Control Buttons

Equipment Information Section

3.6.1 Active Equipment Table

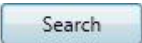
This table is a list of the equipment that you wish to gather information about that is on the rig. The very top of the table displays the count of the number of pieces of equipment setup in the well file.

To change the information of a screen, click on the row you wish to modify; and the edit existing equipment section will be filled with the information for that piece of equipment. Change the cell(s) you need to update

and click  and your changes will be updated.

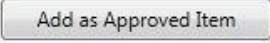
3.6.2 Equipment Information Section

Equipment is added through the search feature of the master data list. The search term is entered in the master list filter field and the  button is pressed. The more defined the search term is the fewer results will be returned for you to choose among. To add a piece of equipment, you need to activate it by clicking on its' row in the search table.

Master List Filter: 

Code	Description
62-8410	414 Centrifuge
62-8411	518 Centrifuge
62-8447	Mark I Centrifuge
62-8449	Mark II Centrifuge
NC-008	518 Euro Centrifuge
NC-009	1850 Centrifuge
NC-011	350A Centrifuge

The Blue Highlight shows when the row is Active

Then press  or , pressing either of them will fill in the data cells of the edit existing equipment section with the information from the master data list. You need to enter any information that is missing in the cells and pull-downs. It is preferable to use “Approved” items whenever possible. However, if you need to add rig equipment from another manufacturer, search for that type of equipment in the master list and add as an unapproved item. For example, your rig has Brant shakers. Search for shakers, add one of the M-I SWACO products as an “Un-Approved Item” and change the information describing it.

Edit Existing Equipment

Equipment Code: Description:

Alt Name:

Serial No.: Taxable: ☐

Rental Price: Stand-by Price:

Discount %:

Equip. Type: Equip. Price Category:

Hazardous Area Classification: General Information:

Property to recalculate:

When you **Add as Un-Approved Item**, you will be able to modify the cells that are locked in by an add approved item transaction:

- Equipment Code – for approved equipment this is the M-I SWACO product number
- Description – for approved Items this is the official M-I SWACO Name.

Fields that are updateable in both approved and un-approved item adds:

- Alt Name
- Serial Number
- Taxable – will M-I SWACO collect taxes on the item
- Rental Price – cost per day of this item, set to “0” if rig equipment or no rental charge
- Stand By Price – cost per day of this item if it is not being used, this field is not always used it can depend on the contract with the operator.
- Discount– how much of a discount off the price is the client receiving
- General Information – useful to record the condition of the equipment and how well it is maintained.
Any other comments or additional information about the equipment you feel it is necessary to record

The drop down menu boxes in the edit existing equipment section are driven from the application data and have choices limited to those data lists.

The screenshot shows a form with the following fields and dropdowns:

- Equip. Type:** A dropdown menu currently showing "Centrifuge".
- Hazardous Area Classification:** A dropdown menu currently showing "Zone 2".
- Equip. Price Category:** A dropdown menu currently showing "Drilling Fluids Solids Remo".
- Property to recalculate:** A dropdown menu currently showing "Discharge Weight".

Two text boxes with arrows pointing to the dropdowns provide additional context:

- A box pointing to the "Equip. Type" and "Equip. Price Category" dropdowns states: "Equipment Type and Equipment Price Category are determined by the piece of equipment itself."
- A box pointing to the "Hazardous Area Classification" dropdown states: "This drop down will only be visible when the Equipment type is set to 'Centrifuge'. You have the choice to have ONE-TRAX calculate either the discharge weight of the solids or the flow rate of the discharge."

The Hazardous Area Classification comes from the manual of the equipment and shows which zone the equipment is rated for:

- Zone 0 – an explosive atmosphere is consistently present or present for long periods of time
- Zone 1 – an explosive atmosphere is occasionally present
- Zone 2 – an explosive atmosphere is rarely present or present for only a short time

Drop Down allowing you to choose which type of equipment to add the extra property too.

Extra properties table – shows which properties you have added.

Equip. Type: Vacuum System

Description	Unit
Vacuum Size	
Hose Size	in
Hose Length	ft
Vertical Lift	ft
Number 45° Bends	
Number 90° Bends	
Average Hg Reading	

Description	Unit
Hours Ran	Hour

Description:

Unit:

Add

Cancel

This table shows the properties currently tracked for that type of equipment.

Enter description and unit you wish to track in these two fields. Once they are filled in the button will become active, press it and the new property will be added to the Extra Properties table.

Cancel

will leave the window and return to the equipment setup tab.

3.7 LITHOLOGY / WELL SURVEY / FLUID SYSTEMS TAB

This multifunction tab contains much useful information. It consists of three main sections relating to the areas named in the tab.

OTWS 2.0 - TBA Gusher 2 Pass 3.sdf

View Daily Information Quick Open

General Intervals Personnel Setup Product Setup Screens Setup Equipment Setup **Lithology / Well Survey / Fluid Systems** Pits Setup Loss Setup Plan Data Utility

Lithology

Starting Depth (ft)	Lithology Type	% Sand
200	IgnBas	3
500	Dolomite	2
800	Sand	100
2000	Anhypr	0.5

Well Survey

☒ Use Well Survey

Starting Depth (ft)	Inclination	Azimuth
0	0	
800	0.25	226
5002	35	226
8002	45	226

Fluid System

Operational Mode	Fluid System
Drilling	Spud Mud
Drilling	KCL/Polymer
Drilling	VERSACLEAN
Completion	Calcium Chloride

Well Survey Import: Import, Export, Graph

File Output: Export

English / Español / русский / Portuguese English

Costs - 26-Apr-2014 Day Chemicals 0.00 Personnel 0.00 Screens 0.00 Equipment 0.00 Total 0.00

3.7.1 Lithology Section

While it is not essential to input lithology, this information can be very useful in visualizations – particularly in the SnapShot. It can demonstrate the correlation between depth, tight spots, or changes in rate of penetration with formation type. Formation layers are defined by TVD, type, and sand content.

OTWS 2.0 - TBA Gusher 2 Pass 3.sdf

View Daily Information Quick Open

General Intervals Product Setup Screens Setup Equipment Setup **Personnel**

Lithology

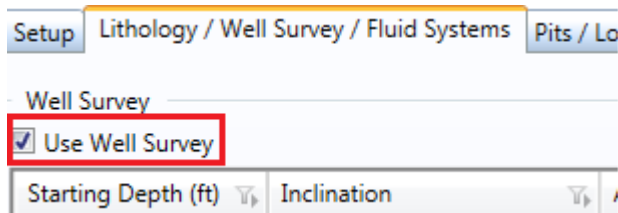
Starting Depth (ft)	Lithology Type	% Sand
200	IgnBas	3.00
800	Sand	100.00
5000	Dolomite	2.00
8000	Anhypr	0.50

the
be

3.7.2 Well Survey

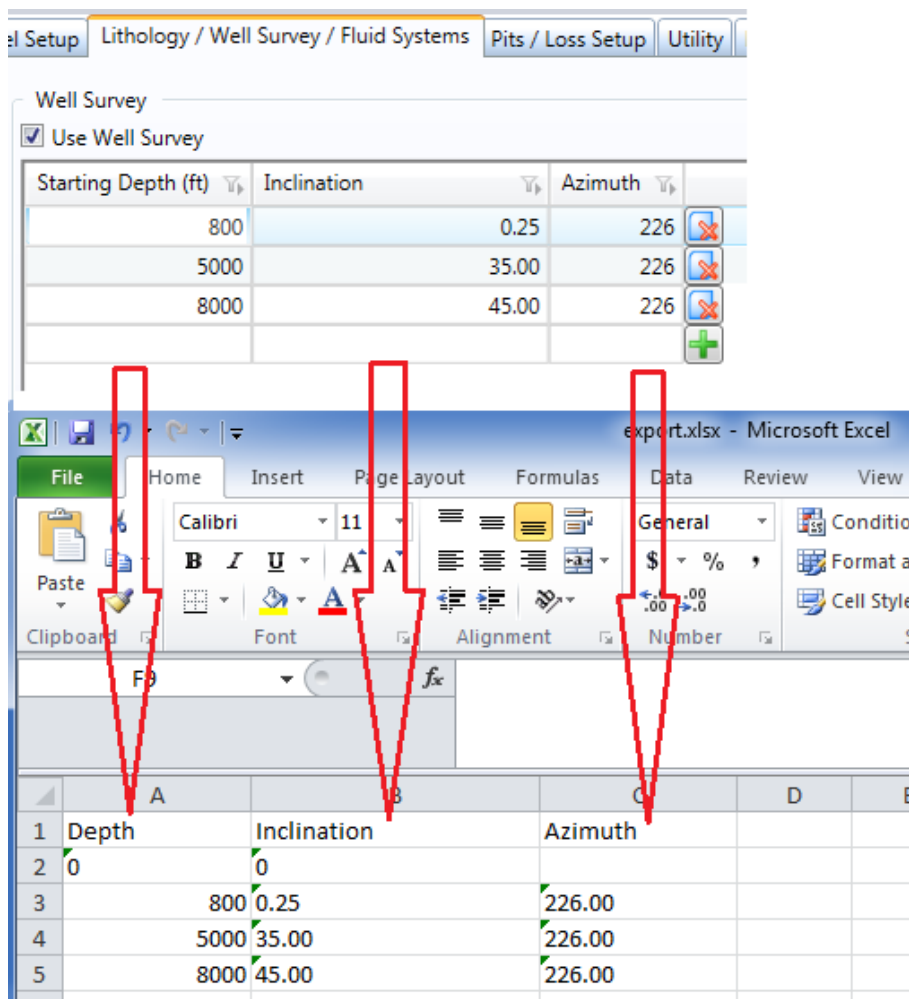
Well survey data are the planned or actual measured surveys for a well. The well survey data inputs are depth, inclination, and azimuth. Well survey data can be entered manually or imported from an Excel file.

IMPORTANT: For more precise hydraulic calculations in ONE-TRAX, check the use well survey check box located at the top of the well survey table.



The screenshot shows the ONE-TRAX software interface for the Well Survey table. The 'Well Survey' tab is active. A red rectangular box highlights the 'Use Well Survey' checkbox, which is checked. Below the checkbox, the table headers 'Starting Depth (ft)' and 'Inclination' are visible.

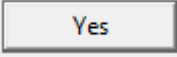
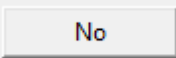
If you receive an Excel file with directional data, you can easily import the information. The key is to have the columns of the Excel file match the columns of the well survey table in ONE-TRAX. An example is shown below. To insure the file imports correctly, the top cell in the column must have the names shown and the worksheet must be named WellSurvey.



The depth, inclination, and azimuth columns in the Excel file will be imported into the corresponding columns in ONE-TRAX – starting depth, inclination and azimuth.

File Input
Import

Once you have edited an Excel file to match the format shown in the example, click , navigate to the file you want to import and click open. If there was well survey information previously entered

or uploaded, a warning message will pop up. Click  to overwrite previously entered information or  to abandon the import.

3.7.3 Fluid Systems

All fluid systems that will be used in the project have to be added to the fluid system table. If you do not set up a fluid systems you will not be able to use it in daily or recap reports.

To set up a fluid system:

1. Pick a fluid system name from a drop down menu
2. Indicate the operational mode the fluid will be used in – drilling or completion.

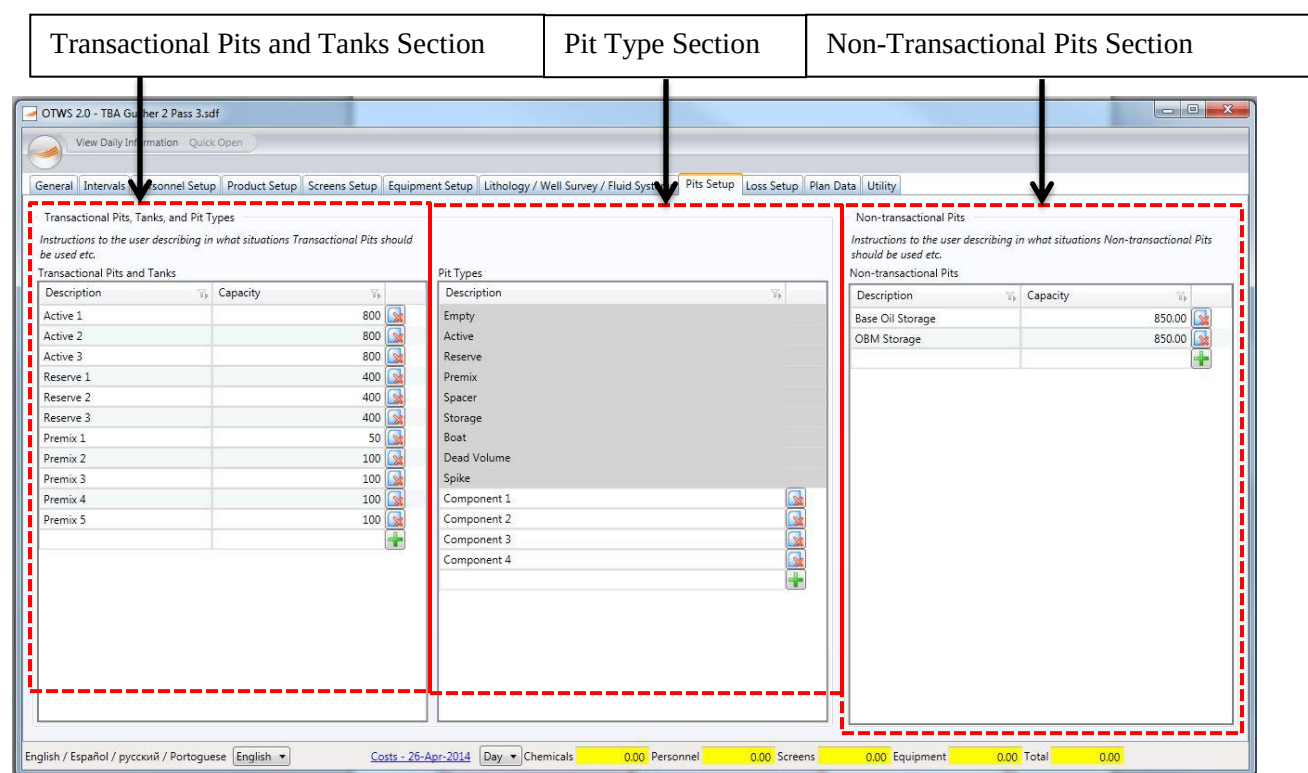
As many fluid systems as necessary can be added in the table.

Fluid System		
Operational Mode	Fluid System	
Drilling	Spud Mud	
Drilling	KCL/Polymer	
Drilling	VERSACLEAN	
Completion	Calcium Chloride	
		

3.8 PIT SETUP TAB

This tab is where information relating to the pits on the rig are set up. It contains 3 Sections:

1. Transactional Pits, Tanks, and Pit Types – These are pits and storage tanks that ONE-TRAX will calculate volumes for. Their volumes and product concentrations will be affected by add chemical and add whole mud transaction. You may also move mud between these pits using the transfer function. You may add as many pits as necessary to describe your pit system.
2. Pit Types – the descriptions of the fluid pits
3. Non-transactional Pits – this section is designed to display information you may find useful or to put it on daily reports when using an Excel based template. These pits ARE NOT affected by any volume accounting transactions.



3.8.1 Transactional Pits and Tanks Section

Pits and Tanks Table

Enter the pit description and the pit capacity for every pit you will be using on the project. The list created in this table will be used on the daily information view in the volume accounting tab and the displacements tab. In the picture below you see the information entered on the project information view (left) and how it is seen on the daily information view (right).

Prev
25-Apr-2014
Next

General
Pumps/Bits
Fluid Checks
Geome

Pits, Tanks, and Pit Types

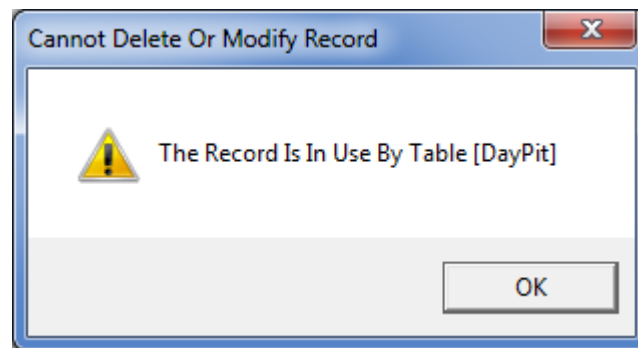
Pits and Tanks

Description	Capacity
Active 1	800.00
Active 2	800.00
Active 3	800.00
Reserve 1	400.00
Reserve 2	400.00
Reserve 3	400.00
Premix 1	50.00
Premix 2	100.00
Premix 3	100.00
Premix 4	100.00
Premix5	100.00

Pits

Pit Description	Capacity
Active 1	800
Active 2	800
Active 3	800
Reserve 1	400
Reserve 2	400
Reserve 3	400
Premix 1	50
Premix 2	100
Premix 3	100
Premix 4	100
Premix5	100

Note that you will not be able to delete the pit description and capacity line if the pit has already been used to enter daily information. In this case you will receive a message similar to the following:



If you edit pit description or capacity at any time during the project, the information will also be changed in all daily records you have previously created.

If you enter two identical descriptions, the second record will be highlighted in red and a hover help box will be generated as shown in the picture below.

Pits, Tanks, and Pit Types

Pits and Tanks

Description	Capacity	
Active 1	800.00	
Active 2	800.00	
Active 3	800.00	
Reserve 1	400.00	
Reserve 2	400.00	
Reserve 3	400.00	
Premix 1	50.00	
Premix 2	100.00	
Premix 3	100.00	
Premix 4	100.00	
Premix5	100.00	
* Active 1	800.00	
		

Pit Description is Invalid

3.8.2 Pits Type Section

The pit types table has a predefined set of pit types in gray. To add a pit type that is not in the predefined set, click on a blank description field and enter the pit type. Note that you will not be able to add a pit type that already exists in the list.

In the picture, the descriptions with the gray background are not updateable and come from the application data. Those with a white background are user defined for the project. In this example, these pits are defined for use in a displacement train. You may assign any description desired to the pit types you create.

Pit Types	
Description	
Empty	
Active	
Reserve	
Premix	
Spacer	
Storage	
Boat	
Dead Volume	
Spike	
Component 1	
Component 2	
Component 3	
Component 4	
	

The list in the pit types table is used to generate a new pit type drop down list in the volume accounting tab. In the picture below you see the information entered on project information view (left) and how it is used on daily information view (right).

Pit Types
Description
Empty
Active
Reserve
Premix
Spacer
Storage
Boat
Dead Volume
Spike

Change Pit Type

Pit: Premix5

Current Pit Type:

New Pit Type:

Empty
Active
Reserve
Premix
Spacer
Storage
Boat
Dead Volume
Spike

3.8.3 Non-Transactional Pits Section

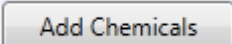
The non-transactional pits section is used to display and track pits that are not part of the system that ONE-TRAX uses in its' volume accounting transactions. You will not be able to add, transfer, or dump mud from these pits using a transaction. The information on volumes in these pits are merely for display purposes only. These pits can be shown on reports designed using an Excel template. You can enter the balance in these tanks in the non-transactional pit section on the volume accounting tab.

Non-transactional Pits

Instructions to the user describing in what situations Non-transactional Pits should be used etc.

Non-transactional Pits

Description	Capacity	
Base Oil Storage	850.00	
OBM Storage	850.00	
		

For Example, the company man wants the base oil volume on the daily report and your project does not charge for base oil when it is in storage. If you add the base oil to a "Transactional" pit with pit type storage, your well will have a charge created. If you add it to a "Non-Transactional" pit you will not be charged. The charge will be created when you use the base oil in your fluid by adding it to a "Transactional" Pit. Since the volume in the "Non-Transactional" pit is not changed by the  transaction, you will manually adjust the volume in the non-transactional pit and ONE-TRAX will adjust the volume in the transactional pits.

3.9 LOSS SETUP TAB

3.9.1 Loss Setup

Loss data categories need to be set up for the drilling mode and the completion mode separately.

HELPFUL HINT

Loss categories are not shared by different Operational Modes. You need to be set up for Loss Categories for both Completion and Drilling.

1. From dropdown list select the operational mode you want to set up first – drilling or completion.

The screenshot shows the 'Loss Data Categories' setup screen in the ONE-TRAX WELLSITE software. The 'Operational Mode' dropdown is set to 'Drilling'. The 'Pre-Defined Sets' dropdown is open, showing a list of categories: Evaporation Related, Hydro, Mexico, M-I SWACO, Russia, Statoil, and UK. The 'Use Pre-Defined Set' button is visible. Below the dropdowns is a table with columns 'Category' and 'Description'. The table lists several categories: Shakers - Mud Or, RISER, DUMPED, and CENTRIFUGES, each with a corresponding description.

Category	Description
Shakers - Mud Or	Fluid wetting the cuttings discharged by Shakers
RISER	
DUMPED	
CENTRIFUGES	

2. You can use a pre-defined set or select loss categories one by one.

Pre-Defined Loss Set – a set of loss categories placed together for a specific purpose in the application data. Most commonly created so the loss categories match those desired by a particular operator.

- If you want to use a pre-defined set:
 - Choose a predefined set from a drop down menu. The pre-defined sets are linked to the mode they were created for.
 - To add a pre-defined set for the drilling mode called evaporation related. Choose drilling in the operational mode drop down and evaporation related in the pre-defined sets drop down.
- The same process is used to add loss categories for completion operations.

3.10 PLAN DATA TAB

The plan data tab display the project plan. This information is not manually entered in ONE-TRAX; it is imported from an Excel file. The normal workflow is for the project engineer to fill in a plan template for the project from the mud program and email it to the mud engineer so it can be uploaded into ONE-TRAX.

HELPFUL HINT

Plan Data is required if the project was 15 days or longer, or if the project is billed in US dollars and the cost was greater than \$150,000.00.

Interval #	Fluid System 1	Fluid System 2	Bit Size	TD	TVD	Days	Fluid System 1 Volume Built	Fluid System 2 Volume Built	Fluid System 1 Cost	Fluid System 2 Cost	Total Cost
1	Spud Mud		26	800	800	2					10000.000
2	KCL/Polymer		17.5	5000	4500	4					150000.000
3	VERSACLEAN		12.25	8000	6000	6					75000.000
4	Calcium Chloride		8.5	8000	6000	8					50000.000

Day #	Depth	Cumulative Cost
1	0	
2	400	
3	800	10000.000
4	800	
5	1400	
6	1800	
7	2600	16000.000
8	3400	
9	4200	
10	4600	
11	5000	
12	5000	
13	5400	235000.000
14	5800	

To Create a Well Plan for Import into ONE-TRAX:

- Click **Export (Empty)** to open the blank PlanDataTemplate.xlsx file. You will only have to save the template in your computer once, since all projects use the same template. Do not change the names of the worksheets in the template – the file can only be uploaded into ONE-TRAX if the worksheet names are correct.
- Open the template using Excel.
- Fill out the template with the information from the mud program. There are 2 worksheets that are required to be filled in, “Plan and Interval” and “Days vs. Depth”.
- When finished use the file tab, **File**, then the **Save As** command. Give the file a new name and chose where you want it to be stored.

To Import a Well Plan into ONE-TRAX:

- Click **Import**, this will open a navigation window. Navigate to the location your Excel file is stored, click on the file name, and press **Open**. ONE-TRAX will import the information into plan data.

3.11 UTILITY TAB

Recalculate Inventory and Cost

will make ONE-TRAX recalculate the inventory transactions. This should only be used to fix errors, or rarely when ONE-TRAX is upgraded.

4 DAILY INFORMATION VIEW

The daily information view is where data relating to the operations performed each day are entered. On this view you will create the daily record, which is the basis of the daily reports. This section of the manual will discuss how to enter fluid checks, equipment usage, geometry, and other information concerned with daily rig operations.

4.1 STARTING A NEW DAY

CREATING THE FIRST DAY OF THE PROJECT

After you fill out and set up the well information, you will be able to create a new day and start recording daily operational information. To create the **very first day** for the project click [View Daily Information](#) on the top of the screen. The “Create New Day” window will pop up. The current date will automatically be generated as the report date, but it can be changed if necessary to match your actual first report date. It is very important to assign the right default interval to the day you are creating because it will affect your interval costs, interval product usage, etc. The default interval can be changed in the general tab, daily view, after the day has been created, but it is a best practice to set the right default interval when creating a new day.

In order to create a new day, you have to select a default fluid system from the drop down list. Items in this list are generated from the fluid systems list in the “Lithology/Well Survey/Fluids Systems” tab. Note: only fluid systems that have the same operational mode as the selected default interval will be shown. If you do not see the fluid system you are using for the interval in the drop down list, check the operational mode set for the fluid system and make sure it matches the interval’s operational mode.

Create New Day

Date: 04-Aug-2014 15

Report #: 1

Default Interval: 0

Default Fluid System: [Dropdown]

☐ Copy Data From Prior Day?:

Warning Message

To 'Create New Day' you must select a 'Default Fluid System', which is setup in the 'Lithology / Well Survey / Fluid Systems' tab.

After creating the first day you will not be able to change the following settings, are you sure you want to proceed?

- ☐ Offshore
- ☐ Unit Set

Create New Day Cancel

Before you click  to create your very first day make sure you have chosen the right unit set and indicated if the well is offshore; you will do this by checking the  **Is Offshore?** box in the general tab, project view. You will not be able to change these two parameters after the first day is created. If you have made an error, you will have to create a new well file in order to make the change.

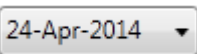
CREATING A NEW DAY

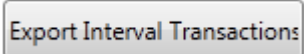
After the first day has been created, new days are created by clicking  on the upper information ribbon, see [Upper Daily Information Ribbon](#) for more details. Clicking the button will cause a create new day window to open. The date and report number will be set automatically to the next value after the previous day. The  **Copy Data From Prior Day?:** box will be checked automatically, this lets you copy most of the information from the previous day. Change the default interval number and/or default fluid system if necessary and click .

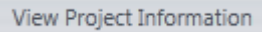
4.2 UPPER DAILY INFORMATION RIBBON



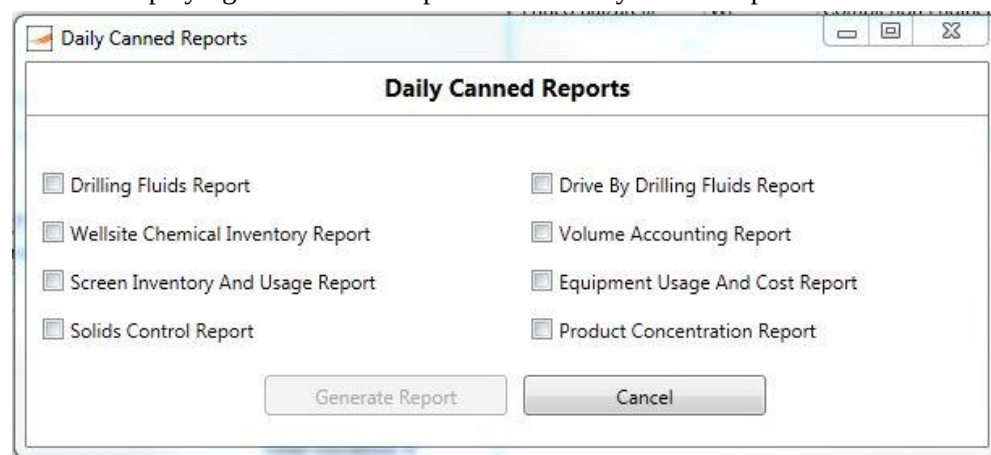
The upper information ribbon, shown above, will be displayed on every tab of the daily information view. You

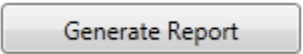
can navigate between dates using either the green arrows,  , or the date drop down menu, .

 opens up a project summary that shows a list of all the project dates along with some information such as interval, depth, daily activity, bit size, costs, etc. You can open any day on the list by clicking the date button on the appropriate row. You can also export all information to Excel using . This is a very useful tool that gives you a quick access to other days' information without leaving your current date.

There is information displayed about the current date on the upper right of the information bar. Operational mode shows the operational mode assigned to current interval. Double check the information presented. If the wrong operational mode is assigned to the interval you can change it by clicking  and going to intervals tab. Interval and fluid system shown are chosen when you create the new day. You can change those parameters in the general tab on the daily information view. Present activity is entered on the general tab as well.

You will also see the  button on this ribbon. Clicking this will open the “Daily Canned Reports” window displaying the available reports. In the daily canned reports window click the checkbox(s) for the reports you wish to print and press



. A new window displaying the chosen reports will open; from here you can print or save them.

100% PDF

After your report is exported, you can return to the report choice menu by using the Windows command in the upper right corner of the report window . To close the report choice window, press



Print Daily



Mi SWACO
A Schlumberger Company

4.3 GENERAL TAB

4.3.1 Current Information

Present Activity:	Drilling Vertical	Report #:	8
Default Interval:	1 - Change Default to Interval # 0 Interval # 2		
Default Fluid System:	MEGADRIL (w/MEGAMUL) ▼		
Operator Rep:	Rick Harless / Ron Fawley		
Contractor Rep:	Paul Slutz / Jerry Bannia		
Rig Telephone #:	724-954-3383	M-I Contact 1:	330-283-3177
Warehouse Telephone #:	732-773-0740	M-I Contact 2:	412-627-6897

Fill out the section to the best of your knowledge. Make sure you have selected the right interval number and fluid system for the day.

The interval can be changed by clicking the correct interval number next to “Change Default Interval”.

HELPFUL HINT	Changing the default interval will not change the interval assigned to transactions already created, but only those created after it is changed. If your well changes interval during the day, you may enter all the transaction for the earlier interval, then change the default interval, and enter the transaction from the newer interval. This will assign the transactions to the proper interval without having to use the “Manage Transactions” feature on the volume accounting tab.
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The default fluid system can be changed with the drop down menu. The fluids are limited to those defined to the operational mode for the interval.

4.3.2 M-I SWACO Representative

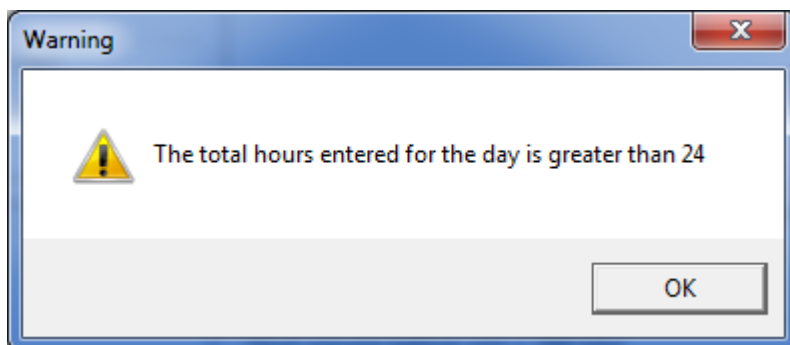
M-I SWACO Representatives						
Representative ▼	Service Lines ▼	Description ▼	Item No. ▼	Daily Charge Amount ▼	Daily Charge Qty ▼	
Miles Johnson ▼	DS	Mud Engineer	R0081108	\$200.00	1	
Doug Fryman ▼	DS	Mud Engineer	R0081108	\$200.00	1	
▼						
Total Personnel: 2						
Daily Personnel Cost: 400.00						

Only personnel that you are charging the operator for the current day should be present in this table, it is good practice to remove non active personnel from this table. Removing them will not remove them from the project, only from this date. To add personnel, select the name you wish to add from the drop down menu. Daily charge quantity is the only updateable field in this table. Daily charge quantity “1” will be used most often. If you need to change any other columns in the table, go to the personnel setup tab, project information view.

4.3.3 Time Distribution Table

Time Distribution		
Activity	Hours	
Rig Up,Service	2.00	
Drilling	22.00	
		
Total Hrs: 24		

Select the activity from the drop down menu and enter its' duration. Make sure that the total hours add up to 24! If the total number of hours is less than 24 hours, the “Total Hrs:” at the bottom of the table will remain red, it will turn black once the total hours are 24. If total hours are more than 24 hours you will get a warning message shown below. If the exact wording of the rig activity is not in the drop down list, chose the closest match possible.



4.3.4 Operational and Fluid Problems

Operational Problems

Problem Encountered ▾	Duration (hrs) ▾	
		+
Total Duration: 0		

Fluid Problems

Problem Encountered ▾	Duration (hrs) ▾	
		+
Total Duration: 0		

Operational and fluid problems encountered during your shift have to be documented! This information is very valuable for project engineers and is used to produce more accurate tenders, to prepare contingency plans for future projects, to improve our service to our customers, etc. From a drop down menu select an event that best describes the problem encountered and enter your best estimate of the duration of the problem.

4.3.5 Remarks and Comments

Remarks entered here are used for recap and daily reports. To add additional comments click

[Additional Comments](#)

4.4 PUMPS/BITS TAB

The information entered in this tab is used for hydraulics calculations and it is very important to enter the information as fully and accurately as possible.

4.4.1 Pump Data Table

Make and Model	Pump Type	Liner Diameter	Stroke Length	Rod Diam Duplex	Eff%	Pump Rate Stroke/Min	Connected to Riser	
EMSCO F-1300 TRIPLE	TRI-PLEX	6.5000	12.00		97.00	100.00	☐	
EMSCO F-1300 TRIPLE	TRI-PLEX	6.5000	12.00		97.00	100.00	☐	
EMSCO F-1300 TRIPLE	TRI-PLEX	6.0000	12.00		97.00		☐	
							☐	

Column only active if well is designated as offshore.

The pump data section shows the main parameters for the mud pumps. You will be able to add as many pumps as necessary, but note that some report templates are designed to generate information for the pumps in first two rows only. Select the make and model from the drop down menu, and the liner diameter and stroke diameter will be filled out automatically, you can overwrite these values if necessary. Fill out the rest of the information, and indicate if the pump is connected to riser. Pump data is important to calculate flow rate, time and strokes for bottoms up and total circulation.

HELPFUL HINT

If the pump is not in the drop down list, you can simply enter the information by typing it in the fields provided.

4.4.2 Bit Information Section

Bit Information		TFA Input (Nozzles)	
Bit Size (in):	17.500	Size (1/32"):	Quantity:
Bit Number:	2	12.00	X 3
Bit Type:	PDC	13.00	X 2
Bit Mfg:	SMITH-GRUNER		X
Bit Name:	Bit 2		TFA: 0.590

To fill in the bit information section enter current bit size and number, select the bit manufacturer and type from the drop down menus. Bit number is used to keep track of number of bits used for the current project. Bit name is an alternative name that the bit is known by for easier identification. Bit TFA can be entered manually, or the TFA will be calculated automatically when nozzle sizes and number of nozzles are entered. The nozzle size number is a fraction. For example, the bit above has three 12/32" nozzles and two 13/32" nozzles.

4.4.3 Drilling Data Section

HELPFUL HINT	The information in this section is very important in calculating the Hydraulic results. The more accurate the information is, the better the calculations. Always enter as much as possible.
--------------	---

Drilling Data

Rotary (rpm):	10	Pump Flow Rate (gal/min):	1003.165
Rotating Hours:	4	Pump Pressure (psi):	4500
Weight On Bit (lbf):	15000	dp MWD (psi):	
ROP (ft/hr):	50	dp Motor (psi):	
Riser Pump Flow (gal/min):	0	Motor (rpm):	
Bottomhole Circulating Temp (°F):	143	Surface Temp (°F):	68
Lithology:	Anhypri		

Most of the information required in these fields is obtained from the IADC report in the Doghouse or the well monitoring software such as Pason. Information about the MWD tool's pressure drop can be obtained from the MWD team.

If there is a mud motor in the BHA, enter the parameters here.

There are 2 temperature fields that should be entered here, these figures are used in the hydraulics results calculations:

Bottomhole Circulating Temperature

Surface Temperature

4.4.4 Extra Pressure Loss Information Section

Extra Pressure Loss Information

Surface Code:

Surface Pressure Loss (psi):

MWD Ref Flow Rate (gal/min):

On / Off Bottom (psi):

Surface Annular Pressure (psi):

In the surface code drop down menu choose the option that best describes your well. Correct surface code selection is important for correct pressure loss calculations. If is chosen in the drop down menu, the field will remain active, enter your surface loss here. If another entry is chosen in the drop down, this field will not accept entry as there is a surface pressure loss figure associated with that rig type. Enter figures in the remaining fields if available.

4.4.5 Hydraulic Results

To obtain hydraulics results click on the bottom of the tab. If any information is missing for calculations, an error message listing the missing items will pop up. Even though the calculation results might be generated after you click , for full and most accurate hydraulic calculations enter the missing inputs and press again. Below is the hydraulics results window. The five yellow cells showing pressure losses cannot be edited on this page

Hydraulics Results

SecNo	Length	SecMD	HoleDia	PipeOD	PipeID	DSLoss	AnnPL	ECD	AnnTemp	AnnPV	AnnYP	AnnVCritical	AnnAV	AnnRe	AnnLSYP
1	304.00	304.00	19.50	5.00	4.28	114.59	1.02	11.67	111.96	25.00	20.37	317.79	69.21	1,028.99	8.32
2	6,305.00	6,609.00	12.55	5.00	4.28	2,253.55	31.28	11.74	132.91	25.29	16.46	379.37	185.47	2,333.12	6.02
3	146.00	6,755.00	12.55	9.50	3.00	125.31	3.61	11.76	132.94	25.38	16.44	342.91	365.25	1,977.83	6.00
4	268.00	7,023.00	12.55	8.63	7.78	78.75	8.51	11.77	132.89	25.56	16.41	342.93	295.61	1,976.51	5.97

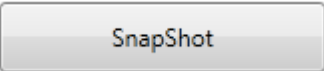
DS Loss	Ann Loss	Bit Loss	Surface Equip Loss	Total Loss
2688	45	2123	149	5023

Snapshot

Hole Cleaning Parameters

Cuttings Size (in)	Cuttings s.g.	Cuttings Dia (in)	Cuttings Diameter
Medium Large	2.6	0.12	0.12

The “Reynolds Number” for each section of the well is calculated and displayed in the column labelled AnnRe. A Reynolds number greater than 2000 is an indicator that the fluid is in turbulent flow, below 2000 the flow regime is laminar. The annular velocity needed in each section of the well to transition from laminar to turbulent flow is displayed in the column labelled AnnVCritical.

The  will open a window showing a drilling snapshot of the current drilling conditions. A number of useful properties are plotted. The hole cleaning index will be more accurate if the proper cuttings size is entered in the Hole Cleaning Parameters section of the hydraulics results window. Chose the proper size from the drop down menu, once the size is chosen the information fields will be filled in. The figures in this field may be overwritten if they are incorrect.

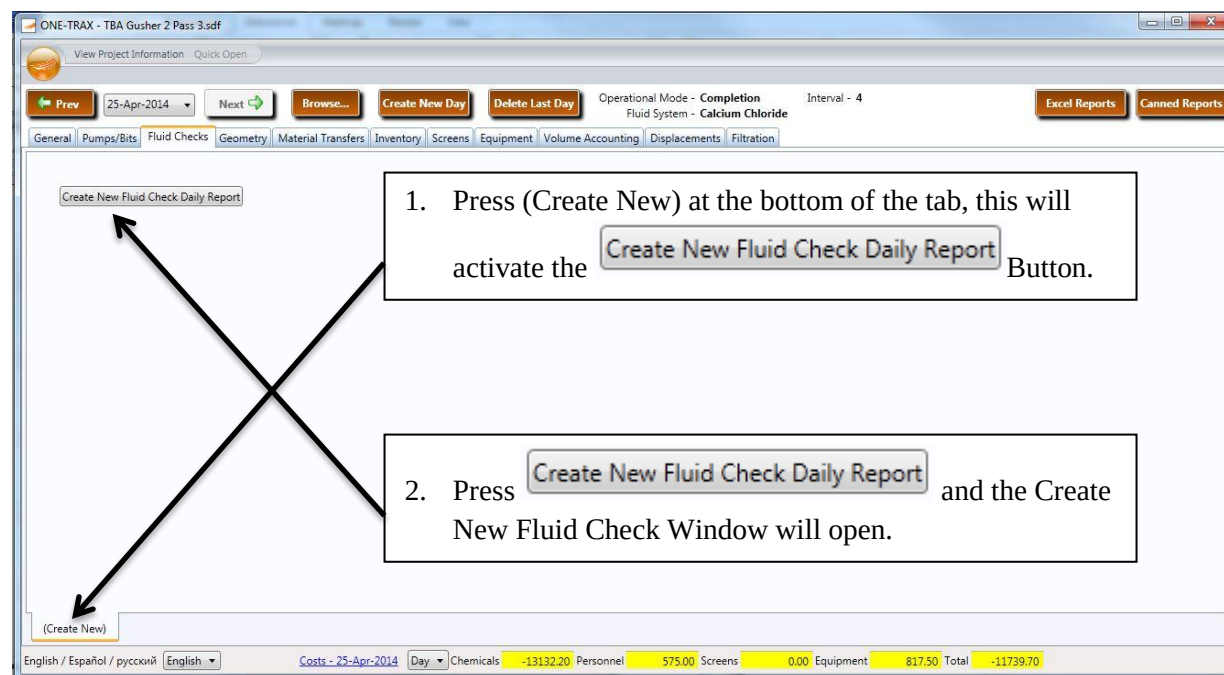
Hole Cleaning Parameters

Cuttings Size (in)	Cuttings s.g.	Cuttings Dia (in)	Cuttings Diameter
Medium	2.6	0.09	0.09

- Small
- Medium Small
- Medium
- Medium Large
- Large
- Extra Large

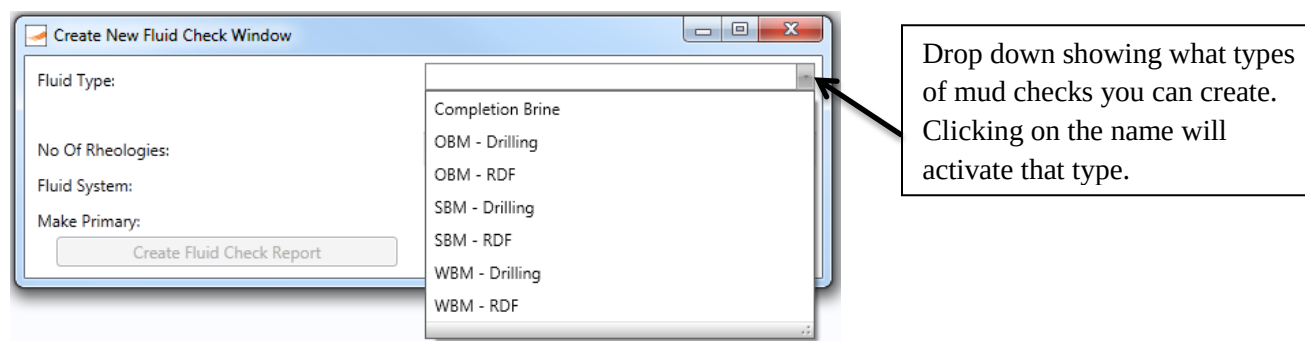
4.5 FLUID CHECKS TAB

This tab is where the fluid checks are created and entered for all types of fluids used in ONE-TRAX. When the tab is first opened it will appear as seen below, except the **Create New Fluid Check Daily Report** button will not be visible.



4.5.1 Create New Fluid Check Window

To create a new fluid check follow the steps shown in the picture above. After step two a window will open allowing you to choose what type of mud the check is created for and also enter certain other information about the mud check. The first step is to choose the type of fluid the check will be made on. Each mud will have two types, drilling and RDF; as well as completion brine..

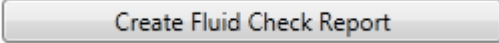


Once the type of mud is chosen the second pull down will become active. There is a different piece of information entered here for the different types of muds. The dropdown will show the available choices for each type.

MUD	INFORMATION ENTERED
Completion Brine	Base Fluid
OBM and SBM	Internal Phase Salt
WBM	Make Up Water

After the second drop down, you will enter the number of rheologies you will run on the mud. The default is 1, but some operators will ask for a rheology at three different fluid temperatures. You can change the number to how many are required. Rheologies ran at varying temperatures will improve hydraulics results. The next step is to choose the fluid system. This drop down will display the fluids with the chosen type which you added on the Lithology / Well Survey / Fluid System tab.

Drop down showing
Fluids with the chosen
type that are active in the
well file.

The make primary checkbox will default to on if this is the first mud check of the day. If you make multiple mud checks you can designate which one is the primary by checking the box or designating it with the button on the fluid check entry tab. Below is an example of a fully filled in window. Once all required data is entered the  will become active. Push it and a blank mud check will be created.

will close the window without creating a fluid check.

4.5.2 Fluid Properties Entry Window

Here is an example of the fluid properties entry form. The example is for WBM, there will be some differences in properties entered for Synthetic or Oil based mud. This is another “busy” screen but it is where some of the most important information in ONE-TRAX is entered. The graphic indicates the major sections that will be discussed in greater detail.

The screenshot shows the Fluid Properties Entry Window with the following sections highlighted by callouts:

- Upper Information Display:** Located at the top left, containing tabs for General, Pumps/Bits, Fluid Checks, Geometry, Volume Accounting, Material Transfers, Inventory, Equipment, Screens, Displacements, and Filtration.
- Upper Control Buttons:** Located at the top right, containing buttons for No Of Rheologies, Make Primary, Copy To Clipboard, Delete This Fluid Check Report, Print Daily, WITSML Export, and OpenWells Export.
- Extra Fluid Properties Table:** Located on the right side, containing a table for Extra Fluid Properties with columns for Property, Unit, Check 1, Check 2, Check 3, and Check 4.
- Fluid Properties and information entry section:** The main table area with columns for Property, Unit, Check 1, Check 2, Check 3, and Check 4. It contains various fluid properties like Sample From, Time Taken, Flow Line Temp, Depth, TVD, Fluid Weight, Fluid Temp, Funnel Viscosity, Rheology Temp, R600, R300, R200, R100, R6, R3, PV, YP, 10s Gel, 10m Gel, 30m Gel, and API Fluid Loss.
- Fluid Check Report Tabs:** Located at the bottom left, containing tabs for (Create New), WBM - Drilling - Seawater - KCL/Polymer - Primary, OBM - Drilling - CaCl2 - VERSACLEAN, and Completion Brine - CaCl2 - Calcium Chloride.
- Lower Control Buttons:** Located at the bottom right, containing buttons for Manage Extra Fluid Properties, Display Solids Analysis Page, and Daily Property Specifications.

To insure you enter all required data, you will need to scroll down.

4.5.3 Fluid Check Report Tab

Fluid Check Report Tab

The first subsection we will cover is the fluid check report tab. The first tab of the day will be opened when you create it with the create new fluid check report window. At the bottom of the tab it displays all the relevant information entered about the mud. Each tab is limited to 4 checks but a second mud report tab for the day can be created by using the (Create New) tab. It is possible to create other types of mud reports on the same date, for example if you will displace from WBM to SBM, you can have checks for both types of mud. The example below contains an OBM, WBM, and Completion Brine mud check.

The screenshot shows the Fluid Check Report Tab with the following tabs:

- (Create New)
- WBM - Drilling - Seawater - KCL/Polymer - Primary
- OBM - Drilling - CaCl2 - VERSACLEAN
- Completion Brine - CaCl2 - Calcium Chloride

4.5.4 Fluid Properties and Information Section

In this section, you will enter all the information about the mud from the mud check. Some of the cells may be left blank depending on whether or not that information is determined. For Example, you will most likely not run both a API Fluid Loss and a HTHP fluid loss test.

Property	Unit	Check 1	Check 2	Check 3	Check 4
Sample From		Active 1			
Time Taken		09:00			
Flow Line Temp	°F	90			
Depth	ft	5002			
TVD	ft	4747			
Fluid Weight	lb/gal	10.5			
Fluid Temp	°F	86			
Funnel Viscosity	s/qt	45			
Rheology Temp	°F	120			
R600		50			
R300		38			
R200		26			
R100		22			
R6		17			
R3		14			
PV	cP	12			
YP	lb/100ft ²	26			
10s Gel	lb/100ft ²	14			
10m Gel	lb/100ft ²	16			
30m Gel	lb/100ft ²	18			
API Fluid Loss	cc/30min	4.5			
HTHP Fluid Loss	cc/30min				
HTHP FL Temp	°F				
Cake API	1/32"	2			
Cake HTHP	1/32"				
Solids %Vol	%vol	10			
NAF Base Fluid	%vol	0			
Water	%vol	90			
Sand	%vol	.5			
MBT	lb/bbl	5.0			
pH		9.0			
pH Temp	°F	85			
Alkal Mud (Pm)		0.4			
Pf		0.2			
Mf		0.3			
Chlorides	mg/L	24000			
Hardness Ca	mg/L	450			
Retort Mud Wt@Room Temp	lb/gal	10.5			
Room Temp.	°F	86			
K+ (From KCl)	mg/L	20000			
Wt Material SG	sg	4.2			
NAF Base Fluid SG	sg	.7			
Frac Bent	~0.1111	.1111			
Chem Conc	lb/bbl	16			
Drill Solids SG	sg	2.6			
Is Weighted	Y/N	Y			

To the left is the form for entering information determined by the mud check

Enter the Density of the fluid poured into the retort cell, at the indicated temperature.

Frac Bent – is the ratio of the activity of the formation clay vs. the commercial bentonite (M-I Gel). This value is not always 0.1111 so further tests may be required to calculate the correct value for your area.

Chem Conc – is the amount of chemicals in your fluid without counting weighting material, salts, or liquids items. These concentrations can be viewed using [Product Concentrations](#) on the volume accounting tab. The product concentrations will affect the calculations of drill solids in your mud. If you leave the value at “0”, you will get higher drill solids than you actually have.

Weighted fluid – it is important to correctly indicate whether the fluid contains a weighing agent or not. It is used in solids analysis calculations.

HELPFUL HINT

When using OBM or SBM, the Retort Mud Weight and Temperature is very important. If the Mud weight is unadjusted for Expansion and Contraction at different temperatures, the solids analysis calculations will result in higher drill solids values than is the actual case.

In ONE-TRAX, it is necessary to open another form to view the results of the solid analysis calculations.

Press **Display Solids Analysis Page** and the solids analysis section of the mud check will appear. If there is a problem with any of the values, for example a negative LGS concentration, a red outline will appear around that cell. It will be necessary to double check your entries to find the error. You do so by pressing

Hide Solids Analysis Page

and returning to fluid properties entry section.

NaCl	%vol	0.62			
NaCl Concentration	lb/bbl	7.32			
KCl	%vol	0			
KCl Concentration	lb/bbl				
LGS	%	9.36			
LGS Concentration	lb/bbl	85.18			
Bentonite	%	0.44			
Bentonite Concentration	lb/bbl	3.98			
Drill Solids	%	7.93			
Drill Solids Concentration	lb/bbl	72.21			
HGS	%	N/A			
HGS Concentration	lb/bbl	N/A			
Inert/React		5.349			
Avg SG Solids	sg	2.6			
Carbonate	m mole/L	1.21			
BiCarbonate	m mole/L	0.08			

4.5.5 Upper Information Display

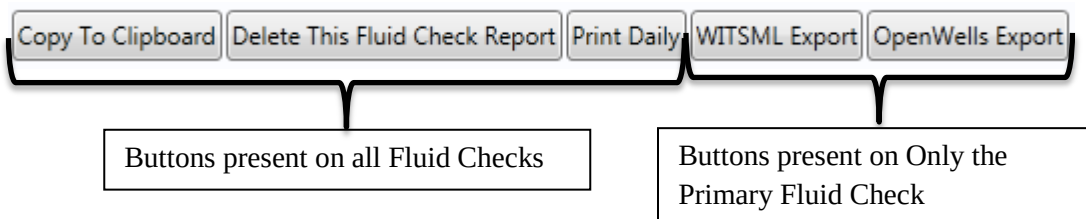
Fluid System: **KCL/Polymer** Emulsified Oil: No Of Rheologies: **1**

This section of the tab displays and allows you to change certain pieces of information about the mud check. The drop down of the fluid system will allow you to change the mud system of that mud check, but is limited to the same type of base fluid. Thus, you can change from one WBM system to another WBM, as long as the fluid system has been defined in the Lithology / Well Survey /Fluid Systems tab.

The emulsified oil drop down will allow you to assign what type of oil is being used for your OBM/SBM system. The choices are limited to those defined in the application data.

The “No of Rheologies” will allow you to expand the fluids properties section to accept a mud check with additional rheologies. The maximum you can run is 3. When the drop down is changed to add additional information the fluid properties section will expand. Not all the additional fields will accept input, only those for the viscometer readings and the gels.

4.5.1 Upper Control Buttons



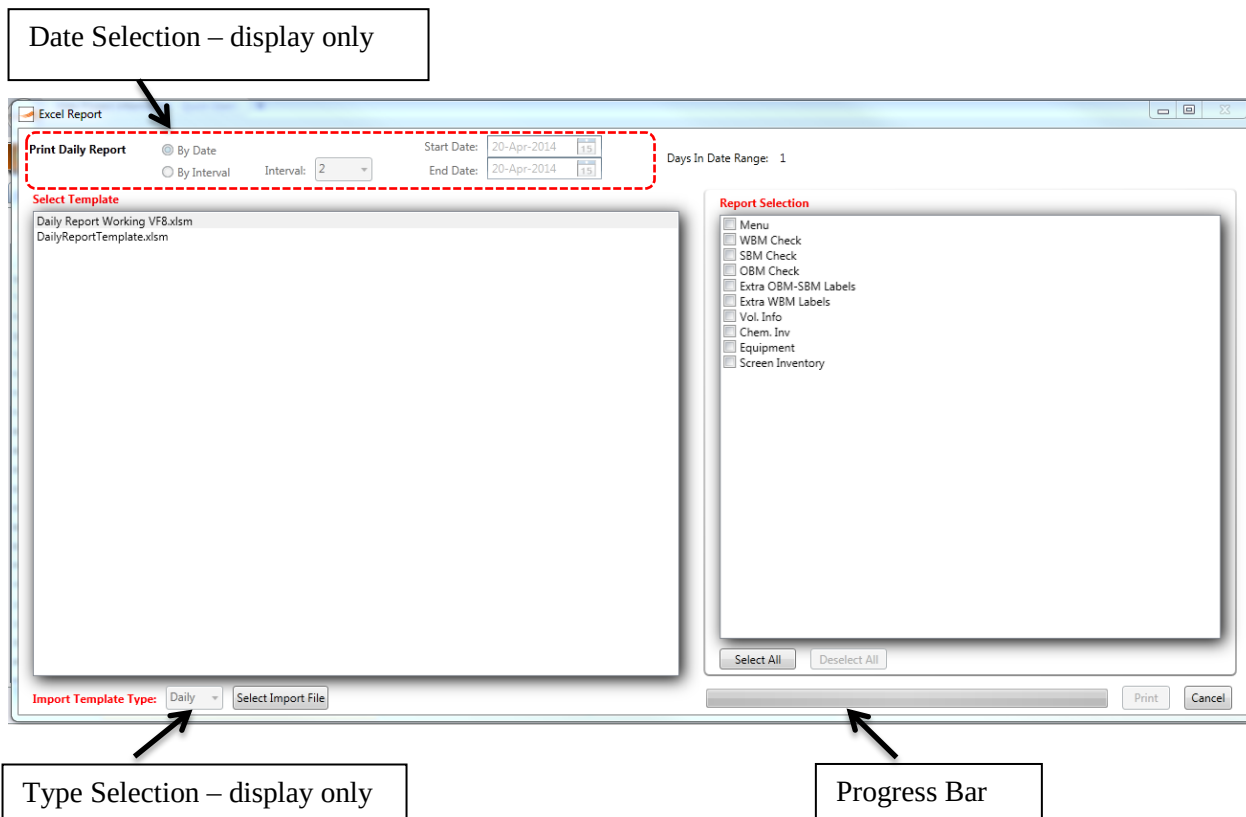
This group consists of buttons that perform functions managing the active fluid check. There are 4 control buttons for all fluid checks but the primary check will have 2 additional buttons that are used for exporting the fluid check. When the fluid check is the primary for the day there are 6 buttons, but only 4 for the non-primary checks.

Make Primary this will change the mud check to the primary one for the day. It is only active if there is more than one fluid check report tab for the day and if the tab that is active is not the primary one. It is also important to note, that only the primary mud check may be exported via WITSML or OpenWells. The primary mud check is also the one whose properties are used to calculate hydraulics.

Copy To Clipboard will allow you to copy the fluid properties and paste them in other software programs such as Word or Excel. It will only copy the active fluid properties to the clipboard. In other words, you will need to **Display Solids Analysis Page** and **Copy To Clipboard** then paste again to get the solids analysis section into the outside program as well

Delete This Fluid Check Report removes the active fluid check report tab.

Print Daily opens the dialogue window called Excel Report. This window allows you to print customized reports created through Excel Templates. This window is used in other places in ONE-TRAX so you will notice that some of the information is locked. You cannot change the date or the template type.



You can import additional templates, if you receive a template your project engineer requires you use for this well, it can be stored so that ONE-TRAX can use it by pressing **Select Import File** this will open up a dialogue box where you will need to find and highlight the new template. Once the file is highlighted, press **Open** and it will be placed in the select template section and can be used to create reports.

To print a report using an excel template see [4.5.2 How to Make a Daily Report Using Templates](#).

WITSML Export

WITSML Export will export the primary mud check and certain well information in the WITSML format. This functionality was included in ONE-TRAX at the request of certain operators so that our information could easily be included in their reports. Pressing the button will open the WITSML export window. Certain fields will be automatically filled in but you will fill in the entry fields and check boxes required by the client. The export field will be in the WITSML format and have an .xml file extension.

The radial buttons in the WITSML section **Report V13.1** **EOG Report Package** are used to designate whether you will use the EOG report package or not. The EOG report package is designed to meet the needs of a specific client and clicking it will result in the proper report check boxes being checked. Using the report

V1.3.1 will allow you to choose which reports you wish to export. You may click

Select All

The Max Checks drop down **Max Checks # 4** allows you to determine how many mud checks you will export. Four is the maximum number of mud checks. Changing the number will create additional columns in the extra fluid check properties section. When the max checks drop down is set to a value greater than 1, the

Copy button in the extra fluid check properties section will become active. This button will copy the values in the previous check to that check.

WITSML Selection Section

Extra Fluid Check Properties Section

The **Create Report** button will open a navigation dialogue box allowing you to choose a location to store your report and to name it.

Exit closes the WITSML export window and returns to the active fluid check tab.

OpenWells Export

OpenWells Export

is another informational export designed to get data to clients so they can include it in their reports. Pressing the button will open the open wells export window which allows you to fill in what information will be exported. The export will be a text file and have a .txt extension.

Output Format – there are several output formats created to conform the output too operator requirements.

Output Selection – drop down allowing you to choose the format of some variables that will be output.

Select/Order Output Section

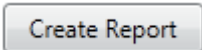
When the output format choice is custom, you can choose which field are exported by checking the boxes of the ones you want output. You can also choose the ordering of the output fields by entering the desired order

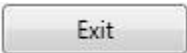
in the print order column. **Select All** will

check all the property boxes and **Unselect All** will remove all the checkmarks.

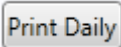
Open Wells Properties Section

In this section you will enter the information as defined in the row headings on the right side. You may enter information for up to 4 fluid checks.

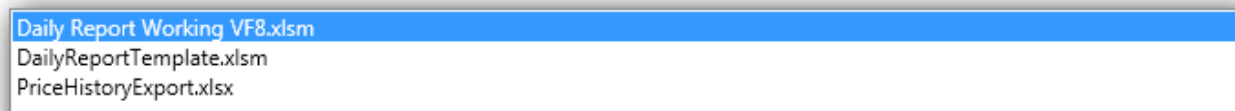
The  button will open a navigation dialogue box allowing you to choose a location to store your report and to name it.

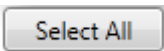
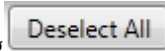
 closes the open wells export window and returns to the active fluid check tab.

4.5.2 How to Make Daily Reports Using Templates

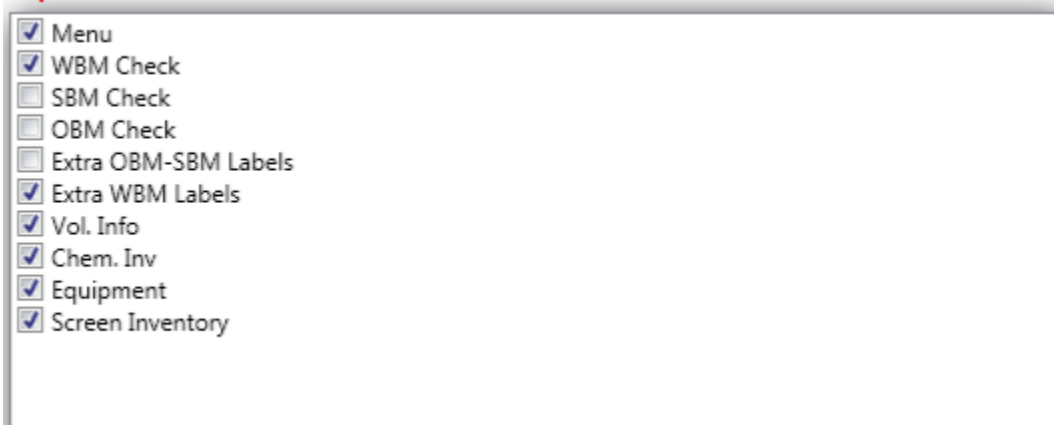
1. Open the Excel report window by pressing 
2. Choose the report template for the report you wish to print. You do this by clicking on the template name in the select template section. This will activate this template and indicate this by highlighting it.

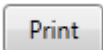
Select Template



3. This will show which reports are available in this template in the report selection section. You may choose which reports to create by checking the boxes. You may select all by clicking . You may remove all check marks by clicking .

Report Selection



4. Click  and Excel will open two workbooks. The Daily_Workbook contains the information that populates the fields in your report workbook. The report workbook is made up of pieces of information about the well and is titled WellNameLOGIT#Date.
5. Activate the menu worksheet in your report workbook by clicking on it.



1 - Run First

6. Press the button the menu, this will run the macros that populate the data fields in the report workbook.
7. Clean up the reports, especially the check worksheet itself by removing the extra slashes and/or @ present.

MUD PROPERTIES				
Sample From	Active 1 @ 09:00			
FlowLine Temp °F	90			
Depth/TVD ft	5002 / 4747	/	/	/
Mud Weight /Temp lb/gal @ °F	10.5 @ 86	@	@	@
Funnel Viscosity s/qt	45			
Rheology Temp °F	120			
R600/R300	50 / 38	/	/	/
R200/R100	26 / 22	/	/	/
R6/R3	17 / 14	/	/	/
PV cP	12			
YP lb/100ft²	26			
10s/10m/30m Gel	14 / 16 / 18	/ /	/ /	/ /
API Fluid Loss cc/30min	4.5			
HTHP Fluid Loss cc/30min				
Cake APT/HT 1/32"	2 /	/	/	/
Solids %vol	10			
Oil/Water	2.6			
Sand %vol	0.5			
MBT lb/bbl	5			
pH / Temp	9 @ 85	@	@	@
Alkal Mud (Pm)	9			
Pf/Mf	0.2 / 0.3	/	/	/
Chlorides mg/L	24000			
Hardness (Ca++)	450			
Mud delivered bbl	900			
Mud on location bbl	400			
Daily loss bbl	2741.5			
Net loss bbl	4364			
Net loss cost USD				
Cleanup cost USD				

8. Save your reports.

Cancel

will close the Excel report window and return to ONE-TRAX.

4.5.3 Extra Fluid Properties Table

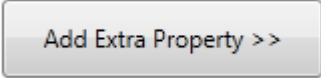
This table allows you to add extra fluid properties to your mud report. These properties are controlled through the application data but there are extra properties defined that allow you to add your own property description and track it on the fluid reports (described later in this section). For each extra property you have added there is a cell allowing you to add the value for each check.

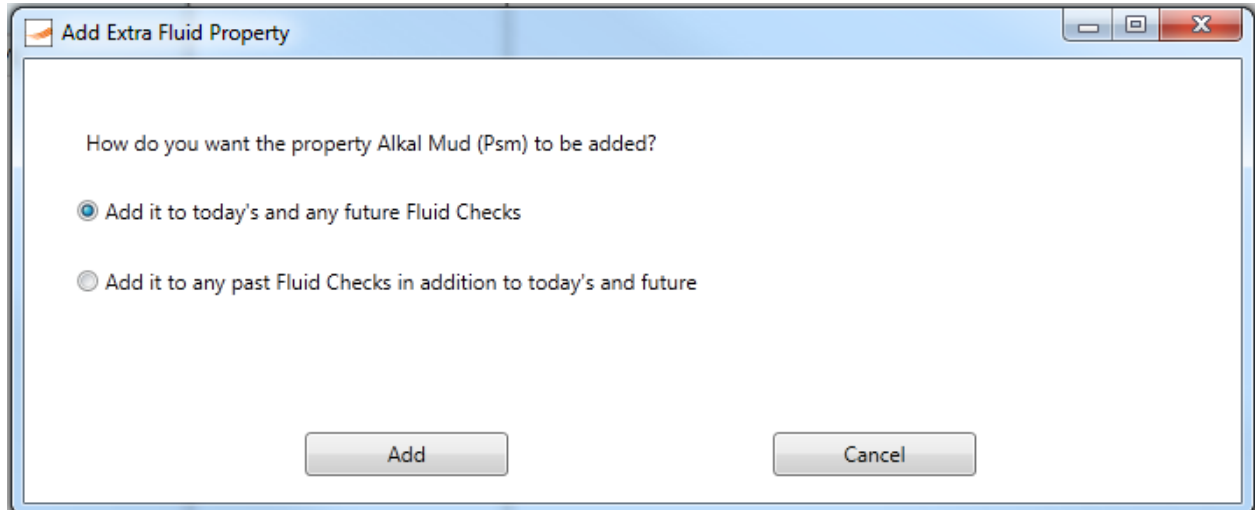
There is also a column labelled “Print Order”, this allows you to determine in what order these properties will appear on the daily report. If the print order column is left blank or has a “0” entry, the property will not be printed on the report.

Extra Fluid Properties						
Property	Unit	Check 1	Check 2	Check 3	Check 4	Print Order ▲
AgNO3 0.282N	vol/vol	1.5	1.6			1
Magnesium	mg/L	0.6	0.7			2

Extra properties are added by pressing the **Manage Extra Fluid Properties** button. This will open the extra fluid properties window where you will add them to your report.

The first step to adding is to find the property you wish to add. You do this by typing a search term in the field at the top and pressing . This will bring up all properties containing that search term in the table below it. Choose the property to add by clicking on it and

pressing . This will open the add extra fluid property window on which you will need to indicate whether the property is to be added to Mud checks for this date forward only or to include the earlier mud checks.



The dialog box titled "Add Extra Fluid Property" contains the following text and options:

How do you want the property Alkal Mud (Psm) to be added?

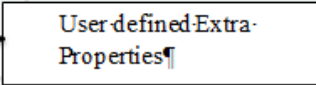
- ☒ Add it to today's and any future Fluid Checks
- ☐ Add it to any past Fluid Checks in addition to today's and future

At the bottom are two buttons: "Add" and "Cancel".

Once you have chosen,  will add that property to the extra fluid properties

Custom Defined Extra Fluid Properties

There are 10 extra fluid check properties that are included in the application data that are not locked and allow you to use them as you see fit. These are added in the same manner as the other properties. Once in the extra properties table you are able to type into the property cell to give it a unique name.

Extra 1	Custom Property 1	
Extra 10	Custom Property 10	
Extra 2	Custom Property 2	
Extra 3	Custom Property 3	
Extra 4	Custom Property 4	
Extra 5	Custom Property 5	
Extra 6	Custom Property 6	
Extra 7	Custom Property 7	
Extra 8	Custom Property 8	
Extra 9	Custom Property 9	

These can be used to display information on the daily reports. For Example, the project is using OBM and the operator wishes to see the volume accounting for this fluid summarized on the mud report. You are able to do this by adding customized extra properties and then entering the information in the extra properties table.

Extra Fluid Properties

Property	Unit	Check 1	Check 2	Check 3	Check 4	Print Order
Mud Delivered	bbl	2116				1
Mud on Location	bbl	1802				2
Daily Loss	bbl	30				3
Net Loss	bbl	315				4
Net Loss Cost	\$225.47/bbl	\$71023.05				5

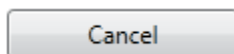
Print order tells ONE-TRAX in what order you want you extra properties to be exported so they can be printed. A print order of "0" or blank tells ONE-TRAX not to export the data for that property. See **Important Information** below the next graphic for discussion of reasons not to delete no longer needed extra properties.

Here is an example of the extra properties on the daily mud report. Note how the properties follow the print order entered on the extra fluid properties table.

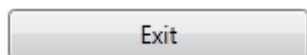
UNCON Water		% Vol	20	
Synthetic/Water Ratio			70/30	
Alkal Mud (Psm)			1	
Cl- Whole Mud	mg/L		69000	
Salt	%Wt		30.16	
Lime	lb/bbl		1.3	
Emul Stability			415	
WPS	mg/l		-	
Mud Delivered	bbls		2116.	
Mud on Location	bbls		1802.	
Daily Loss	bbls		30.	
Net Loss	bbls		315	
Net Loss Cost	225.47		71023.05	
Cleanup Cost	\$			
Reserve Volume	bbl		498	
REMARKS AND TREATMENT				
MW 10.8 Visc 76. Used MI Wate to slug pipe for tripping. Received SBM chemicals.				

IMPORTANT INFORMATION: If you use “Extra 1” for one property and no longer need it, do not delete it or change it. Doing that will delete it from all prior mud reports and you will be unable to reprint them. It is better to change its print order to “0” and add an unused extra property for your new purpose.

If you change the description of the property, it will change it for all past mud reports as well. Changing the print order is a better method, it will allow you to recreate past reports correctly. If you add a user defined extra property to the table, it will show up without the property field displayed, you will type the description you want to see on the mud report and it will be saved once you leave that cell.



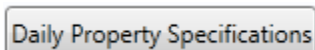
will return to the extra fluid properties window without adding it.



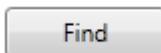
The button on the extra fluid properties window will close it and return to the fluid check tab.

4.5.4 Daily Properties Specifications

This is where you enter the parameters you wish the mud to be maintained within. Pressing



will open the daily property specifications window. It is similar to the extra fluid properties window. You will enter the search term of the property you wish to add in the search field and press

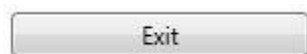


Add to Daily >>

. This will populate the table below with properties matching the term. Highlight it and press which will move it to the properties table. The property you have added will have no value for min value, max value, or comment. You will enter this information and it will be saved once you exit that field. On the mud check, anytime the chosen property is outside of the defined parameters, it will be outlined in red.

Properties

Property	Unit	Min Value	Max Value	Comment
Fluid Weight	lb/gal			
HTHP Fluid Loss	cc/30min	25.0000	30.0000	
PV	cP	20.0000	25.0000	
YP	lb/100ft ²	3.0000	5.0000	



closes the daily property specifications window and returns to the active fluid check.

4.6 GEOMETRY TAB

This tab displays and accepts information relating to the well bore. The information you enter here is used to calculate hole volumes. The tab is divided into three main sections:

The screenshot shows the ONE-TRAX Wellsite Software interface for the Geometry Tab. The interface is divided into three main sections:

- Diagram Section:** Located on the left, it displays a vertical wellbore diagram with depth markers (20.5, 20, 13.375, 6000, 12.553, 7023) and a casing diagram with depth markers (20.5, 20, 13.375, 5002).
- Geometry Section:** Located in the center, it contains two tables:

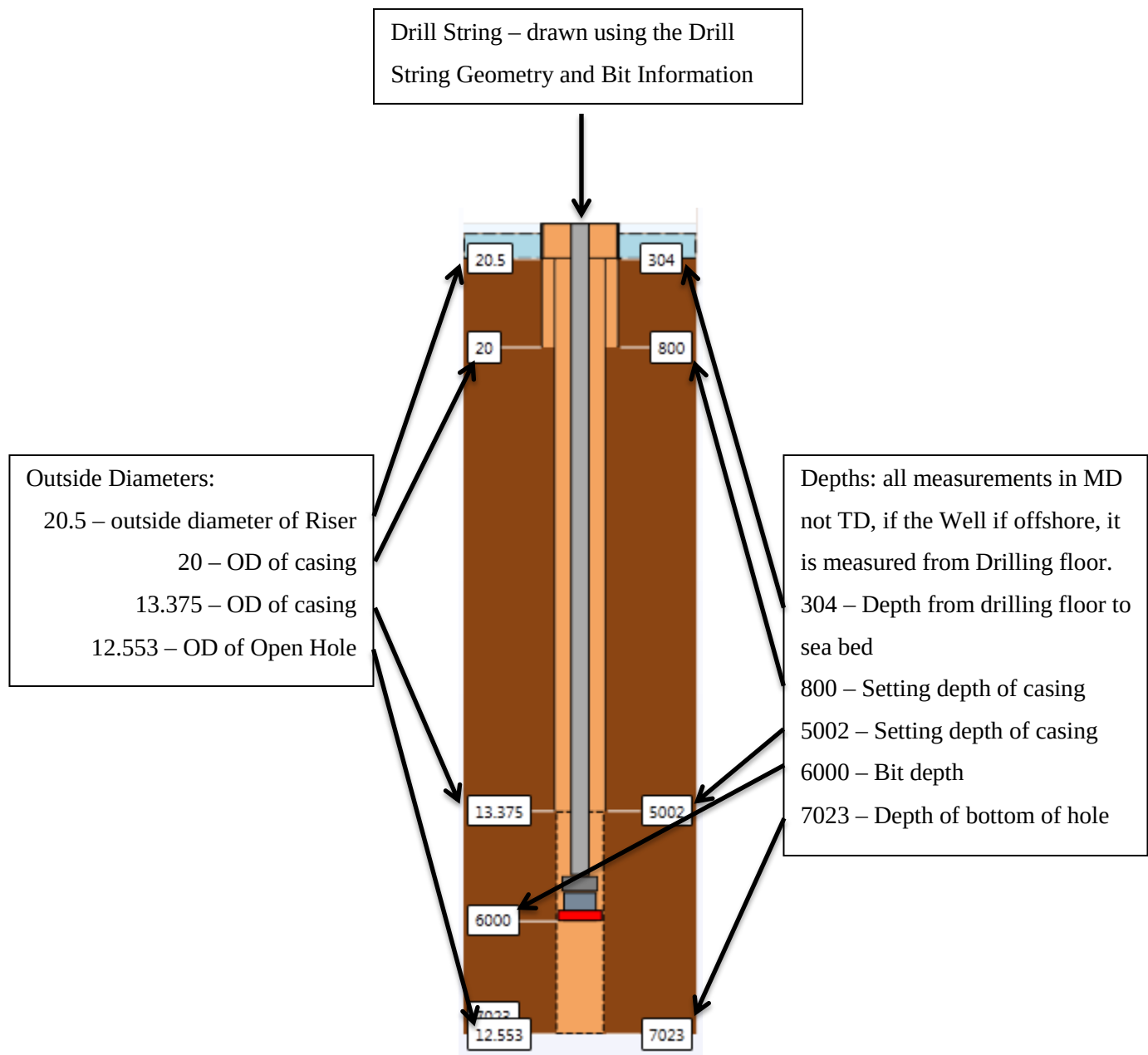
Wellbore Geometry						
Type	Casing OD in	Casing ID in	Depth ft	TVD ft	Top (liner only)	Hole Size in
Casing	20	19.124	800	800	0	28.4816
Casing	13.375	12.615	5002	4747	0	18.3542
Open Hole	0	0	7023	6188	0	12.5525

Drill String Geometry						
Type	Length ft	Pipe Od in	Pipe Id in	Tool Jt OD in	Tool Jt ID in	T J Length in
Drill Pipe	5586	5	4.276	6.25	3	34
Drill Collar	146	9.5	3	0	0	0
Drill Collar	268	8.625	7.775	0	0	0
- Information Section:** Located on the right, it contains various input fields for well parameters:
 - Air Gap: 84
 - Water Depth (ft): 220
 - Riserless Drilling? ☐
 - Riser OD: 20.5
 - Riser ID: 19.5
 - Riser Length: 304
 - Depth (ft): 7023
 - Bit Depth (ft): 6000
 - Bit Size (in): 12.25
 - % Washout: 5
 - PBTD-MD (ft):
 - PBTD-TVD (ft):
 - Inclination: 35
 - Azimuth: 226
 - Daily Length Drilled (ft): 2.021
 - Daily Volume Drilled (bbl): 309.3
 - Reset Start Depth for Sidetrack ☐
 - Start Depth (ft): 5002
 - Additional Length Drilled (ft):
 - Diameter (in):
 - Recalculate button
 - Down Hole Tools section with Tool and Depth (ft) fields.
 - Import VH button
 - Hole Volume Detail button
 - Hydraulics Result button

At the bottom, there is a summary bar showing costs for Chemicals (273380.00), Personnel (2175.00), Screens (3000.00), Equipment (2257.50), and a Total (281712.50).

4.6.1 Diagram Section

The diagram section gives a visual representation of the data entered in the geometry section, the information section, and other tabs of ONE-TRAX.



4.6.2 Geometry Section

Displays the wellbore and drill string geometry in tabular form, It is here that you enter most of the information that will be shown in the diagram section.

Information is entered by typing directly into the table with the exception of the type and hole size columns. The types are entered via a drop down menu and are limited to the choices in the application data. Hole size is calculated from the drill bit size and washout percentage for that

portion of the hole. The  buttons bring up some common ODs and IDs for the type code chosen. Pressing it will open a window

Wellbore Geometry

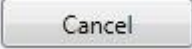
Type	Casing OD in	Casing ID in	Depth ft	TVD ft	Top (liner only)	Hole Size in	
Casing ▾	20.0000 	19.124	800	800		28.4816	
Casing ▾	13.3750 	12.615	5002	4747		18.3542	
Open Hole ▾	0.0000 		7023	6188		12.5525	
▾							

Drill String Geometry

Type	Length ft	Pipe Od in	Pipe Id	Tool Jt OD in	Tool Jt ID	T J Length in	
Drill Pipe ▾	5586	5.0000 	4.276	6.2500 	3	34	
Drill Collar ▾	146	9.5000 	3	0.0000 			
Drill Collar ▾	268	8.6250 	7.775	0.0000 			
▾							

Simply chose the row matching what you have, this will highlight it and press

Select

The information will be transferred to the table.  will close the window without transferring the data. Once the information is in the table, you can overwrite if necessary.

The length field on the first entry of the drill String geometry table is calculated from the depth of the drill bit. You may see a negative number here if you bit is on the surface and a drill string is entered.

Drill String Geometry

Type	Length ft	Pipe Od in	Pipe Id	Tool Jt OD in	Tool Jt ID	T J Length in	
Drill Pipe ▾	* -414	5.0000 	4.276	6.2500 	3	34	
Drill Collar ▾	146	9.5000 	3	0.0000 			
Drill Collar ▾	268	8.6250 	7.775	0.0000 			
▾							

In this example, there is a negative length of -414 ft., because that is what is necessary to have the drill bit be at “0” depth and still have the drill collars in the drill string shown.

4.6.3 Information Section

In the information section you will enter details about the well. Many of the fields are displaying information entered in other tabs of ONE-TRAX or calculated from data you input.

Air Gap:	84	Water Depth (ft):	220										
☐ Riserless Drilling?													
Riser OD:	20.5	Riser ID:	19.5										
		Riser Length:	304										
Depth (ft):	7023	Bit Depth (ft):	7023										
TVD (ft):	6188	Bit Size (in):	12.25										
PBTD-MD (ft):		% Washout:	5										
PBTD-TVD (ft):		(increase in OH volume):											
Inclination:	35.00	Washout Hole Size (in):	12.5525										
Azimuth:	226.00												
Daily Length Drilled (ft):	2021												
Daily Volume Drilled (bbl):	309.3												
☐ Reset Start Depth													
Start Depth (ft):	5002												
Additional Length Drilled (ft):													
Diameter (in):													
Recalculate													
<table border="1"> <thead> <tr> <th colspan="2">Down Hole Tools</th> </tr> <tr> <th>Tool</th> <th>Depth (ft):</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> </tr> <tr> <td></td> <td></td> </tr> <tr> <td></td> <td></td> </tr> </tbody> </table>				Down Hole Tools		Tool	Depth (ft):						
Down Hole Tools													
Tool	Depth (ft):												
Import VH Hole Volume Detail Hydraulics Result													

Top Section

The top of the graphic shows fields that only apply to offshore projects. These fields will not be visible unless the well is setup as an offshore project. The

☐ Riserless Drilling? will be checked while the well is being drilled without the Riser in place. Once you uncheck the box, you will enter the OD, ID and Length of the Riser. This is important in calculations for the mud in the hole. The air gap and water depth fields come from information entered in the general tab on

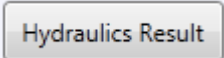
the project view.

Left Column

The depth and TVD fields are displayed from the information you entered in the geometry section.

PBTD (Plug Back Total Depth) MD and TVD are entered in these fields. They will only be entered if the well has plugged back.

The inclination and azimuth will be available from downhole tools or logging runs. The inclination is very important as it is used in the hole cleaning calculations that are viewed through

the  button.

Right Column

Enter the bit depth and the diagram will be updated.

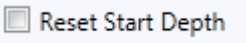
Bit size is displayed from the Pump/ Bits tab.

If you enter a washout percentage, ONE-TRAX will calculate the increased hole diameter which will be used in all hole related Calculations. If the washout is left at zero, these calculations will be made for a gauge hole.

Bottom Section

The daily depth and volume drilled are calculated from the information enter in the geometry section. This information is used in equipment usage calculations.

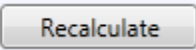
Drilling a Sidetrack

It is easy to record information for a sidetrack in ONE-TRAX. Check the  checkbox. Enter the starting depth of the sidetrack in the start depth field. When a new ending depth of the hole is entered the footage and amount drills will be calculated form the depth entered.

The additional length and diameter field are used if a side track is drilled without using the reset start depth function. Enter the length and diameter of the extra hole, and that hole volume will be added to the daily totals.

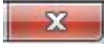
The down hole tools section is an informational area where you enter the tools that are in the well and what depth they are located.

Control Buttons

 is used to have ONE-TRAX recalculate the volumes drilled for the day.

 is used to import a drill string Geometry that has been exported from Virtual Hydraulics.

Hole Volume Detail

opens a window giving more detailed information on the Hole volumes. It also displays stroke calculations and circulating times based on the information entered for the mud pumps on the Pumps/Bits tab. There is not an exit control on this window and it is closed by using the Windows  button in the upper right corner.

Hole Volume Results

SecNo	Length	HoleDiameter	PipeOD	PipeID	TooljointOD	Description	DSVol	AnnVol	PipeDisplacement
1	304.00	19.50	3.50	2.99	0.00	CT/Riser	2.64	108.68	0.97
2	7,516.00	8.54	3.50	2.99	0.00	CT/Casing	65.36	442.43	24.08
3	182.00	8.54	0.00	0.00	0.00	Casing	0.00	12.88	0.00

Total DS Length (ft)	Drill String (bbl)	Annulus Volume (bbl)	Below Bit Volume (bbl)	Total Volume (bbl)	Bottom Up Stks	Bottom Up Time	Tubing Displacement (bbl)
7820	68.01	551.11	12.88	632	4614.72	23.07	25.05
					Total Circ Stks	Total Circ Time	
					5292.02	26.46	

4.6.4 Hydraulics Results

Hydraulics Result

To obtain hydraulics results click on the bottom of the tab. If any information is missing for calculations, an error message will be generated listing the missing items. For most accurate hydraulic calculations enter the missing inputs and press

Hydraulics Result

again. Below is the hydraulics results window. The five yellow cells showing pressure losses cannot be edited on this page

The screenshot shows the 'Hydraulics Results' window. It contains a table with 17 columns: SecNo, Length, SecMD, HoleDia, PipeOD, PipeID, DSLoss, AnnPL, ECD, AnnTemp, AnnPV, AnnVP, AnnVCritical, AnnAV, AnnRe, and AnnLSYP. Below the table are summary statistics for DS Loss, Ann Loss, Bit Loss, Surface Equip Loss, and Total Loss. To the right is the 'Hole Cleaning Parameters' section with fields for Cuttings Size, Cuttings s.g., Cuttings Dia, and Cuttings Diameter.

SecNo	Length	SecMD	HoleDia	PipeOD	PipeID	DSLoss	AnnPL	ECD	AnnTemp	AnnPV	AnnVP	AnnVCritical	AnnAV	AnnRe	AnnLSYP
1	304.00	304.00	19.50	5.00	4.28	114.59	1.02	11.67	111.96	25.00	20.37	317.79	69.21	1,028.99	8.32
2	6,305.00	6,609.00	12.55	5.00	4.28	2,253.55	31.28	11.74	132.91	25.29	16.46	379.37	185.47	2,333.12	6.02
3	146.00	6,755.00	12.55	9.50	3.00	125.31	3.61	11.76	132.94	25.38	16.44	342.91	365.25	1,977.83	6.00
4	268.00	7,023.00	12.55	8.63	7.78	78.75	8.51	11.77	132.89	25.56	16.41	342.93	295.61	1,976.51	5.97

DS Loss	Ann Loss	Bit Loss	Surface Equip Loss	Total Loss
2688	45	2123	149	5023

Hole Cleaning Parameters			
Cuttings Size (in)	Cuttings s.g.	Cuttings Dia (in)	Cuttings Diameter
Medium Large	2.6	0.12	0.12

The Reynolds Number for each section of the well is calculated and displayed in the column labelled AnnRe. A Reynolds Number greater than 2000 is an indicator that the fluid is in Turbulent flow, below 2000 the flow regime is Laminar. The annular velocity needed in each section of the well to transition from laminar to turbulent flow is displayed in the column labelled AnnVCritical.

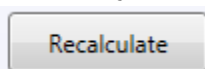


The **Snapshot** will open a window showing a drilling snapshot of the current drilling conditions. A number of useful properties are plotted. The hole cleaning index will be more accurate if the proper cuttings size is entered in the hole cleaning parameters section of the hydraulics results window. Chose the proper size from the drop down menu, once the size is chosen the information fields will be filled in. The figures in this field may be overwritten if they are incorrect.

Transfer Ticket Summary

Type	Ticket #	Ordered By	Received By	Action
* Receive from Warehouse	*			+
Receive From Other Well				+
Receive from Warehouse				
Return To Other Well				
Return To Warehouse				

After you enter all the information in the transfer ticket Summary table, click

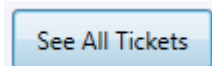


on the bottom of the tab to update product inventory.

4.7.2 Ticket Browser

The transfer ticket summary table in the material transfer tab shows only the tickets for the

current date. If you need to find information entered any other day click



. The

material transfer-ticket browser window, shown below, will pop up.

Material Transfer - Ticket Browser

Item Filter:

Transfer Ticket Summary

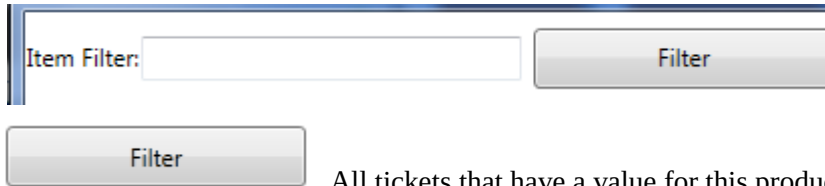
Date	Type	Ticket No.
09-Jul-2013	Receive From Other Well	No Ticket
10-Jul-2013	Receive from Warehouse	1445750
11-Jul-2013	Receive from Warehouse	1446042
12-Jul-2013	Receive from Warehouse	1447533

Selected Ticket Number -1445750

Item No.	Product I	Alt Name	Locat	UO	Price	Qty From Ticket	Actual Quantit
M0003502	ALUMINUM S	AlumSt	Imported	25	76.5	5	5
M0003026	CEDAR FIBER	C/F SS	Imported	40	6.08	65	65
R0088304	FREIGHT - PA	Freight	Imported	1	1	623	623

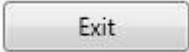
When first opened the **Transfer Ticket Summary** table lists every single ticket you entered up to the current date. If you click on the ticket you are interested in, all products on this ticket will be generated in the table on the right side of the window. You will not be able to edit any information in this window. To edit actual quantity, quantity from the ticket, or any information in transfer ticket summary table, navigate to the day the ticket was created and edit it in material transfer tab by clicking on the ticket you want to edit.

The item filter lets you find a ticket containing a certain product. Enter the product name or item number in the item filter field and click



The screenshot shows a software window titled 'Item Filter'. It contains a text input field with the label 'Item Filter:' to its left. To the right of the input field is a button labeled 'Filter'. Below the input field, there is another button labeled 'Filter'.

. All tickets that have a value for this product will be generated in the transfer ticket summary table. Do not enter a full product name in search field; this will limit your search results. If, for example, you are looking for Poly-Plus and enter “Poly Plus” (no hyphen) or “Polyplus”, no results will be generated, because in master list it is spelled as POLY-PLUS. But if you enter part of the product name, Poly, for example, you will have search results that will contain poly-plus. DO NOT HIT THE SPACE AFTER THE PRODUCT NAME IS ENTERED, it will be read as part of the product name and incorrect search results will be generated.

Click  to close the window.

4.8 INVENTORY TAB

The inventory tab shows all products that have been setup to use in the project. These items are entered via the product setup tab.

The unexpanded view of the inventory table shows the following information:

- Product name
- Alternative name the product is known by
- Item Number
- Locator – domestic or import
- Gross price per unit
- Used for fluid
- Used other
- Final quantity

To see expanded inventory view, check the ☒ **Show Expanded View** box. In addition to all information in unexpanded table, the expanded table shows the following information:

- Start quantity
- Received quantity
- Returned quantity
- Quantity on order
- Daily product quantity used
- Daily product cost
- Option to exclude the product from all the reports

There are only three parameters you will be able to enter or edit in this table – the *Used Other* and *On Order* fields, and the exclude from print check box.

☒ Show Expanded View

Product Name	Alt Name	Item No	Locator	Gross Price	Start	Received	Returned	Used For Fluid	Used Other	On Order	Final	Daily Used	Daily Cost	Exclude from Print
M-I-X II MEDIUM	M MDX2	M0002155	Imported	15.26	0	0	0	0	0	0	0	0	0	☐
OIL ABSORBANT (OIL DRY) SURFOOT	Oil Dry	M0005340	Imported	8.59	0	0	0	0	0	0	0	0	0	☐
OTHER SERVICE - PASS-THROUGH	OtherSC	R0088306	Imported	1.06	-250	0	-600	0	350	0	0	350	371	☐
PIPE-LAX	PipeLax	M0003631	Imported	160.27	0	0	0	0	0	0	0	0	0	☐
PLATINUM D-D	PLAT DD	M0008170	Imported	45.79	0	0	0	0	0	0	0	0	0	☐
POLYPAC R	PacR	M0002204	Imported	124.02	0	0	0	0	0	0	0	0	0	☐
POLY-PLUS	Poly+	M0003657	Imported	57.24	0	0	0	0	0	0	0	0	0	☐
SAFE-CARB 40	S/C 40	M0003047	Imported	7.96	0	0	0	0	0	0	0	0	0	☐



Used other is used to enter the quantity of a product that was used but was not added to the fluid via a volume accounting transaction. For example, shrink wrapping, pallets, damaged sacks, products like Clean-Up, etc.

☒ Show Expanded View

Product Name	Alt Name	Item No	Locator	Gross Price	Start	Received	Returned	Used For Fluid	Used Other	On Order	Final	Daily Used	Daily Cost	Exclude from Print
M-I-X II MEDIUM	M MDX2	M0002155	Imported	15.26	0	0	0	0	0	0	0	0	0	☐
OIL ABSORBANT (OIL DRY) SURFOOT	Oil Dry	M0005340	Imported	8.59	0	0	0	0	0	0	0	0	0	☐
OTHER SERVICE - PASS-THROUGH	OtherSC	R0088306	Imported	1.06	-250	0	-600	0	350	0	0	350	371	☐
PIPE-LAX	PipeLax	M0003631	Imported	160.27	0	0	0	0	0	0	0	0	0	☐
PLATINUM D-D	PLAT DD	M0008170	Imported	45.79	0	0	0	0	0	0	0	0	0	☐
POLYPAC R	PacR	M0002204	Imported	124.02	0	0	0	0	0	0	0	0	0	☐
POLY-PLUS	Poly+	M0003657	Imported	57.24	0	0	0	0	0	0	0	0	0	☐
SAFE-CARB 40	S/C 40	M0003047	Imported	7.96	0	0	0	0	0	0	0	0	0	☐



On Order is used if the product has been ordered and has not been delivered yet and the client wishes to see a list of ordered products in the report. The entered number will not affect your final inventory; it is for display on reports only.

☒ Show Expanded View

Product Name	Alt Name	Item No	Locator	Gross Price	Start	Received	Returned	Used For Fluid	Used Other	On Order	Final	Daily Used	Daily Cost	Exclude from Print
M-I-X II MEDIUM	M MDX2	M0002155	Imported	15.26	0	0	0	0	0	0	0	0	0	☐
OIL ABSORBANT (OIL DRY) SURFOOT	Oil Dry	M0005340	Imported	8.59	0	0	0	0	0	0	0	0	0	☐
OTHER SERVICE - PASS-THROUGH	OtherSC	R0088306	Imported	1.06	-250	0	-600	0	350	0	0	350	371	☐
PIPE-LAX	PipeLax	M0003631	Imported	160.27	0	0	0	0	0	0	0	0	0	☐
PLATINUM D-D	PLAT DD	M0008170	Imported	45.79	0	0	0	0	0	0	0	0	0	☐
POLYPAC R	PacR	M0002204	Imported	124.02	0	0	0	0	0	0	0	0	0	☐
POLY-PLUS	Poly+	M0003657	Imported	57.24	0	0	0	0	0	0	0	0	0	☐
SAFE-CARB 40	S/C 40	M0003047	Imported	7.96	0	0	0	0	0	0	0	0	0	☐



Exclude from Print only shown in expanded view, if checked, it excludes the product from all reports.

Once you are done entering or editing the data in the table, click

Recalculate

to update the product inventory.

Chemical Weight (lb):

46105.86

Chemical Weight shows total weight of all products currently on the location. It is calculated based on quantity of the products on hand and cannot be entered or edited manually.

4.9 SCREENS TAB

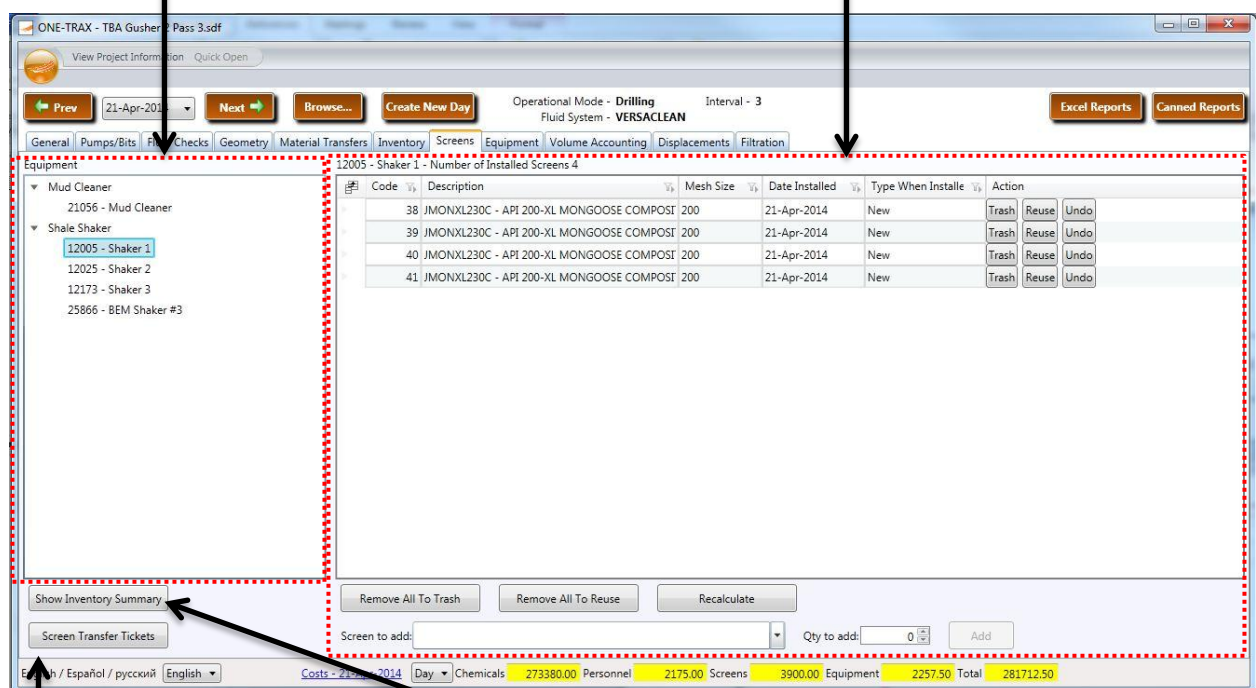
Use the screens tab to view the screen inventory summary; enter, search, and view screen transfer tickets; to view screens currently installed screens, and to keep track of screen installation and removal.

The screens tab has four main sections:

1. Equipment list
2. Installed screens table
3. Screen inventory summary
4. Screen transfer tickets

Equipment List – will only show Shakers and Mud Cleaners

Installed Screens Table – shows screens installed on the active piece of equipment.



Screen Transfer Tickets – Delivery and Return tickets of Screens

Screen Inventory Summary

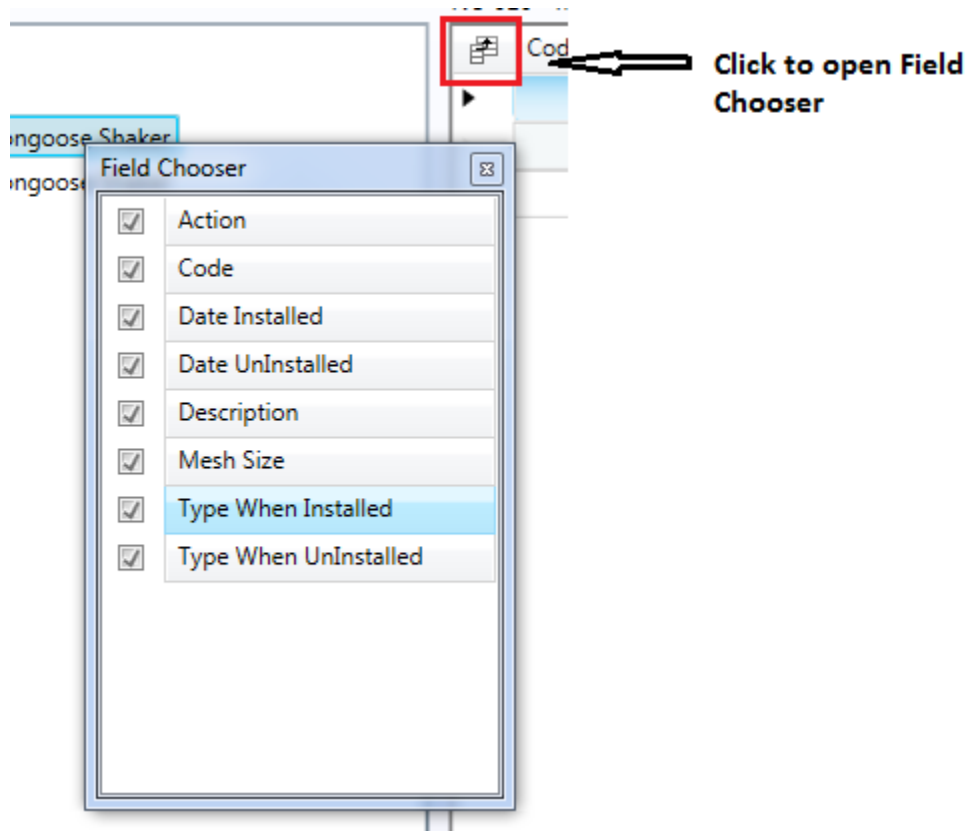
4.9.1 Equipment List

The equipment list located on the left side of the page lists pieces of equipment on the project. Equipment is added on the equipment setup tab. The equipment list will include M-I SWACO and non M-I SWACO equipment. The list can be viewed expanded and unexpanded. The unexpanded list has equipment type only; the expanded view shows the equipment's unique serial number and its' description. To expand the list click on grey triangle next to the equipment type or double click on the equipment type.

4.9.2 Installed Screens Table

On the right side of the screens tab you will see a table containing information about screens currently installed on each piece of equipment. This table is referred to as the installed screens table. The header of the table will always show serial number, description, and number of installed screens for the equipment you chose from the expanded equipment list. Equipment that has its information currently displayed in the table will be highlighted blue in the table on the left side of the tab, as shown in the picture.

To choose the information you want to be displayed in the installed screens table, click  and the field chooser window will pop up. Check the boxes of the fields you want to see in the table.



Installing a Screen

1. From the equipment list select the equipment that the screens are being installed on
2. Select the screen from "Screen to add" dropdown list in the bottom of the screen

Screen to add: Qty to add:

3. Select the quantity of this type of screens to be installed
4. Click .

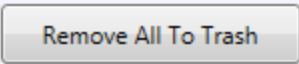
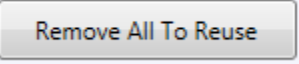
Each screen will be added in a separate row. Screens with the same description are differentiated by a unique value in the code field. It is very important to choose a right screen for a transaction in order to keep a correct screens inventory. You will be able to add as many screens to the equipment as required. You will not be able to define the screen position in ONE-TRAX.

To Remove a Screen

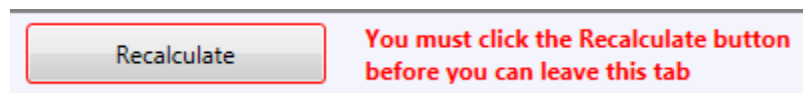
To remove a single screen or **remove screens one by one**, repeat the steps describe below for each screen:

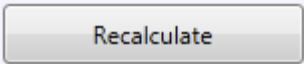
1. From the equipment list select the equipment that the screen is uninstalled from
2. In the installed screens table highlight the screen you want to remove by clicking on it; the screen will be highlighted blue
3. In the action column choose:
 -  if the screen is uninstalled and disposed. **The screen will be removed from screens inventory.**
 -  if the screen is uninstalled and will be reused. **The screen is returned to screens inventory as a used screen.**
 -  to undo the installation. Use this function if you had dressed the shaker with a screen but it was not used. This will return the screen to the inventory as it was before being installed; new if it was a new screen, used if it was a used screen. If you use the reuse button to remove a screen that was not installed, ONE-TRAX will change its' status to used in the inventory.

Remove all screens from the equipment:

1. From the equipment list select the equipment that the screens are uninstalled from
2. On the bottom of the page choose:
 -  if all the screens are uninsulated and disposed. Screens will be removed from screens inventory.
 -  if all screens are uninstalled and will be reused. Screens will be returned to inventory as used.

After each transaction you will see a following message:



Click  to update screen inventory.

4.9.3 Screen Inventory Summary

Show Inventory Summary

Click in the lower left corner and the screen inventory summary window will open.

Screen Inventory Summary

Screen Cd	Description	Cost				Old							DT - Qty	
		Final	Gross Price	Daily	Cumulative	Start	Receiv ed	Transfer From Equipme nt	Returne d	Dispos ed From Storag e	Used	Final	New	Used
JMONXL200E	Mongoose XL200# magnum pt	0	0.00			8		0		2	6	2		
W100TB080H	VSM100 SCALPING 80 MESH SWNG()	0	0.00			6					3	3		

Daily Cost: 0Cumulative Cost: 0

Exit

The screen inventory summary is for informational display only. You will not be able to delete, add, or edit information in this table. Information available from screen inventory summary:

1. Screen code
2. Screen Description
3. New screen information for the certain screen model:
 - Start - starting quantity for the day
 - Received – quantity of new screens received for the day
 - Returned – quantity of new screens returned for the day
 - Used – quantity of new screens installed this day
 - Final – final quantity of new screens for the day
4. Screen cost information for the certain screen model:
 - Gross price – cost, without discount, of each screen of this type
 - Daily – daily cost of each screen type
 - Cumulative – cumulative cost each screen type for the project
5. Old screen information for the certain screen model:
 - Start - staring quantity for the day
 - Received – quantity of old screens received for the day
 - Transferred from equipment – quantity of old screens removed from the equipment this day
 - Returned – quantity of old screens returned for the day
 - Disposed from storage – quantity of old screens disposed that day
 - Used – quantity of old screens installed this day
 - Final – final quantity of old screens for the day
6. DT-Qty – Delivered Quantity
 - New – delivered quantity of new screens
 - Used– delivered quantity of used screens

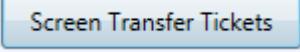
7. RT-Qty – Returned Quantity
 - New – returned quantity of new screens
 - Used – returned quantity of used screens

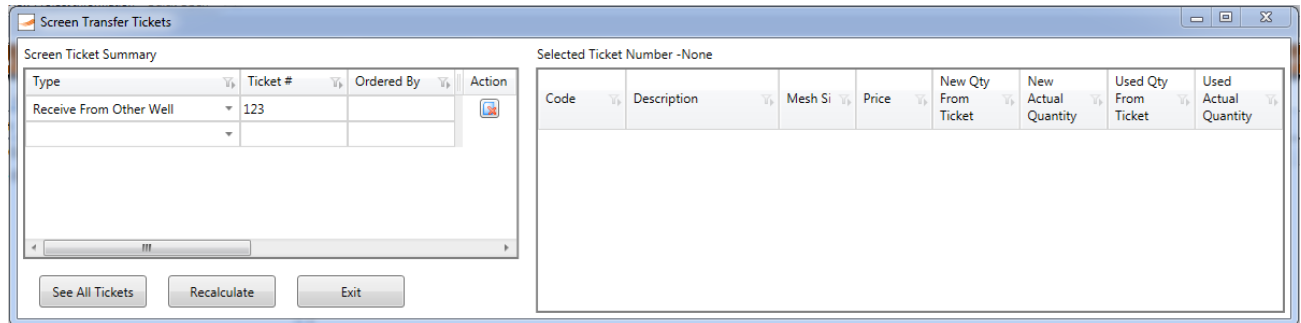
Screens daily cost and cumulative cost are displayed in the bottom of the page.

Daily Cost: 1625 Cumulative Cost: 4875

Click  to close the page and return to the screens tab.

4.9.4 Screen Transfer Tickets

Click  on the bottom of the page and the **Screen Transfer Tickets** window will pop up. This window is used for receiving and returning screens. Here you enter information from the ticket you receive when screens are delivered to the rig or a ticket you create when screens are returned to the warehouse or transferred to another rig.



Type	Ticket #	Ordered By	Action
Receive From Other Well	123		

See All Tickets Recalculate Exit

Code	Description	Mesh Si	Price	New Qty From Ticket	New Actual Quantity	Used Qty From Ticket	Used Actual Quantity
------	-------------	---------	-------	---------------------	---------------------	----------------------	----------------------

4.9.5 Screen Ticket Summary Table

Enter the receipt/return ticket information in ONE-TRAX as soon as the screens are received or returned! Ticket information is entered in **Screen Ticket Summary** table shown below.

Type	Ticket #	Ordered By	Action
* Receive From Other Well	*		
Receive From Warehouse			
Return To Other Well			
Return To Warehouse			

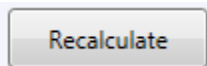
See All Tickets Recalculate Exit

To enter the ticket information:

1. Chose the transaction type from the Type drop down list. You have four options: Receive from Other Well, Receive from Warehouse, Return to Other Well, and Return to Warehouse.
2. Enter the Ticket Number
3. Enter the name of the person who ordered the delivery/return
4. Enter the name of the person who received delivery
5. Enter the Trucking Company information (Ship via)
6. Choose a Warehouse from the dropdown list (if applicable). To add a warehouse name to the dropdown list go to View Project Information, General Tab and click

Warehouse Code Setup
7. Add any comments you would like to keep in the system regarding the ticket or any screen on the ticket. For example, the rig name that screens were transferred from/to.
8. Once you enter the transaction type and ticket number, a list of all active screens will be generated in the table on the right side of the page. Find the screen model received/returned in the list and enter Quantity of the screen from the Ticket and the Actual Quantity of the screen you counted. Make sure you enter the correct quantity in the correct column – quantity of New screens or Quantity of Used screens.
9. Repeat step 8 for every single screen model on the ticket. If you do not see the screen model you received/returned in the list of active screens, go to the Screen Setup tab and [add the screen model](#) to the active product list following the instructions in this manual.

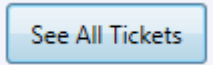
After you have entered all the information in the screen ticket summary table, hit



on the bottom of the page to update screen inventory.

4.9.6 Screen Transfer Ticket Browser

The screen transfer tickets window shows only the transactions for the current date. If you need

to find information entered any other day click . The **Screen Transfer-Ticket Browser** window will open.

When the window first opens, the transfer ticket summary table lists every single ticket you entered up to the current date. It has exactly the same information you entered in the screen transfer tickets window discussed above. If you click on the ticket you want to review, all screens on that ticket will be shown in the table on the right side of the window. Above the table you will see the selected ticket number. You will not be able to edit any information in this window. To edit quantities or any other information, navigate to the day the ticket was created and edit it in the screen transfer tickets window by clicking on the ticket you want to edit. If you need to edit information regarding the screens themselves go to the screen setup tab.

The item filter lets you find a ticket containing a certain screen type. Enter the screen description

or screen code in the **Item Filter** field and click . All tickets that have matching information will be shown in the transfer ticket summary table. Entering a full screen name or full screen code in the search field will limit your search results. Enter the first three or four letters or numbers. If for example, you are looking for Mongoose XL200, entering “mong” to generate more search results. **DO NOT HIT SPACE AFTER YOU ENTER SCREEN NAME OR PART OF THE NAME.** It will be read as the part of the name and no results will be

4.10 EQUIPMENT TAB

In the equipment tab you will enter charges for the equipment, the equipment parameters, and keep track of equipment performance characteristics.

ONE-TRAX - TBA Gusher 2 Pass 3.sdf

View Project Information Quick Open

Prev 21-Apr-2014 Next Browse... Create New Day

Operational Mode - Drilling Fluid System - VERSACLEAN Interval - 3

Excel Reports Canned Reports

General Pumps/Bits Fluid Checks Geometry Material Transfers Inventory Screens Equipment Volume Accounting Displacements Filtration

Equipment Performance

- Centrifuge
 - 50258 - Centrifuge 1
 - 50138 - Centrifuge 2
 - 50258 - Centrifuge 1
- Vacuum System
 - V120 - Vacuum
- Cuttings Container
 - C0000 - Cuttings Box
- Mud Cleaner
 - 21056 - Mud Cleaner
- Shale Shaker
 - 12005 - Shaker 1
 - 12025 - Shaker 2
 - 12173 - Shaker 3
 - 25866 - BEM Shaker #3
- Others
 - A12345 - Auger
 - HVD338 - D-Gasser
 - 10KUC02 - Choke
 - 3408 - Storage Container
 - 2456 - Storage Container

12005 - Shaker 1

#	Property	Unit	Value
1	Size, Specifications		
2	Daily Running Hours		12
3	Well Total Hours		24
4	Mud On Cuttings	bbl/bbl	2.5
5	Daily Volume Discarded (Sludge)	bbl	324.8
6	Volume of Mud On Solids Discarded	bbl	232
7	Well Total Volume Discarded	bbl	1,097.1
8	Well Total Vol. of Mud On Solids Discarded	bbl	668.5
9	% of Cuttings Removed by this Equipment	%	30
10	Deck Angle	deg	+3
11	G-Force	G's	4
12	Vibration Angle		
13	Type of Motion	linear	
14	Flow Over Shaker	gal/min	15
15	Discard Rate	gal/min	45
16	Average Beach Length	in	20

Equipment Utilization

Set all to Full Charge Set all to Standby Charge Set all to No Charge

Serial N	Description	Used	Charge	Daily t	Cum t
50258	Centrifuge 1	1	Full Charge	350.00	2,090.00
50138	Centrifuge 2	1	Full Charge	350.00	2,090.00
V120	Vacuum	1	Full Charge	570.00	1,995.00
VG-4568	Verti G	1	Full Charge	807.50	4,512.50
A12345	Auger	1	Full Charge	50.00	170.00
F2424	DE Filter	1	Stand By	30.00	210.00
C0000	Cuttings Box	1	Full Charge	20.00	140.00
12005	Shaker 1	0	Full Charge	0.00	0.00
12025	Shaker 2	0	Full Charge	0.00	0.00
12173	Shaker 3	0	Full Charge	0.00	0.00
21056	Mud Cleaner	0	Full Charge	0.00	0.00
HVD338	D-Gasser	0	Full Charge	0.00	0.00
10KUC02	Choke	0	Full Charge	0.00	0.00
3408	Storage Container	1	Full Charge	20.00	140.00
2456	Storage Container	1	Full Charge	20.00	140.00
2346	Storage Container	1	Full Charge	20.00	140.00
345	Storage Container	1	Full Charge	20.00	140.00

Cuttings Return Enter Remarks Recalculate

English / Español / português / English

Costs - 21-Apr-2014 Day Chemicals 273380.00 Personnel 2175.00 Screens 3900.00 Equipment 2257.50 Total 281712.50

Equipment List

Equipment Parameters and Performance Characteristics

Equipment Utilization

There are three main parts to equipment tab:

1. Equipment list
2. Equipment properties
3. Equipment utilization

4.10.1 Equipment List

The equipment list shows all equipment set up on the equipment setup tab, project information view. Note that, even if the equipment is provided by other company, you should include it in the equipment list! The list can be viewed expanded or unexpanded. In the picture below you see an unexpanded view; to expand the list click on grey triangles next to the equipment name or double click on the equipment type.

Equipment Performance

- ▶ Centrifuge
- ▶ Auger
- ▶ Vacuum System
- ▶ Mud Cleaner
- ▶ Shale Shaker
- ▶ Others
- ▶ Cuttings Dryer
- ▶ Filtration Equipment



UNEXPANDED VIEW

Clicking on the arrow besides the equipment type opens the expanded list. This shows the equipment code and alternate name. Each piece of equipment, even if code and alternate name match, will be listed separately. If a piece of equipment is missing on the expanded list, you will need to add it on the equipment setup tab.

- ▼ Centrifuge
 - 50258 - Centrifuge 1
 - 50138 - Centrifuge 2



EXPANDED VIEW

- ▼ Vacuum System
 - V120 - Vacuum
- ▼ Cuttings Container
 - C0000 - Cuttings Box
- ▼ Mud Cleaner
 - 21056 - Mud Cleaner
- ▼ Shale Shaker
 - 12005 - Shaker 1
 - 12025 - Shaker 2
 - 12173 - Shaker 3
 - 25866 - BEM Shaker #3
- ▼ Others
 - A12345 - Auger
 - HVD338 - D-Gasser
 - 10KUC02 - Choke
 - 3408 - Storage Container

4.10.2 Equipment Properties Table

This is a great tool that not only lets you keep track of your equipment performance characteristics, but is also used for more precise volume accounting. Volume of mud discharged on solids will be calculated based on the data you enter. This calculated volume is used to create an [equipment loss transaction](#) that subtracts the volume lost with cuttings from the active system, resulting in more accurate volume accounting.

To see equipment properties click on the piece of equipment in the equipment list. The table containing equipment parameters and performance characteristics will be generated. This table has different predefined sets of parameters for different groups of equipment. To [add an extra property](#) you would like to track, go to the equipment setup tab.

The piece of equipment that currently has its properties' displayed in the table will be highlighted blue, and the equipment code and alternate name will be displayed the top of the table, as shown in the picture below. In the property table there are white cells and yellow cells. Values in yellow cells are calculated, they cannot be edited manually. Enter the information required in white cells.



50258 - Centrifuge 1			
#	Property	Unit	Value
1	Size, Specifications		518
2	Daily Running Hours		19
3	Well Total Hours		19

Predefined properties of the equipment:

Centrifuge Properties:

50258 - Centrifuge 1			
#	Property	Unit	Value
1	Size , Specifications		
2	Daily Running Hours		
3	Well Total Hours		
4	Mud On Cuttings	bbl/bbl	
5	Density - In	lb/gal	
6	Density - Out	lb/gal	
7	Density - Discard	lb/gal	
8	Flow Rate - In	gal/min	
9	Flow Rate - Out	gal/min	
10	Flow Rate - Discard	gal/min	
11	Closed-Loop Centrifuging?	Y/N	N
12	Daily Volume Processed	bbl	
13	Daily Volume Recovered	bbl	
14	Daily Volume Discarded (Sludge)	bbl	
15	Volume of Mud On Solids Discarded	bbl	
16	Volume of Solids Discarded	bbl	
17	Avg. Density of Solids Discarded	lb/gal	
18	Discard Mass Rate	ton/hour	
19	Daily Mass Discarded	ton	
20	Well Total Volume Processed	bbl	
21	Well Total Volume Recovered	bbl	
22	Well Total Volume Discarded	bbl	
23	Well Total Vol. of Mud On Solids Discarded	bbl	
24	Well Total Mass Discarded	ton	
25	Bowl Speed	RPM	
26	G-Force	G's	
27	Main Drive Torque	amp	
28	Differential Speed	RPM	
29	Feed Pump Torque	amp	
30	Vibration Angle		
31	Weir Setting		

1. Size, Specification – name or description identifying this piece of equipment
2. Hours Run – hours the centrifuge had been in use in past 24 hours

3. Total Hours – (calculated value) cumulative hours equipment has been in use in this project
4. Mud on Cuttings - the ratio of volume of mud to volume of cuttings in the discharge. If for example, the value of mud on cuttings is 1, it means the discharge is made up of equal parts mud and cuttings, 50% mud and 50% cuttings. Let's look at another example. If mud on cuttings is 0.8, that means 44% mud and 56 % cuttings. [MudV/CutV=0.8 – for every 0.8 barrels of mud there is 1 barrel of cuttings, total volume is 1.8. To determine % of mud by volume, calculate what % of 1.8 is 0.8 (0.8/1.8)]
5. WT In – weight of mud entering the centrifuge
6. WT Out – weight of mud discharged from centrifuge
7. Discard WT – weight of cuttings discarded from centrifuge (can be calculated by ONE-TRAX)
8. Flow rate – rate fluid is entering the centrifuge
9. Flow Rate Out – rate fluid is exiting the centrifuge (can be calculated by ONE-TRAX)
10. Discard Flow Rate - (calculated value) rate the solids are exiting the centrifuge.
11. Close Loop System – in a closed loop system, the effluent of the on centrifuge is further processed by another.
12. Volume Recovered – (calculated value) total volume of mud recovered from the centrifuge
13. Volume In – Centrifuge – (calculated value) total volume of mud entering the centrifuge
14. Vol Discharged – (calculated value) total volume discharged from the centrifuge
15. Vol Mud On Solids Discharged – (calculated value) total volume of mud discharged with the cuttings
16. Volume of Solids Discharged – (calculated value) total volume of solids discharged
17. Discard Mass rate – (calculated value) total mass of cuttings discarded in one hour
18. Mass Discharge – (calculated value) total mass of cuttings discharged
19. Average Density of Discharge - (calculated value) the density of the discharge
20. Total Volume In – Centrifuge – (calculated value) cumulative volume of mud ran through the centrifuge during the project
21. Total Volume Recovered - (calculated value) cumulative volume of mud recovered during the project

-
- 22. Total Vol Discharged - (calculated value) cumulative volume discharged during the project
 - 23. Total Volume mud on Solids Discharged - (calculated value) cumulative volume of mud discharged with solids during the project
 - 24. Total Mass Discharge - (calculated value) cumulative mass of cuttings and mud discharged during the project
 - 25. Bowl Speed – how fast the bowl is rotating
 - 26. G-Force – g-force generated by the centrifuge bowl
 - 27. Main Drive Torque - only available on hydraulic and VFD centrifuges
 - 28. Differential Speed – speed differential between the main drive and back drive
 - 29. Feed Pump Torque - only available on hydraulic and VFD centrifuges
 - 30. Vibration Angle
 - 31. Weir Setting

Shaker Properties:

12005 - Shaker 1			
#	Property	Unit	Value
1	Size , Specifications		
2	Daily Running Hours		
3	Well Total Hours		
4	Mud On Cuttings	bbl/bbl	
5	Daily Volume Discarded (Sludge)	bbl	
6	Volume of Mud On Solids Discarded	bbl	
7	Well Total Volume Discarded	bbl	
8	Well Total Vol. of Mud On Solids Discarded	bbl	
9	% of Cuttings Removed by this Equipment	%	
10	Deck Angle	deg	
11	G-Force	G's	
12	Vibration Angle		
13	Type of Motion		
14	Flow Over Shaker	gal/min	
15	Discard Rate	gal/min	
16	Average Beach Length	in	

1. Size, Specification – name or description identifying this piece of equipment
2. Hours Run – hours the shaker has been in use in past 24 hours
3. Total Hours – (calculated value) cumulative hours equipment has been in use in this project
4. Mud on Cuttings - the ratio of volume of mud to volume of cuttings in the discharge. If for example, the value of mud on cuttings is 1, it means the discharge is made up of equal parts mud and cuttings, 50% mud and 50% cuttings. Let's look at another example. If mud on cuttings is 0.8, that means 44% mud and 56 % cuttings. [MudV/CutV=0.8 – for every 0.8 barrels of mud there is 1 barrel of cuttings, total volume is 1.8. To determine % of mud by volume, calculate what % of 1.8 is 0.8 (0.8/1.8)]
5. Vol Discharged – (calculated value) total volume of solids discharged
6. Vol Mud On Solids Discharged – (calculated value) total volume of mud discharged with the cuttings
7. Total Vol Discharged – (calculated value) total volume of mud and cuttings discharged from the shaker

8. Total Volume mud on Solids Discharged – (calculated value) cumulative volume of mud discharged during the project
9. % of cuttings removed by this equipment – estimate of what percent of overall cuttings being discharged are processed by this shaker
10. Deck Angle - angle the shakers deck is set at, usually -3 to +3 degrees
11. G-Force
12. Vibration Angle
13. Type of motion
14. Flow over shaker
15. Discharge rate
16. Average beach length - If 75% screen coverage is achieved, the beach length will be 25%

Percent of Cuttings Removed by this Equipment field:

General	Pumps/Bits	Fluid Checks	Geometry	Material Transfers	Inventory	Screens	Equipment	Volume Accounting	Displacement
Equipment Performance									
Centrifuge 50258 - Centrifuge 1 50138 - Centrifuge 2 50258 - Centrifuge 1 Vacuum System V120 - Vacuum Cuttings Container C0000 - Cuttings Box Mud Cleaner 21056 - Mud Cleaner Shale Shaker 12005 - Shaker 1 12025 - Shaker 2 12173 - Shaker 3 25866 - BEM Shaker #3 Others									
12005 - Shaker 1									
#	Property	Unit	Value						
1	Size , Specifications								
2	Daily Running Hours		12						
3	Well Total Hours		24						
4	Mud On Cuttings	bbl/bbl	2.5						
5	Daily Volume Discarded (Sludge)	bbl	324.8						
6	Volume of Mud On Solids Discarded	bbl	232						
7	Well Total Volume Discarded	bbl	1,097.1						
8	Well Total Vol. of Mud On Solids Discarded	bbl	668.5						
9	% of Cuttings Removed by this Equipment	%	30						
10	Deck Angle	deg	+3						
11	G-Force	G's	4						
12	Vibration Angle								
13	Type of Motion		linear						
14	Flow Over Shaker	gal/min	15						
15	Discard Rate	gal/min	45						
16	Average Beach Length	in	20						

The field labeled “% of Cuttings Removed by this Equipment” is where the estimated the percentage of cuttings (drilled) removed by a particular piece of equipment.

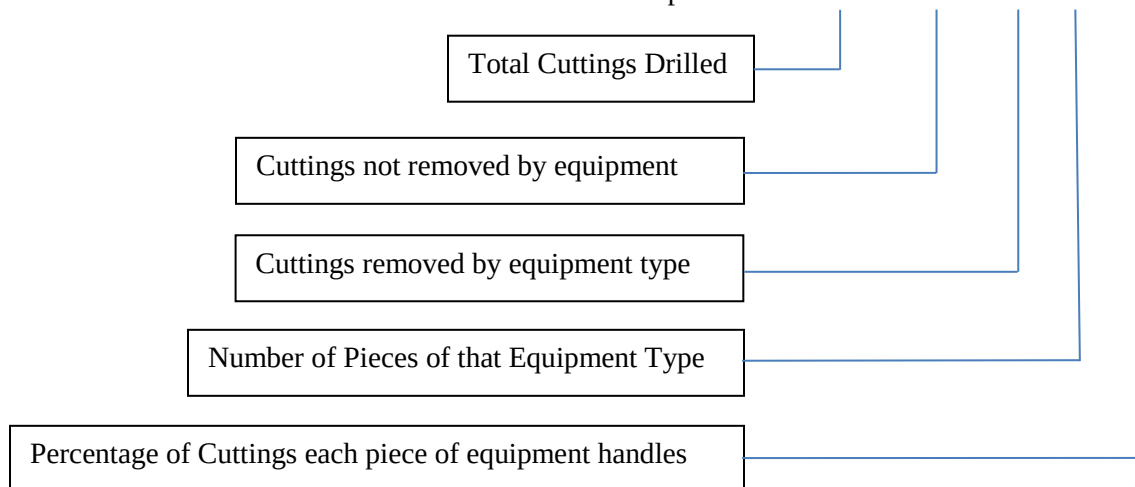
EXAMPLES:

In the example screenshot above, if all 3 shakers are run and we estimate that the entire shaker system removes 90% of the mass of cuttings drilled, then we can say that each shaker will remove 30% of the cuttings drilled (90% divided by 3 shakers). Therefore, the value entered for “% of Cuttings Removed by this Equipment” will be 30%, for each shaker.

A system with 3 shakers that removes 75% of the cuttings drilled, if all shakers are run, then the percentage of cuttings each shaker handles will be 25%.

A system with 4 shakers that removes 75% of the cuttings drilled, if all shakers are run, then the percentage of cuttings each shaker handles will be 18.75%.

Calculation of example: $100\% - 25\% = 75\% / 4 = 18.75\%$



In the example above, ONE-TRAX would take the percentage entered for each shaker (as explained above) and the value of mud on cuttings entered for each shaker to calculate the amount of *fluid lost over the shakers*. A fluid lost transaction (with the total value of *fluid lost over the shakers*) will be automatically created by the software using the **Create Equipment Loss Transaction** button on the volume accounting tab.

Mud Cleaner Properties:

21056 - Mud Cleaner			
#	Property	Unit	Value
1	Size , Specifications		
2	Daily Running Hours		
3	Well Total Hours		
4	Mud On Cuttings	bbl/bbl	
5	Daily Volume Discarded (Sludge)	bbl	
6	Volume of Mud On Solids Discarded	bbl	
7	Well Total Volume Discarded	bbl	
8	Well Total Vol. of Mud On Solids Discarded	bbl	
9	Pressure	psi	
10	Deck Angle	deg	
11	G-Force	G's	
12	Vibration Angle		
13	Type of Motion		
14	Flow Over Shaker	gal/min	
15	Cone Size	in	
16	Cones Used		
17	Input Head	ft	

1. Size, Specification – name or description identifying this piece of equipment
2. Hours Run – hours the Mud Cleaner has been in use in past 24 hours
3. Total Hours – (calculated value) cumulative hours equipment has been in use in this project
4. Mud on Cuttings - the ratio of volume of mud to volume of cuttings in the discharge. If for example, the value of mud on cuttings is 1, it means the discharge is made up of equal parts mud and cuttings, 50% mud and 50% cuttings. Let's look at another example. If mud on cuttings is 0.8, that means 44% mud and 56 % cuttings. $[MudV/CutV=0.8 - \text{for every } 0.8 \text{ barrels of mud there is } 1 \text{ barrel of cuttings, total volume is } 1.8. \text{ To determine \% of mud by volume, calculate what \% of } 1.8 \text{ is } 0.8 (0.8/1.8)]$
5. Vol Discharged – total volume of solids discharged
6. Vol Mud On Solids Discharged – (calculated value) total volume of mud discharged with the cuttings
7. Total Vol Discharged – (calculated value) total volume of mud and cuttings discharged
8. Total Volume mud on Solids Discharged – (calculated value) cumulative volume of mud discharged during the project
9. Pressure – what pressure is going to the micro cyclones

10. Deck Angle - angle the shakers deck is set at, usually -3 to +3 degrees
11. G-Force
12. Vibration Angle
13. Type of motion
14. Flow over shaker
15. Cone Size
16. Cones Used
17. Input Head

Verti-G Properties:

NC-059 - Verti-G (01) High Capacity			
Property	Property	Unit	Value
1	Size , Specifications		
2	Hours Run		
3	Total Hours		0
5	Mud On Cuttings	bbl/bbl	
6	Vol Discharged	bbl	
7	Vol Mud On Solids Discharged	bbl	
8	% of Cuttings Removed by this Equipment		
9	Volume Recovered	bbl	
10	Total Volume Recovered	bbl	
11	Main Drive G Force	G's	
12	Back Drive Differential		
13	Screen Size		
14	Main Drive Torque	amp	
15	Back Drive Torque	amp	
16	Feed Auger Torque	amp	
17	Vibration		
18	Screen Flushed		
19	By-Pass Type		
20	No. Times By-Pass Engaged		

1. Size, Specification – name or description identifying this piece of equipment
2. Hours Run – hours the Verti -G has been in use in past 24 hours
3. Total Hours – (calculated value) cumulative hours equipment has been in use in this project
4. Mud on Cuttings – the ratio of volume of mud to volume of cuttings in the discharge. If for example, the value of mud on cuttings is 1, it means the discharge is made up of equal parts mud and cuttings, 50% mud and 50% cuttings. Let's look at another example. If mud on cuttings is 0.8, that means 44% mud and 56 % cuttings. [MudV/CutV=0.8 – for every 0.8 barrels of mud there is 1 barrel of cuttings, total volume is 1.8. To determine % of mud by volume, calculate what % of 1.8 is 0.8 (0.8/1.8)]
5. Vol Discharged – (calculated value) total volume of mud and solids discharged
6. Vol Mud On Solids Discharged – (calculated value) total volume of mud discharged
7. % of cuttings removed by this equipment

8. Volume Recovered – (calculated value) volume of mud recovered
9. Total Volume Recovered - (calculated value) cumulative volume of mud recovered during the project
10. Main drive G force
11. Back Drive Differential
12. Screen Size
13. Main Drive Torque - available only on MV Verti-G's
14. Back Drive torque – available only on MV Verti-G's
15. Feed auger torque
16. Vibration
17. Screen flushed
18. By-pass type
19. No Times by pass engaged

Sets of properties for other equipment types are very similar to the descriptions above or self-explanatory.

HELPFUL HINT	You will only enter charges for filtration equipment on this tab. All filtration equipment usage data is entered on the Filtration tab.
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4.10.3 Equipment Utilization Table

Equipment Utilization

Serial N	Description	Used	Charge	Daily t	Cum
50258	50258 - Centrifuge 1	1	Stand By	\$230.00	\$230.00
50138	50138 - Centrifuge 2	1	Stand By	\$230.00	\$230.00
V120	V120 - Vacuum	1	Stand By	\$237.50	\$237.50
VG-4568	VG-4568 - Verti G	1	Stand By	\$617.50	\$617.50
A12345	A12345 - Auger	1	Stand By	\$20.00	\$20.00



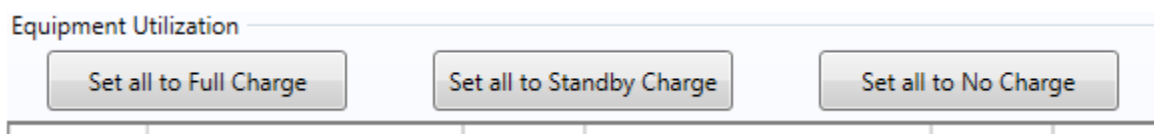

NUMBER OF DAYS TYPE OF CHARGE

The equipment utilization table is used to enter equipment charges. The table lists all equipment added on the equipment setup tab. If you have equipment that has the same description, differentiate them by serial number in first column of the table and make sure you are charging for the right piece of equipment.

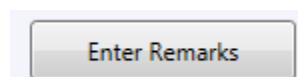
Charging for equipment:

1. Select the type of charge form dropdown list:
 - Set to full charge if the equipment has been used
 - Set to Stand-by if the equipment is present on the location, but has not been used
2. Enter number of days you are charging for in *Used* column of the table.

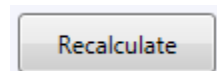
There are three very helpful buttons on the top the equipment utilization table that will save you time when all equipment has the same type of charge. Instead of selecting the type of charge one by one from drop down menu, click one of the buttons show below. This will set all equipment to the same type of charge you select.



4.10.4 Enter Remarks and Recalculate Functions



Click enter remarks to enter comments, remarks, or any other information about the equipment you think it is necessary to keep a record of. These remarks are used in some of solids control equipment daily and recap reports.



Click recalculate after you enter all the information in equipment properties and equipment utilization tables to update the equipment costs and recalculate equipment parameters.

4.11 VOLUME ACCOUNTING TAB

This tab displays the information relating to drilling fluid transactions. The example below shows the tab and its arrangement into five information display sections and function buttons.

The screenshot shows the ONE-TRAX Volume Accounting tab with the following sections highlighted:

- Pit Section:** A table listing pits with columns: Pit Descrip, Capacity, Start Volume bbl, Calc End Volume bbl, Fluid Weight lb/gal, Comment, Pit Type, and End Volume bbl. It includes rows for Active, Reserve, and Premix pits.
- Fluids in Hole Section:** A table showing fluid volumes in the hole, including Total Hole Volume, Volume Not Fluids, and Fluid Volume.
- Total Loss Breakdown Section:** A table showing the breakdown of total loss, including Auger, Boat Clean, DISPLACEMENT, Filtration, and Surface Loss.
- Fluid Volume Section:** A table showing fluid volumes, including Total Weight of Fluid in Pits (lb), Sum of Fluid Volumes (End Value) - bbl, and Active Pits w/o Hole.
- Fluids Balance Section:** A table showing fluid balance, including Start Volume, Base Fluid Added, Water Added, Volume Chemicals, Total Volume Built, Received, Back Load, Transferred In, Transferred Out, Loss and Dumped, and Final Fluid Volume.

4.11.1 Pit Section

The pit section shows information about the pits and mud in them. It is a combination of display and entry fields, and is a good tool for keeping track of your pits and their contents.

Display only fields:

- Pit Description – comes from the information entered in Pits / Loss Setup tab.
- Capacity – also entered on the Pits / Loss Setup tab.
- Start Volume – the end volume of the pit from the prior day or zero on the very first day created.

- Calc. End Volume – this is the volume that ONE-TRAX calculates should be in the pit. This value is not shown for any pit that is defined with the pit type active.

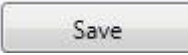
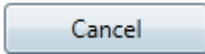
Updateable fields:

- Fluid Weight – entering the density in this field will allow ONE-TRAX to calculate and display the total weight of fluids in pits field at the bottom of this section.
- Comment – this field will allow you to enter notes about that pit. In the example below, the spacers for a displacement have been mixed and labeled in the comment field.

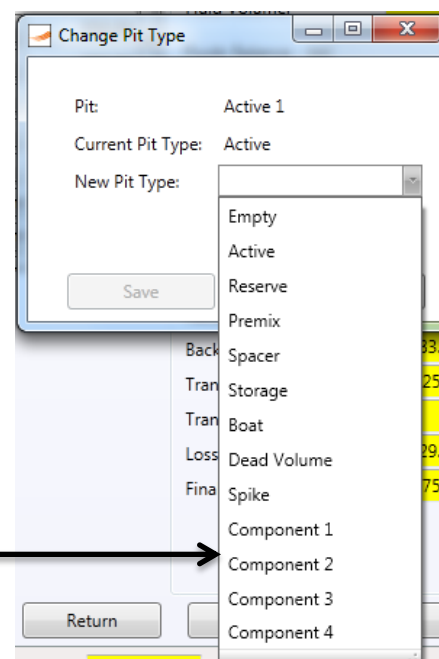
Display Only fields

Updateable Fields

Pits							
Pit Descrip	Capacity	Start Volume bbl	Calc End Volume bbl	Fluid Weight lb/gal	Comment	Pit Type	End Volume bbl
Active 1	800	640		11.50	OBM	Active	200.00
Active 2	800	0	550	11.60	Brine	Reserve	550.00
Active 3	800	0	550	11.60	Brine	Reserve	550.00
Reserve 1	400	0	0			Reserve	0.00
Reserve 2	400	250	0			Reserve	0.00
Reserve 3	400	200	0			Reserve	0.00
Premix 1	50	0	50		Transition Spacer	Component	50.00
Premix 2	100	0	50		Viscous Spacer	Component	50.00
Premix 3	100	0	100		Cleaning Spacer	Component	100.00
Total Weight of Fluids in Pits (lb):			632520				

- Pit Type – this field displays the defined type for that pit. Clicking the button displaying the pit type will open a dialogue box allowing the type to be changed. Pit types are entered on the Pits / Loss Setup tab.  will update the pit type to the new one and  will close the dialogue without saving.

Available Pit Types – from
Pits / Loss Setup tab



- End Volume – the actual volume measured in the Pit.

4.11.2 Fluids in Hole Section

This section displays the volume of the well bore and the fluids filling it. Volume not fluids is defined as something occupying volume in the hole, but is not part of the active system. The section is a combination of display and updatable fields.

Fluids In Hole - bbl				
	Annulus	Total DS	Below	Total
Total Hole Volume:	357.9	15.6		373.5
Volume Not Fluids:				
Fluid Volume:	357.9	15.6		373.5

Total = Annulus + Total DS +
Below

DS – Drill String
Below – Below Bit

Fluid Volume = Total Hole Volume – Volume Not Fluids

- Total Hole Volume is the volume of the well bore calculated from the information entered in the Well Geometry tab.

- Volume Not Fluids is where you will enter the volume in the well that is not part of the active system.
- Examples:

Fluids In Hole - bbl			
	Annulus	Total DS	Below
Total Hole Volume:	146.8	11.2	
Volume Not Fluids:	146.8	11.22	
Fluid Volume:			

On a Land Rig;

you are displacing from WBM to OBM. You need to enter the entire fluid volume in the volume not fluids fields. If the 1589 bbls were not accounted for the fluid volume section, discussed later in the chapter, would be off and you would have 158 bbl more in your active pits than expected.

Fluids In Hole - bbl			
	Annulus	Total DS	Below
Total Hole Volume:	357.9	15.6	
Volume Not Fluids:	357.9		
Fluid Volume:		15.6	

Offshore you

are displacing a well from seawater, which you used to drill the tophole, to mud. You have pumped the mud to the bit, but still have seawater in the annulus at report time. You need to enter 357.9 bbl in the volume not fluids field of the annulus column to account for the volume that is still seawater. If the 357.9 bbls were not accounted for the fluid volume section would be off and you would have 357.9 bbl more in your active pits.

- Fluid Volume shows how much drilling fluid from the active system needs to be accounted for in each portion of the hole.

4.11.3 Fluid Volumes Section

Sum of Fluid Volumes (End Value) - bbl		
	Calc Value:	Actual Values:
Reserve Pits:	1100	1100
Other Pits:	269.9	269.9
Active Pits w/o Hole:		200
Mud Volume In Hole:		550.1
Active Pits w/ Hole:	750	750.1

Not Accounted:

Other Pits are those pits that are not defined as either Active or Reserve

The fluids volume section is strictly display only. It shows the volumes in the pits calculated by ONE-TRAX (Calc Value:) and measured by the mud engineer (Actual Values:). The not accounted fields show the difference between the two volumes. If the value in a not accounted field is not 0, it is necessary to create a transaction(s) to have it display 0.

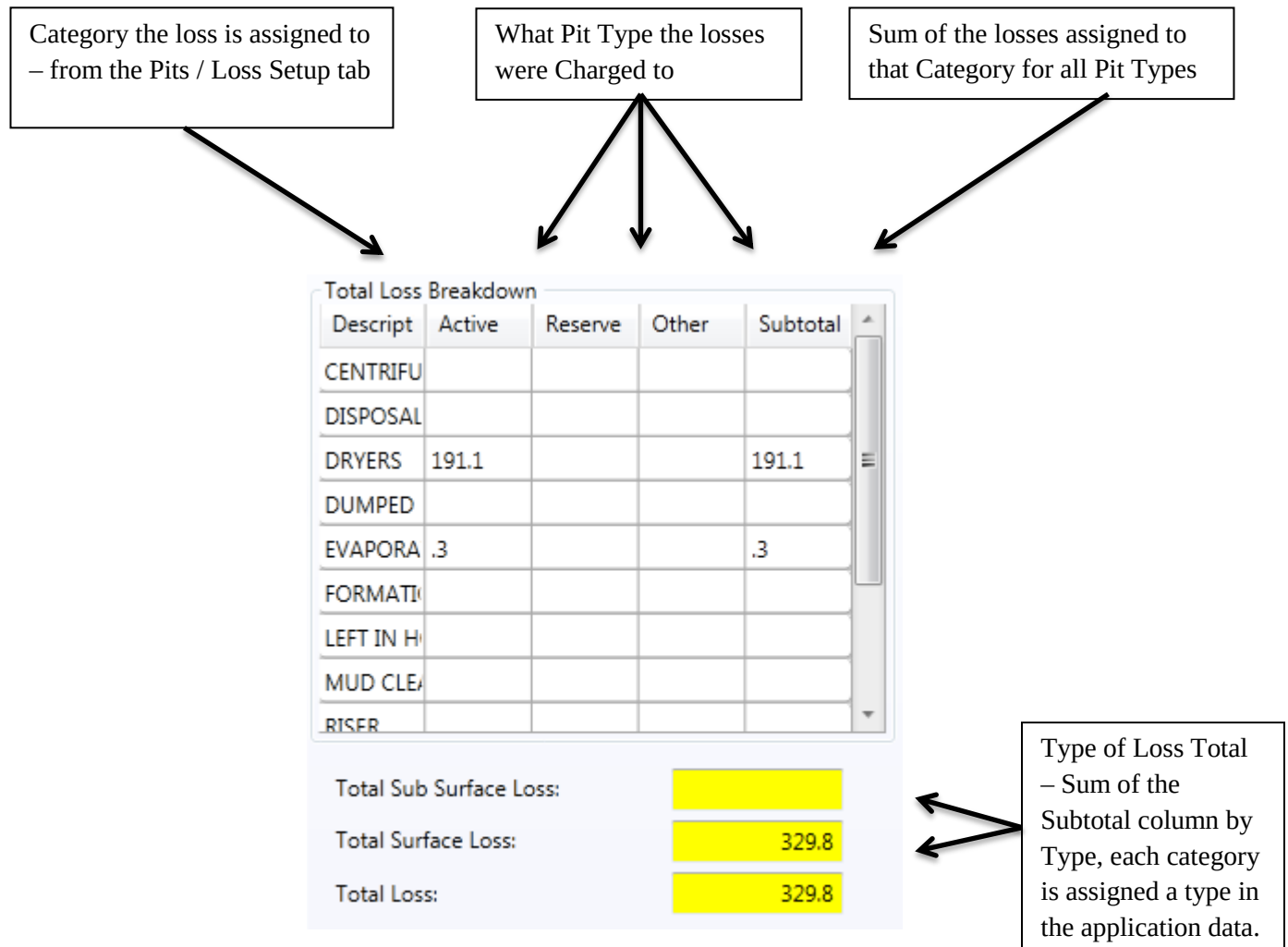
4.11.4 Fluid Balance Section

The fluids balance section is another display only section. It shows all the volume transactions for the day in tabular form.

Fluids Balance - bbl				
	Active System	Reserve Pits	Other Pits	Total
Start Volume:	1713.3	450		2163.3
Base Fluid Added:			50	50
Water Added:			47.4	47.4
Volume Chemicals:		1100	172.6	1272.6
Total Volume Built:		1100	269.9	1369.9
Received:				
Back Load:	883.5	200		1083.5
Transferred In:	250			
Transferred Out:		250		
Loss and Dumped:	329.8			329.8
Final Fluid Volume:	750	1100	269.9	2120

4.11.5 Total Loss Breakdown Section

The total loss breakdown Section displays all the fluid loss transactions for the day in a tabular format.



4.11.1 Non-Transaction Pits Section

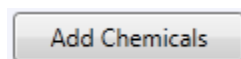
The non-transaction pits Section is where the fluid balance for these pits is entered. These are display only pits and are not used in any volume accounting transactions.

Non-transactional Pits	
Pit Description	End Volume bbl
Base Oil Storage	800.00
OBM Storage	500.00

4.11.2 Function Buttons

The function buttons are used to enter fluid transactions and display useful information.

Add Chemicals Button



is used to add items from inventory to the pits. Clicking will bring up a dialogue box where you can fill in what items, what quantity, and which pit you are adding the items to.

Quantity of Item added - Updateable

Add Chemicals

	Description	Alt Name	Item No	Locator	Size	Unit	Packaging	On Hand	Qty
1	M-I BAR BULK	Bar	M0000003	Domestic	1	TN	N/A	162	2.00
2	M-I GEL BULK	Gel	M0000321	Domestic	1	TN	N/A	4	0.00
3	DF-1 BASE FLUID	DF1	M0001993	Domestic	1	BL	N/A	240	10.00
4	CALCIUM CHLORIDE PELLETS 94-97 PERCENT	CaCl2	M0003212	Domestic	100	LB	BG	20	0.00
5	CALCIUM CHLORIDE BRINE S.G. 1.40	CaCl2 Brine	M0009017	Domestic	1	BL	N/A	3737	0.00
6	CAUSTIC SODA	Soda	M0004024	Domestic	50	LB	BG	64	0.00
7	DUO-VIS	DUO VIS	M0002978	Domestic	50	LB	BG	285	1.00
8	J-GUAR 308N8-GUAR GUM	Guar Gum	M0005469	Domestic	50	LB	BG	40	0.00
9	LIME	LIME	M0004111	Domestic	50	LB	BG	0	0.00
10	VERSAMUL I	Versamul	M0003127	Domestic	200	LT	DM	17	0.00
11	VERSACLEAN SYSTEM 11.1 - 11.5 PPG	SBM	M0016822	Domestic	1	BL	N/A	0	0.00
12	VERSACOAT NS	Versacoat	M0006343	Domestic	200	LT	DM	11	0.00
13	VERSAGEL	VersaGel	M0002679	Domestic	50	LB	BG	50	0.00
14	NUT PLUG	Nut Plug	M0006408	Domestic	50	KG	BG	0	0.00
15	POLYPAC UL	PolyPac	M0002205	Domestic	50	LB	BG	20	0.00
16	POTASSIUM CHLORIDE BRINE-(9.6POUNDS/	KCL Brine	M0001664	Domestic	1	BL	N/A	135	0.00
17	POTASSIUM CHLORIDE (KCL)	KCL Bagged	M0003641	Domestic	50	LB	BG	680	0.00
28	DEEPCLEAN NS	deepclean	M0016425	Domestic	55	GA	DM	0	0.00
19	SAFE-CARB 40	SC 40	M0003047	Domestic	50	LB	BG	0	0.00
21	SODA ASH	Soda Ash	M0004181	Domestic	100	LB	BG	65	0.00
22	KCL / POLYMER PREMIX	KCL Premix	M0003239	Domestic	1	BL	N/A	0	0.00
23	SAFE-VIS E	Safe Vis	M0001686	Domestic	5	GA	CN	8	0.00
24	DIATOMACEOUS EARTH	DE	M0002172	Domestic	50	LB	BG	60	0.00
25	FILTER CARTRIDGE 3 MICRON ABSOLUTE 40	3 micron ABS Cartridge	M0000222	Domestic	1	N/A	EA	80	0.00

Interval: 3

Fluid System: VERSACLEAN

Add Materials To Pit: Active 1 - Active

Add Base Fluid (bbl): 10

Add Water (bbl): 0.00

Volume Of Chemicals (bbl): 2.81

Is Dilution: ☐

Save Cancel

Inventory information – Display Only

Interval: 3

Fluid System: VERSACLEAN

Add Materials To Pit: Active 1 - Active

Add Base Fluid (bbl): 10

Add Water (bbl): 0.00

Volume Of Chemicals (bbl): 2.81

Is Dilution: ☐

Drop down allowing a choice of which Interval to assign the transaction, defaults to that date's interval.

Drop down allowing a choice of which Fluid System to assign the transaction.

Drop down allowing a choice of which Pit to add the items too.

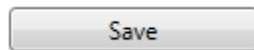
Volume of Items added having a daily cost code of Base Fluid.

Enter the volume of water added.

Calculated Volume of Items added, derived from specific gravity entered in the Product Setup Tab. This does not include base fluid or water.

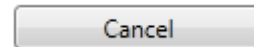
Checkbox indicating a Dilution transaction, see helpful Hint.

HELPFUL HINT	Checking the “Is Dilution” checkbox will allow ONE-TRAX to keep track of when you use an Add Chemical or Add Whole Mud transaction for the purpose of reducing Low Gravity Solids. This information is used in certain recap reports and ONE-TRAX Central. In short, you check the box on if the purpose of the transaction was LGS reduction.
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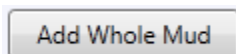
processes the transaction and updates volumes, costs, and inventory. This

button will only be active if all required fields are filled in.

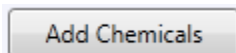


closes the dialogue without processing the transaction.

Add Whole Mud Button



opens a dialogue box very similar to the one opened by the



button. Though they look similar, there are some very important differences between them.

Inventory information –
Display Only

Product concentration field – enter the concentration of each product that is in the mud you are adding to the system.

This does not remove the item from inventory and allows ONE-TRAX to accurately calculate chemical concentrations in the mud.

Add Whole Mud

#	Description	Alt Name	Item No	Locator	Size	Unit	Packaging	Product Conc
1	M-I BAR BULK	Bar	M0000003	Domestic	1	TN	N/A	0.00
2	M-I GEL BULK	Gel	M0000321	Domestic	1	TN	N/A	0.00
3	DF-1 BASE FLUID	DF1	M0001993	Domestic	1	BL	N/A	0.00
4	CALCIUM CHLORIDE PELLETS	CaCl2	M0003212	Domestic	100	LB	BG	0.00
5	CALCIUM CHLORIDE BRINE S.C	CaCl2 Brine	M0009017	Domestic	1	BL	N/A	0.00
6	CAUSTIC SODA	Soda	M0004024	Domestic	50	LB	BG	0.00
7	DUO-VIS	DUO VIS	M0002978	Domestic	50	LB	BG	0.00
8	J-GUAR 308NB-GUAR GUM	Guar Gum	M0005469	Domestic	50	LB	BG	0.00
9	LIME	LIME	M0004111	Domestic	50	LB	BG	0.00
10	VERSAMUL I	Versamul	M0003127	Domestic	200	LT	DM	0.00
11	VERSACLEAN SYSTEM 11.1 - 1	SBM	M0016822	Domestic	1	BL	N/A	0.00
12	VERSACOAT NS	Versacoat	M0006343	Domestic	200	LT	DM	0.00
13	VERSAGEL	VersaGel	M0002679	Domestic	50	LB	BG	0.00
14	NUT PLUG	Nut Plug	M0006408	Domestic	50	KG	BG	0.00
15	POLYPAC UL	PolyPac	M0002205	Domestic	50	LB	BG	0.00
16	POTASSIUM CHLORIDE BRINE	KCL Brine	M0001664	Domestic	1	BL	N/A	0.00
17	POTASSIUM CHLORIDE (KCL)	KCL Bagged	M0003641	Domestic	50	LB	BG	0.00
28	DEEPCLEAN NS	deepclean	M0016425	Domestic	55	GA	DM	0.00
19	SAFE-CARB 40	SC 40	M0003047	Domestic	50	LB	BG	0.00
21	SODA ASH	Soda Ash	M0004181	Domestic	100	LB	BG	0.00
22	KCL / POLYMER PREMIX	KCL Premix	M0003239	Domestic	1	BL	N/A	0.00
23	SAFE-VIS E	Safe Vis	M0001686	Domestic	5	GA	CN	0.00
24	DIATOMACEOUS EARTH	DE	M0002172	Domestic	50	LB	BG	0.00

Interval: 3

Fluid System: VERSACLEAN

Add Whole Mud To: Active 1 - Active

Add Without Charge: ☐

Product For the Whole Mud: M0016822 - VERSACLEAN S

Received From:

Volume (bbl): 100.00

Is Dilution: ☐

Save Cancel

Interval: 3

Fluid System: VERSACLEAN

Add Whole Mud To: Active 1 - Active

Add Without Charge: ☐

Product For the Whole Mud: M0016822 - VERSACLEAN S'

Received From:

Volume (bbl): 100.00

Is Dilution: ☐

Drop down allowing a choice of which Interval to assign the transaction, defaults to that date's interval.

Drop down allowing a choice of which Fluid System to assign the transaction, defaults to the default interval's fluid system.

Drop down allowing a choice of which Pit to add the items too.

Drop down allowing a choice of which Mud to Add. Only Items defined as Whole Mud will be shown. This is from the daily cost code assigned in the Product Setup Tab.

Volume of whole mud added

Checkbox which will allow the mud to be added without cost.

Checkbox indicating a dilution transaction

Add Without Charge: ☐ has an additional feature. When checked it will allow you to add mud without assigning the product for whole mud drop down. This will allow you to add mud to a pit without affecting inventory. For example, if the operator had purchased mud and brought it onto the rig; or you wish to add mud that is rented from M-I SWACO but for which the charge will be added with the used other column on the material transfers tab.

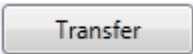
Save

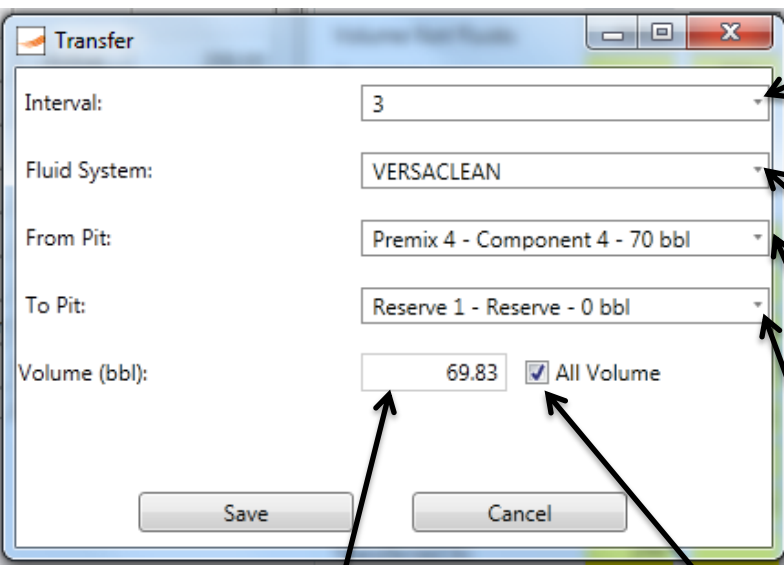
processes the transaction and updates volumes, costs, and inventory. This button will only be active if all required fields are filled in.

Cancel

closes the dialogue without processing the transaction.

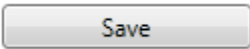
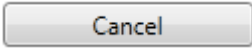
Transfer Button

 is how moving mud between pits is accomplished. It opens a basic dialogue box where the information for the mud transfer is entered.

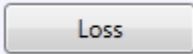


The **Transfer** dialog box contains the following fields and controls:

- Interval:** A drop-down menu showing '3'. Annotation: "Drop down allowing a choice of which Interval to assign the transaction, defaults to that date's interval."
- Fluid System:** A drop-down menu showing 'VERSACLEAN'. Annotation: "Drop down allowing a choice of which Fluid System to assign the transaction, defaults to the default interval's fluid system."
- From Pit:** A drop-down menu showing 'Premix 4 - Component 4 - 70 bbl'. Annotation: "Drop down allowing a choice of which Pit you are transferring from. Shows name of pit, type of pit, and volume in pit."
- To Pit:** A drop-down menu showing 'Reserve 1 - Reserve - 0 bbl'. Annotation: "Drop down allowing a choice of which Pit the mud is being added too."
- Volume (bbl):** A text input field showing '69.83'. Annotation: "How much volume will be transferred"
- All Volume:** A checked checkbox. Annotation: "This check box tells ONE-TRAX to move the entire calculated pit volume. In this example, the exact volume is slightly different from our measured volume. By checking the 'All Volume' checkbox, we are left with an empty pit."
- Buttons:** 'Save' and 'Cancel' buttons at the bottom.

 processes the transaction and updates volumes, costs, and inventory. This button will only be active if all required fields are filled in.  closes the dialogue without processing the transaction.

Loss Button

 can be used to create all mud loss transactions. There is an alternative way to create equipment loss transactions which will be discussed later.

Interval: 3

Fluid System: VERSACLEAN

From Pit: Active 1 - Active - 750 bbl

Loss Type: TRIPPING

Volume (bbl): 10.00 ☐ All Volume

Save Cancel

Drop down allowing a choice of which Interval to assign the loss to.

Drop down allowing a choice of which Fluid system to assign the loss.

Drop down from which the Pit losing the mud is chosen from.

Drop down allowing a choice of which type the loss is. Loss types are defined in the Pits / Loss Setup tab.

How much volume was lost.

This check box tells ONE-TRAX that the entire calculated pit volume was lost. This is very useful when an entire pit is being dumped or returned.

Save processes the transaction and updates volumes, costs, and inventory. This button will only be active if all required fields are filled in. **Cancel** closes the dialogue without processing the transaction.

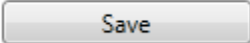
Once the loss transaction is saved it will appear in the total loss breakdown.

Return Button

Return is used when mud is being returned to the warehouse, transferred to another rig or to the next project. It will remove the mud from your pits and inventory but will not create an accounting transaction. For example, if the mud is being returned for credit, in order for the cost to be transacted correctly you will need to use the used other column on the inventory tab.

The screenshot shows a 'Return' dialog box with the following fields and annotations:

- Interval:** A dropdown menu showing '3'. Annotation: 'Drop down allowing you to choose Interval'.
- Fluid System:** A dropdown menu showing 'VERSACLEAN'. Annotation: 'Drop down allowing you to choose fluid system.'.
- From Pit:** A dropdown menu showing 'Active 1 - Active - 750 bbl'. Annotation: 'Drop down allowing a choice of which Pit you are returning.'.
- Returned To:** A dropdown menu showing 'Warehouse'. Annotation: 'Drop down allowing a choice of where the mud is going, controlled through the application data.'.
- Volume (bbl):** A text field showing '750.04'. Annotation: 'How much volume will be returned'.
- All Volume:** A checked checkbox. Annotation: 'Returns entire calculated pit volume'.
- Buttons:** 'Save' and 'Cancel' buttons at the bottom.

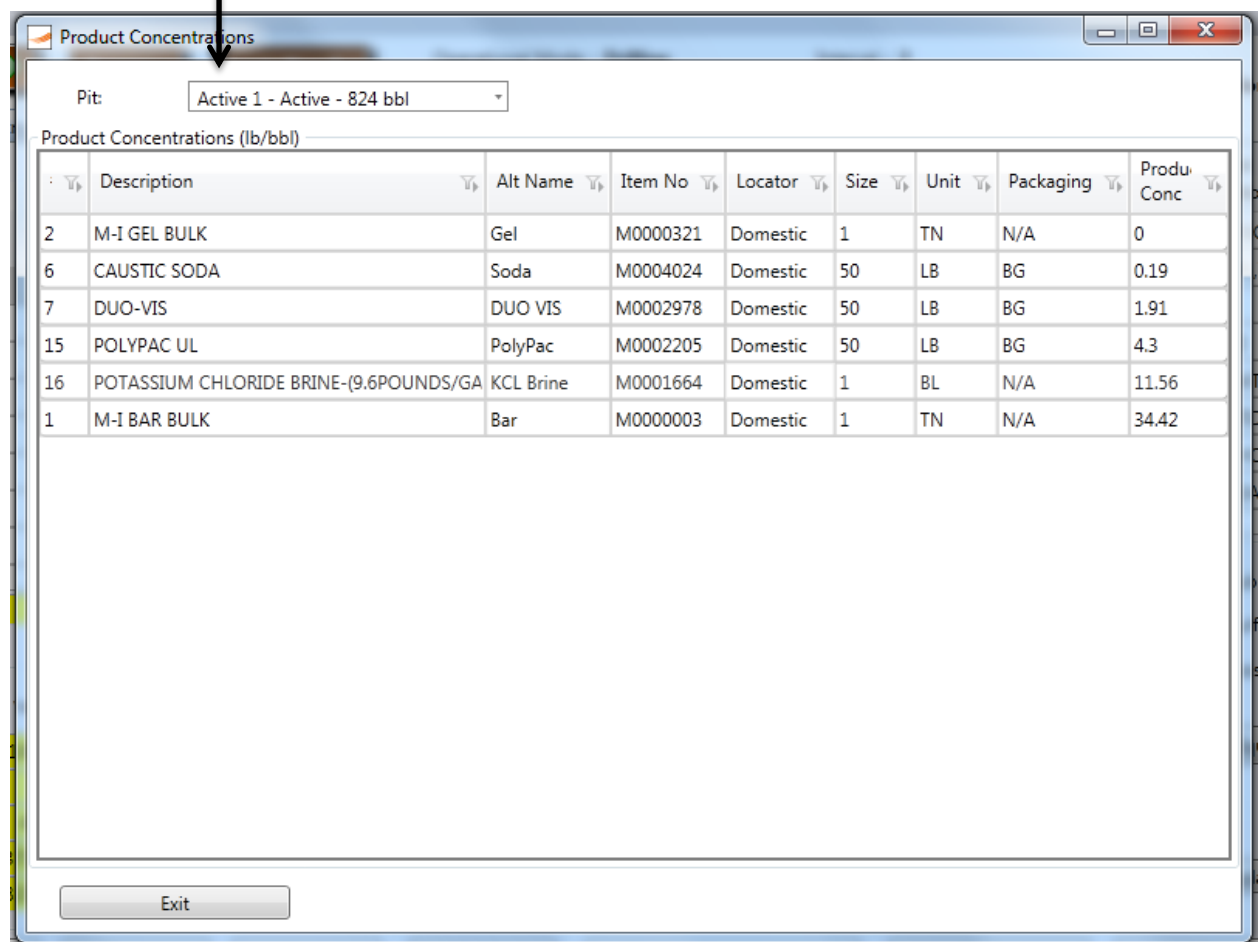
 processes the transaction and updates volumes, costs, and inventory. This button will only be active if all required fields are filled in.  closes the dialogue without processing the transaction.

Once the return is saved it will appear in the back load row in the fluid balance section.

Product Concentrations Button

Product Concentrations opens a window which shows how much of each product is in the mud. Only items with the **Show Conc?** ☒ checked on the product setup tab will appear. This is a strictly informational display and no data can be entered. There is one drop down that allows you to change the pit that is being displayed.

Allows you to change which pit is being viewed.

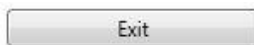
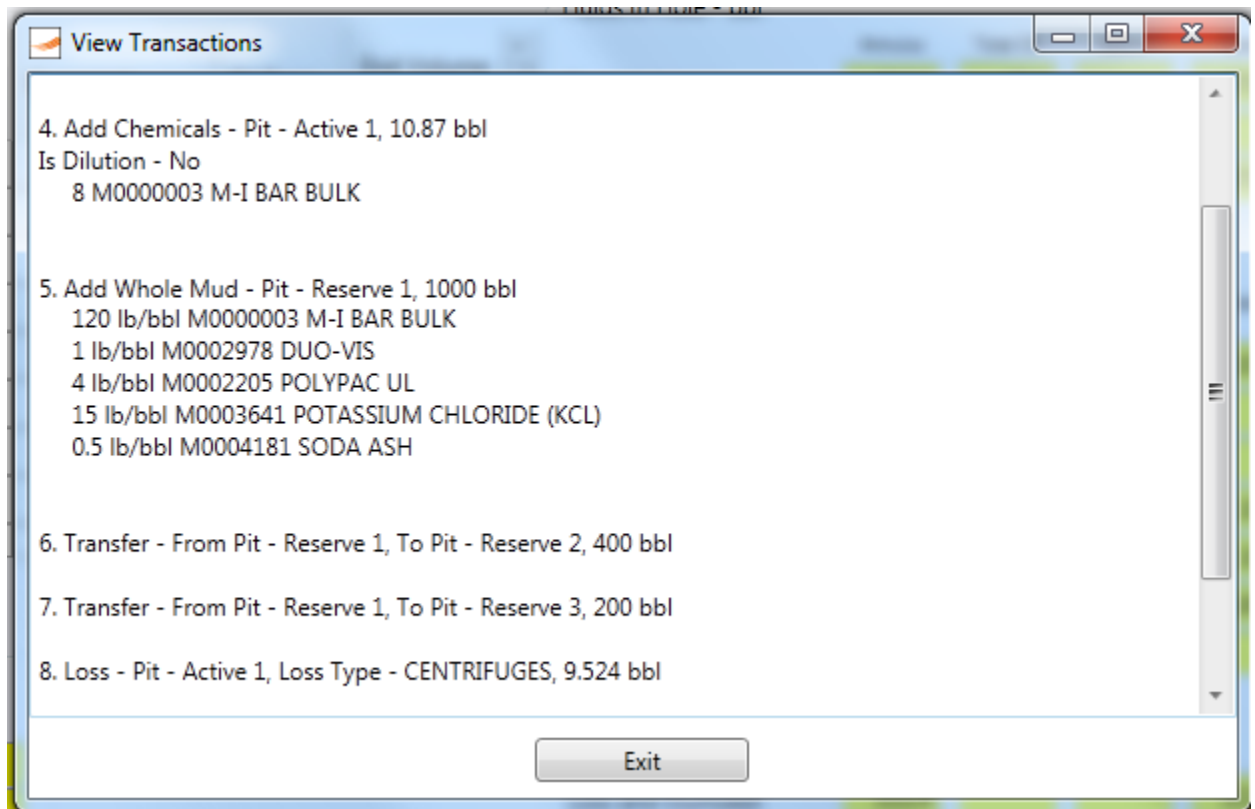


	Description	Alt Name	Item No	Locator	Size	Unit	Packaging	Product Conc
2	M-I GEL BULK	Gel	M0000321	Domestic	1	TN	N/A	0
6	CAUSTIC SODA	Soda	M0004024	Domestic	50	LB	BG	0.19
7	DUO-VIS	DUO VIS	M0002978	Domestic	50	LB	BG	1.91
15	POLYPAC UL	PolyPac	M0002205	Domestic	50	LB	BG	4.3
16	POTASSIUM CHLORIDE BRINE-(9.6POUNDS/GA	KCL Brine	M0001664	Domestic	1	BL	N/A	11.56
1	M-I BAR BULK	Bar	M0000003	Domestic	1	TN	N/A	34.42

Exit closes the dialogue and returns to the volume accounting tab.

View Transactions Button

View Transactions opens an informational display of all fluid transactions processed that day.

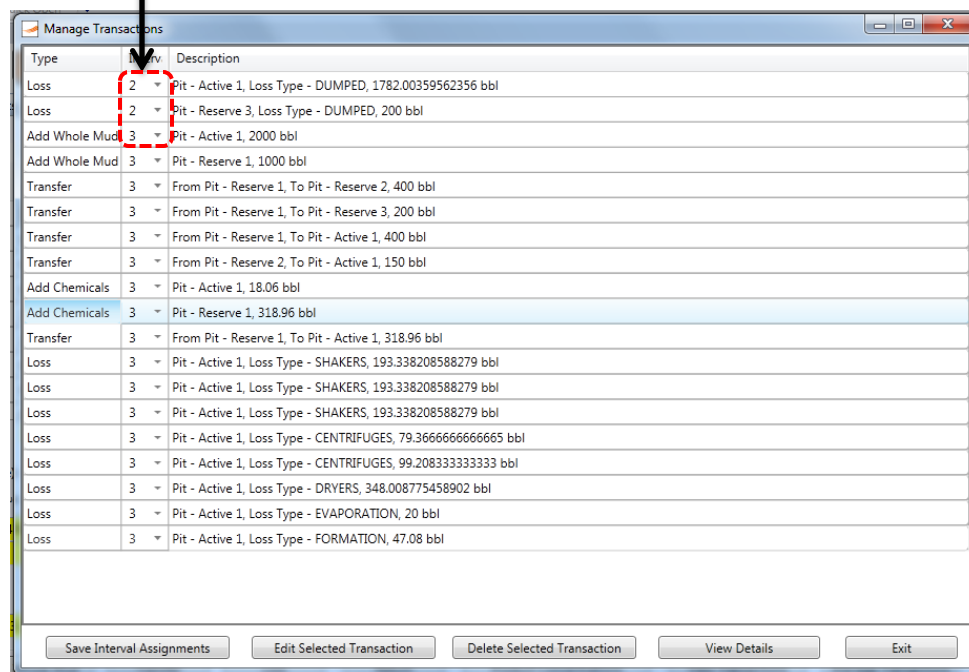


closes the dialogue and returns to the Volume Accounting tab.

Manage Transactions Button

Manage Transactions is a very powerful tool included in ONE-TRAX. With it you can make changes to or delete any fluid transaction. When clicked it opens a window showing a one line summary of each transaction.

This drop down allows you to change the interval the transaction is assigned to. The changes are saved when the Save Interval Assignments button is pressed. In this example, the well has started an OBM interval, but has also disposed of the WBM on the same day. The first two lines show that the dumping of the WBM is assigned to interval 2.

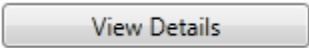


Blue highlighting indicates the active transaction. The transaction is activated by clicking on it with the cursor.

Save Interval Assignments will save any changes made to the which interval a transaction belongs to. If changes have been made, but the button not pressed, the changes will not be saved.

Edit Selected Transaction when pressed it will reopen the active transaction and allow you to make changes. It is only active when a transaction is selected.

Delete Selected Transaction will remove the active transaction from the list. Unlike earlier versions of ONE-TRAX, it is not limited to the last transaction.

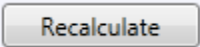
 View Details

opens the view transaction screen, see prior section, so that the transaction can be viewed in more detail.

 Exit

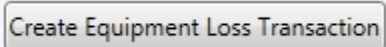
closes the dialogue and returns to the volume accounting tab.

Recalculate Button

 Recalculate

will recalculate volumes and product concentrations using the list of transactions entered that day.

Create Equipment Loss Transaction Button

 Create Equipment Loss Transaction

can be used to create a loss transaction calculated from the information entered on the equipment tab. The loss will be charged off to a loss type corresponding to one of the pieces of equipment on the rig. The loss can only be charged against the active system. If no equipment usage information is entered these figures will not be calculated. Pressing the button opens a dialogue box which allows you to choose which of the calculated losses you wish to use.

What type of equipment the loss will be charged to. This is entered in the Equipment Setup tab.

Identifies the piece of equipment, shows the Equipment Code and Alternate Name from the Equipment Setup tab.

Note - these losses will be created against the Active System

EquipmentType	Equipment	Volume Discharged	Create Transaction?
Cuttings Dryer	NC-059 - Verti G	191.058	☒
Shale Shaker	NC-016 - Shaker 1	46.154	☐
Shale Shaker	NC-016 - Shaker 2	46.154	☐
Shale Shaker	NC-016 - Shaker 3	46.154	☐

Shows volume of mud discharged by each piece of equipment. It is calculated from the information entered in the Equipment tab.

Checkbox which allows you to have ONE-TRAX create the transaction or not.

Create Transactions

Cancel

Create Transactions will create a loss transaction for each create transaction checkbox that has been check. . **Cancel** closes the dialogue without processing the transaction.

Enter PST Button

Enter PST will open the tab to enter the information related to production screen tests. The entry form consists of two tables. The upper table, tests, is where information relating to each test is entered. The lower table, volumes, is where the volume information of each test is entered.

Enter PST Data

Test #	Test Time	Pass?	Sample Location	Cumulative Volume Circulated (ml)	Shaker Screen Size	M-I Swaco Engineer	Screen Provider	Screen Representative	Operator Representative	Fluid Temperature (F)	Fluid Density (lb/gal)	Action
1	0:30	Y	Active 1	3,000.00	500 Micron	Josemaurys V	ResLink		John Able	101	9.40	
2	15:10	Y	Active 1	6,000.00	500 Micron	Josemaurys V	ReLink		John Able	99	9.50	

Volumes tested for Test -1

#	Volume (ml)	Cumulative Volume (ml)	Time/Volume (seconds)	Cumulative Time (seconds)	Action
1	1000	1000	9	9	
2	1000	2000	8	17	
3	1000	3000	7	24	

Exit

Identification Number –
automatically filled in.

Tests

Test #	Test Time	Pass?	Sample Location	Cumulative Volume Circulated (ml)	Shaker Screen Size	M-I Swaco Engineer	Screen Provider	Screen Representative	Operator Representative	Fluid Temperature (F)	Fluid Density (lb/gal)	Action
1	0:30	Y	Active 1	3,000.00	500 Micron	Josemaurys V	ResLink		John Able	101	9.40	
2	15:10	Y	Active 1	6,000.00	500 Micron	Josemaurys V	ReLink		John Able	99	9.50	

The rest of the fields on the table are filled in with information relating to the each test.

- Test Time – when the test was conducted
- Pass – whether the test passed or failed
- Sample Location – drop down limited to defined pits
- Cumulative Volume Circulated – how much volume was circulated for the test
- Shaker Screen Size – Size of the screen used for the test
- M-I SWACO Engineer – engineer conducting the test, drop down limited to personnel added with the Personnel Setup tab.
- Screen Provider – screen manufacturer
- Screen Representative – name of screen manufacturer representative
- Operator Representative – name of operator representative
- Fluid Temperature – temperature of the fluid tested
- Fluid Density – weight of the fluid tested

The lower table, volumes tested, is available for each test in the test table. It is identified by the test number displayed above the table. You can add as many lines as needed to record test

Identification Number –
automatically filled in.

Cumulative Volume of
test, filled in by ONE-
TRAX.

Cumulative Time of test,
filled in by ONE-TRAX.

Volumes tested for Test -1

#	Volume (ml)	Cumulative Volume (ml)	Time/Volume (seconds)	Cumulative Time (seconds)	Action
1	1000	1000	9	9	
2	1000	2000	8	17	
3	1000	3000	7	24	
					

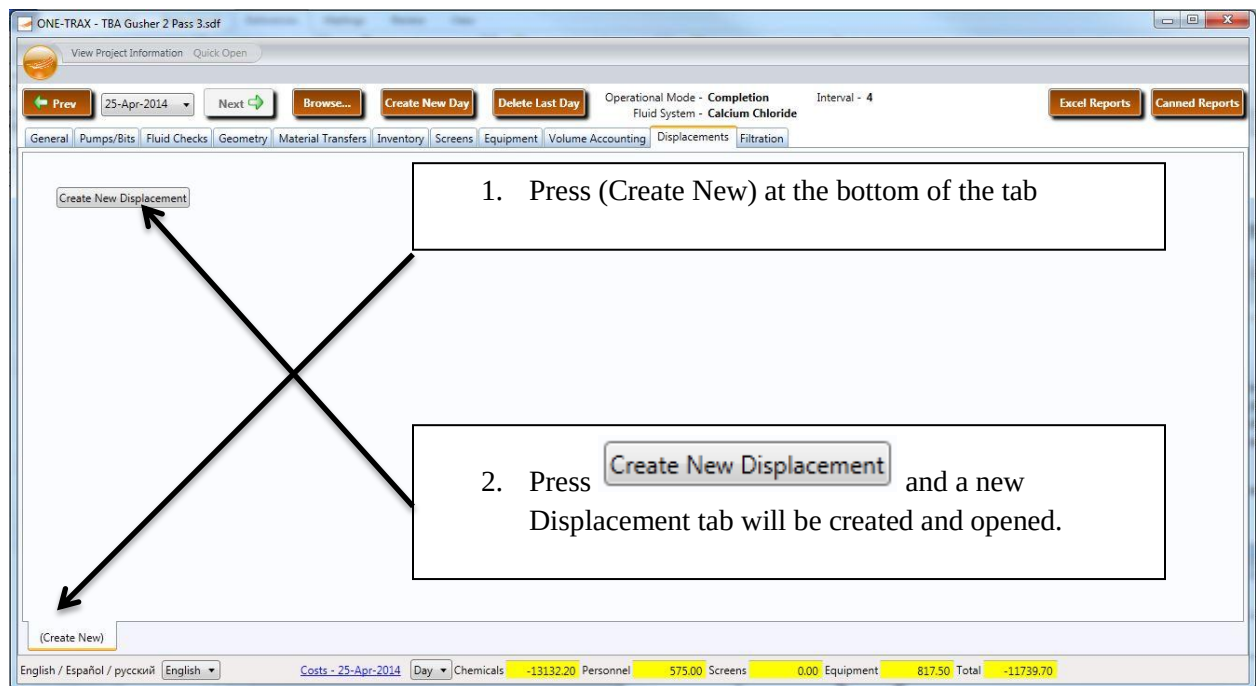
Volume Used for
each portion of the
test.

Time required
for test volume
to flow through
test screen.

4.12 DISPLACEMENTS TAB

The displacement tab can be used anytime the well is displaced from one fluid to another. The tab is most useful when the well is doing a displacement using spacers; for example when changing the well bore from an oil or synthetic based mud to a water based reservoir drill-in fluid or completion brine. It can be used for all displacements, but will be most useful when you want to track information about spacer composition and pumping regime.

The displacements tab has a similar design to the fluid checks tab. A new displacement is created by activating the (Create New) tab and pushing the **Create New Displacement** button.



Below is an example of a displacement. It consists of two parts, information relating to the whole displacement process and information relating to each of the components.

Displacement Section – information about the whole displacement Process.

Displacement Components Section – information about each of the components of the displacement.

Type of Displacement: Indirect Forward

Fluid being Displaced: VERSACLEAN

Fluid being Displaced - Density (lb/gal): 11.60

Displacing Fluid Name: Calcium Chloride

Displacing Fluid Density (lb/gal): 11.50

Pipe Rotation Rate: 30.00

Displacement Interface (bbl): 40.00

Volume Brine Interface Reclaimed (bbl): 0.00

Max Allowable Pump Pressure (psi): 3,700.00

Max Pump Pressure Achieved (psi): 2,780.00

Pump Time For Displacement (hrs): 8.00

Downtime During Displacement (hrs): 0.00

Rig Cleanup Time - Total (hrs): 12.00

Rig Cleanup Time - On-line Only (hrs): 0.00

☐ Balanced?

☐ Short-trip in Mud ☐ Mud Observed on w/s After Displacement

☐ Short-trip in Seawater ☐ Mud Observed on Brushes After Displacement

☐ Short-trip in Brine ☐ Volume Brine Interface Reclaimed

Circulation Data Delete Displacement

(Create New) Indirect Forward - VERSACLEAN

Displacement Components

Component	Pit	Volurr	Name	Description	Action
Base Fluid	Premix 1	50.00	MINERAL OIL	Base Oil	Edit
Transition	Premix 2	50.00		Transition Spacer	Edit
Others	Premix 3	100.10		Cleaning Spacer	Edit
Tail	Premix 4	69.83		Viscous Tail Space	Edit
Others	Active 2	560.00		SeaWater	Edit

Create New Component Delete Selected

Product Concentration

Product	Concentration
SAFE-CARB 40	40.00
SAFE-VIS E	4.21

Component: 2-Transition

Description: Transition Spacer

Pump Rate (gal/min): 50.00

Density (lb/gal): 11.50

Temperature (°F): 80.00

R600: 60.00

R300: 45.00

R200: 25.00

R100: 20.00

R6: 9.00

R3: 8.00

pH: 8.50

YP: 30

LSVP: 7

Recalculate

4.12.1 Displacement Section

This section consists of drop downs, editable fields, and check boxes. Fill in the appropriate information about the displacement. The drop downs and some of the editable fields are known prior to the displacement. The remaining fields and checkboxes will be from observed data and readings during the displacement.

The fluids drop down menus are limited to the mud systems and completions systems defined in the project in the [Lithology / Well Survey / Fluid Systems](#) tab. The types of displacement drop down menu is from the application data.

There are two control buttons in the displacement section.

The Circulation Data button will open the circulation data window in which data recorded during the displacement is recorded. Fill in the tables with the observed results.

The screenshot shows a software window titled "Circulation Data". It contains two data entry tables. The first table, "NTU Data", has columns for Volume, Strokes, NTUs, TSS, and Time. The second table, "Pump Rate vs. Pressure", has columns for Pump Rate and Pressure. Both tables have a single row of data and a green plus icon for adding more rows. An "Exit" button is located at the bottom center of the window.

NTU Data				
Volume	Strokes	NTUs	TSS	Time
250.00	798.00	65.00	0.15	15:00
+				

Pump Rate vs. Pressure	
Pump Rate	Pressure
100.00	2,000.00
+	

Exit

Exit

closes the window and returns to the displacement tab.

Delete Displacement

will delete the displacement tab. It is only active when the displacement has no components.

4.12.2 Displacement Components Section

The displacement components section displays the data about each component.

Displacement Components

Component	Pit	Volume	Name	Description	Action
Base Fluid	Premix 1	50.00	MINERAL OIL	Base Oil	Edit
Transition	Premix 2	50.00		Transition Spacer	Edit
Others	Premix 3	100.10		Cleaning Spacer	Edit
Tail	Premix 4	69.83		Viscous Tail Space	Edit
Others	Active 2	560.00		SeaWater	Edit

Create New Component Delete Selected

Product Concentration

Product	Concentration
SAFE-CARB 40	40.00
SAFE-VIS E	4.21

Component: 2-Transition
Description: Transition Spacer
Pump Rate (gal/min): 50.00

Density (lb/gal): 11.50
Temperature (°F): 80.00
R600: 60.00
R300: 45.00
R200: 25.00
R100: 20.00
R6: 9.00
R3: 8.00
pH: 8.50

YP: 30
LSYP: 7

Recalculate

Before a component may be used, it is necessary to mix it in one of the pits. You have to use the

Add Chemicals

button on the volume accounting tab. Once the components are created,

you add it by pressing the Create New Component button. This will open the create new displacement component window.

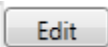
The screenshot shows the 'Create New Displacement Component' window. It contains the following fields and callouts:

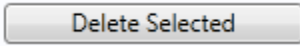
- Component Type:** A dropdown menu with 'Base Fluid' selected. Callout: 'These drop downs are driven by the application data.'
- Component Name:** A text field with 'MINERAL OIL' entered. Callout: 'Description – user defined term to describe component.'
- Description:** A text field with 'Base Oil' entered.
- From Pit:** A dropdown menu with 'Premix 1 - Component 1 - 50 b' selected. Callout: 'Where component is coming from, limited to defined pits.'
- Volume:** A text field with '50.00' and a checkbox labeled 'All Volume'. Callout: 'Volume of component to be used, checking the all volume box will enter calculated end volume and empty the pit.'
- What happened to component?:** Two radio buttons, 'Lost' and 'Reclaimed'. 'Reclaimed' is selected. Callout: 'Radial buttons show disposition of component. If Reclaimed is chosen, the drop down; Reclaimed to Pit, will become Active.'
- Reclaimed to pit:** A dropdown menu with 'Reserve 3 - Reserve - 0 bbl' selected. Callout: 'Pit the component will be sent to upon return from well bore.'
- Buttons:** 'Save' and 'Cancel' buttons at the bottom.

 writes the information to the displacement component section. The product concentration for the component will be filled in with the information from the Volume accounting tab.

HELPFUL HINT	When a component is saved, ONE-TRAX creates the Volume accounting transaction relating to its use.
----------------------------	--

 closes the window without saving the entries.

 will reopen the create new displacement component window with the fields filled in with the information from the component.

 will remove the selected component from the displacement components table.

The right side of the displacement component section contains the rheology information for the component. It also contains the pump rate used for that component. If the component is a breaker, several extra fields will be added to the rheology section.

Rheology Table

Component: 2-Transition
Description: Transition Spacer
Pump Rate (gal/min):

Density (lb/gal):
Temperature (°F):
R600:
R300:
R200:
R100:
R6:
R3:
pH:

YP:
LSYP:

Breaker Rheology Table

Component: 1-Breakers
Description: Breaker

Pump Rate (gal/min):

Delay Time (hrs):

Losses After Breaker (bbl):

Density (lb/gal):

Temperature (°F):

R600:

R300:

R200:

R100:

R6:

R3:

pH:

% Chelant:

YP:

LSYP:

Extra Fields for
use when
component is
Breaker.

Recalculate

will have ONE-TRAX recalculate all volume accounting transactions from the active date.

4.13 FILTRATION TAB

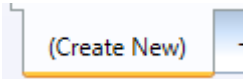
It is drilling fluids engineer's responsibility to collect data from the filtration technician and enter filtration data into ONE-TRAX.

The table included in this chapter lists all the information necessary to complete the filtration tab. Print out or save the table in your computer. Ask the filtration technician to fill out the table for each filtration cycle.

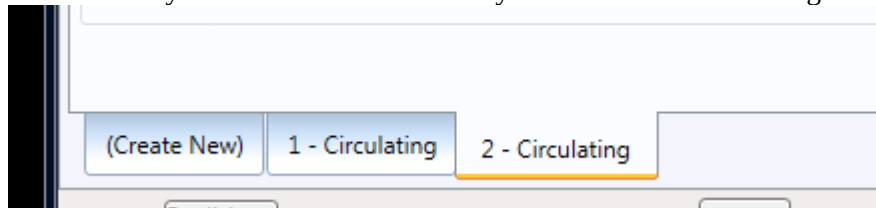
Note: you have to have information about samples taken at suction and discharge for each cycle! In addition to these two samples, you will be able to add the information for up to 10 intermediate samples.

To enter the provided filtration cycle information:

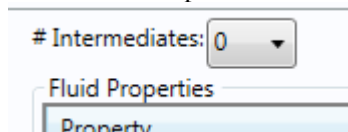
1. Go to Filtration tab

2. Go to  tab in the bottom of the filtration tab. (Create New) tab will be automatically opened if you are creating the very first cycle of the day.

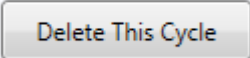
3. Click . Each cycle will be in separate tabs within the filtration tab. Each cycle tab name will contain cycle number and current rig activity. Example:



4. Set the number of intermediates from drop down menu above the fluid properties table (this is the number of samples taken in addition to those taken at the suction and discharge).



5. Enter all information provided in corresponding cells.

6. To delete the cycle tab click  in the tab you want to delete.

Information Required per Filtration Cycle:

1. Cycle Number
2. Rig Activity
3. Fluid System – drop down menu
4. Fluid Weight
5. Suction – drop down menu
6. Discharge – drop down menu
7. Unfilterable Volume
8. Flowrate at the Start of Cycle
9. Flowrate at the End of Cycle
10. Average Flowrate
11. Cycle Start Time
12. Cycle End Time
13. Volume Filtered
14. Number of Cartridges Used
15. Number of DE Sacks Used

Information Required for each Sample taken during the Cycle:

1. NTU's
2. Percentage of Solids

Information required if downtime occurred during the cycle:

1. Downtime Reason – drop down menu
2. Time of Issue
3. Duration of Issue
4. Hours of NPT assigned